TRƯỜNG ĐẠI HỌC FPT

Requirement & Design Specification

Flipped Classroom

Version: 1.0

SWP391_SUM26_G5_Flipped Classroom

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– Hanoi, November 2025 –

Record of Changes

Version	Date	A* M, D	In charge	Change Description
V1.0	31/05/2026	A	Hop, Thuan	Create the initial SRS document, including project overview, system scope, and preliminary requirements.
V1.0	08/06/2026	A	Duong, Cuong	Add functional requirements and update use cases for Blog and Learning Materials features.
V1.0	18/06/2026	A	Thuan, Dung	Add use cases and functional requirements for Assignment and Submission features.
V1.0	25/06/2026	M,D	Duong, Cuong	Revise use case descriptions, update UI screen descriptions, and refine requirement details.
V1.0	04/07/2026	M,D	Dung, Thuan	Update Learning Materials requirements and revise related business processes.
V1.0	08/07/2026	A	Dung, Thuan	Extend functional requirements for file management and assignment submission.
V1.0	15/07/2026	A	Duong, Cuong	Add Quiz and Feedback requirements and update the Authentication section.
V1.0	20/07/2026	M,D	Hop, Thuan	Revise Account Management requirements and update related use cases.
V1.0	22/07/2026	M,D	Cuong, Hop	Update Quiz requirements and revise system specifications.
V1.0	24/07/2026	M,D	All Members	Review, revise, and finalize the SRS document before submission.

*A - Added M - Modified D - Deleted

Contents

Record of Changes **2**

I. Overview **6**

1. System Context 6
2. External Entities 6
3. Business Processes 8
   3.1 Student Assignment & Grading Process 8
   3.2 Class Setup & Lecturer Assign Process 9
   3.3 Student Learning & Quiz Process 10
4. User Requirements 10

4.1 UCs for Admin..... 15
4.2 UCs for Student..... 18
4.3 UCs for Lecturer..... 21
5. System Functionalities..... 23
5.1 Screens Flow..... 23
5.2 Screen Authorization..... 24
5.3 Non-UI Functions..... 25
II. Functional Requirements..... 27
1. User Authentication Features..... 27
1.1 User Login..... 27
1.1.1 Login Screen..... 27
1.2 Password Recovery & Reset..... 28
1.2.1 Forgot Password Screen..... 28
1.2.2 Reset Password Screen..... 30
2. System Administration Features..... 31
2.1 Account Management..... 31
2.1.1 Account Management Screen..... 31
2.1.2 Batch User Import Modal..... 33
2.1.3 Edit User Account Modal..... 35
2.1.4 Export Student List Modal..... 37
2.2 Course Management..... 38
2.2.1 Course Management Screen..... 38
2.2.2 Create / Edit Course Modal..... 40
2.3 Academic Terms Management..... 41
2.3.1 Academic Term Management Screen..... 41
2.3.2 Create / Edit Course Modal..... 43
2.4 Class Management..... 44
2.4.1 Class Management Screen..... 44
3. Student Learning Management System..... 46
3.1. Course Management & Overview..... 46
3.1.1. My Classes..... 46
3.1.2.Class Members..... 47
3.2. Learning Roadmap & Materials..... 48
3.2.1. Student Roadmap..... 48
3.2.2. Learning Materials..... 49
3.2.3. Lesson Discussion..... 51
3.3. Quizzes System & Assignments..... 53
3.3.1. Create Quiz..... 53
3.3.2 Quiz Overview..... 56

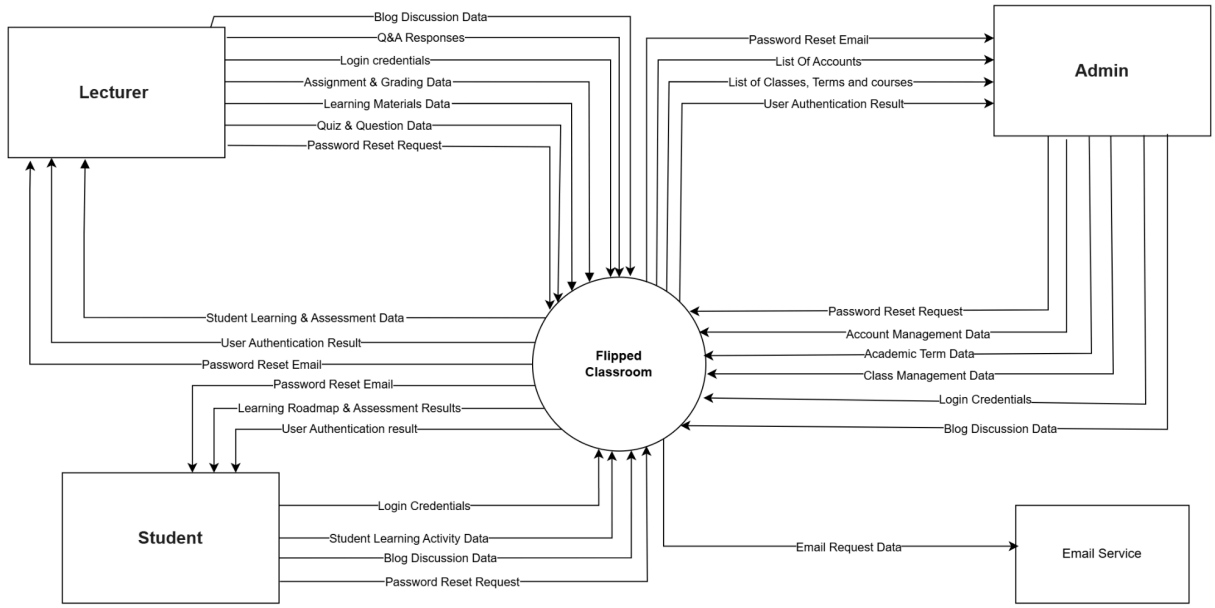
3.3.3. Quiz Player (Quiz Execution)....57
3.3.4. Quiz History Table....58
3.3.5. Quiz Attempt Detail....60
3.3.6. Assignment List and Submission....61
3.4. Class Stream....64
3.4.1 Class Stream Overview....64
4. Blog Management....66
4.1.1 Blog list....66
4.1.2 Create Blog....68
4.1.3 Update Blog....70
4.1.4 Blog Detail....71
5. Lecturer Workspace Management....73
5.1.1 Lecturer Dashboard....73
5.2. Class List & Role Management....74
5.2.1. Student Roster....74
5.2.2. Role Promotion....75
5.3. Learning Materials Management....76
5.3.1. Material List....76
5.3.2. Upload Material....77
5.3.3. Edit Material....79
5.3.4. Chapter Management....80
5.4. Assignment Management....81
5.4.1. Assignment List....81
5.4.2. Create Assignment....82
5.4.3. Edit Assignment....83
5.5. Grading Board....84
5.5.1. Submission Overview....84
5.5.2. Grade Submission....85
5.6. Feedback Support....86
5.6.1. Feedback List....86
5.6.2. Reply Feedback....87
5.7. Student Progress Tracking....88
5.7.1. Progress by Individual....88
5.7.2. Progress by Material....89
III. System Design....90
1. Software Architecture....90
1.1 Overall Architectural Diagram....90
1.2 Component Descriptions....90
2. Code Package Design....92

2.1 Package Diagram..... 92
3. Database Design..... 94
3.1 Database Schema..... 94
3.2 Table Descriptions..... 94
3.2.1 Users..... 94
3.2.3 AcademicTerms..... 95
3.2.3 Courses..... 95
3.2.4 Classes..... 96
3.2.5 ClassStudents..... 96
3.2.6 LearningMaterials..... 97
3.2.7 MaterialCompletions..... 98
3.2.8 QuizAnswers..... 98
3.2.9 QuizAttempts..... 98
3.2.10 QuizOptions..... 99
3.2.11 Quizzes..... 99
3.2.12 Submissions..... 100
3.2.13 SupportFeedbacks..... 100
3.2.14 Comments..... 101
3.2.15 DiscussionReplies..... 101
3.2.16 DiscussionThreads..... 102
3.2.17 QuizQuestions..... 102
3.2.18 Assignments..... 103
3.2.19 Blogs..... 103
3.2.20 ClassSessions..... 104

I. Overview

1. System Context

The Flipped Classroom System is a specialized educational software application designed to transform traditional teaching methods by facilitating pre-class preparation and active in-class learning. The context diagram below illustrates the external entities and system interfaces for release 1.0. The system is expected to evolve over several releases, ultimately connecting to advanced institutional learning analytics and academic management frameworks.



Link Context Diagram

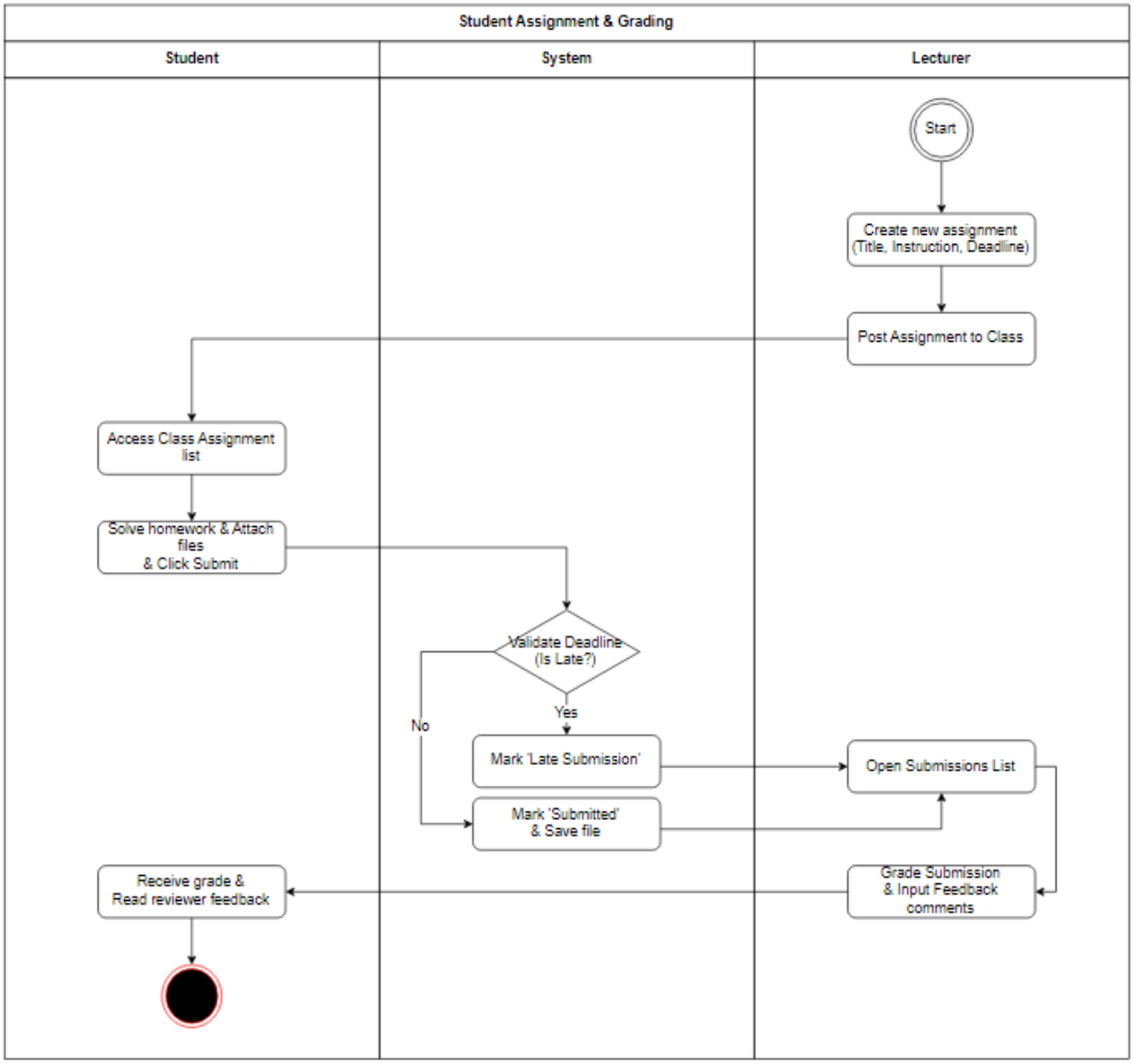
2. External Entities

#	Entity	Description
1	Admin	Responsible for system configuration, user account management, academic term setup, class initialization, and forum moderation. The Admin inputs critical administrative data into the system, including Login Credentials, Account Management Data, Academic Term Data, Class Management Data, Blog Discussion Data, and Password Reset Request. In return, the system processes these inputs and delivers administrative outputs, including User Authentication Result, List Of Accounts, List of Classes, Terms and courses, and Password Reset Email to support system governance and monitoring.
2	Student	The core learners who interact with the system to complete preparatory study tasks, participate in quizzes, submit assignments, and engage in class discussions. Students interact with the platform

		by providing Login Credentials, Student Learning Activity Data (comprising assignment submissions, quiz attempts, material completion markers, and help requests), Blog Discussion Data, and Password Reset Request. In response, the platform provides User Authentication Result, Learning Roadmap & Assessment Results, and Password Reset Email to guide their learning journey.
3	Lecture	Responsible for publishing course materials, creating and managing quizzes, grading assignments, answering student inquiries, and tracking learning progress. Lecturers provide input to the system through Login Credentials, Learning Materials Data, Quiz & Question Data, Assignment & Grading Data, Q&A Responses, Blog Discussion Data, and Password Reset Request. To support their teaching workflow, the system delivers User Authentication Result, Student Learning & Assessment Data, and Password Reset Email back to the lecturer.
4	Email Service	An external third-party SMTP mail delivery system (e.g., Gmail/Brevo SMTP Server) integrated with the platform. The system sends Email Request Data to the Email Service to automatically dispatch password reset link emails to users' inboxes when a password recovery operation is requested.

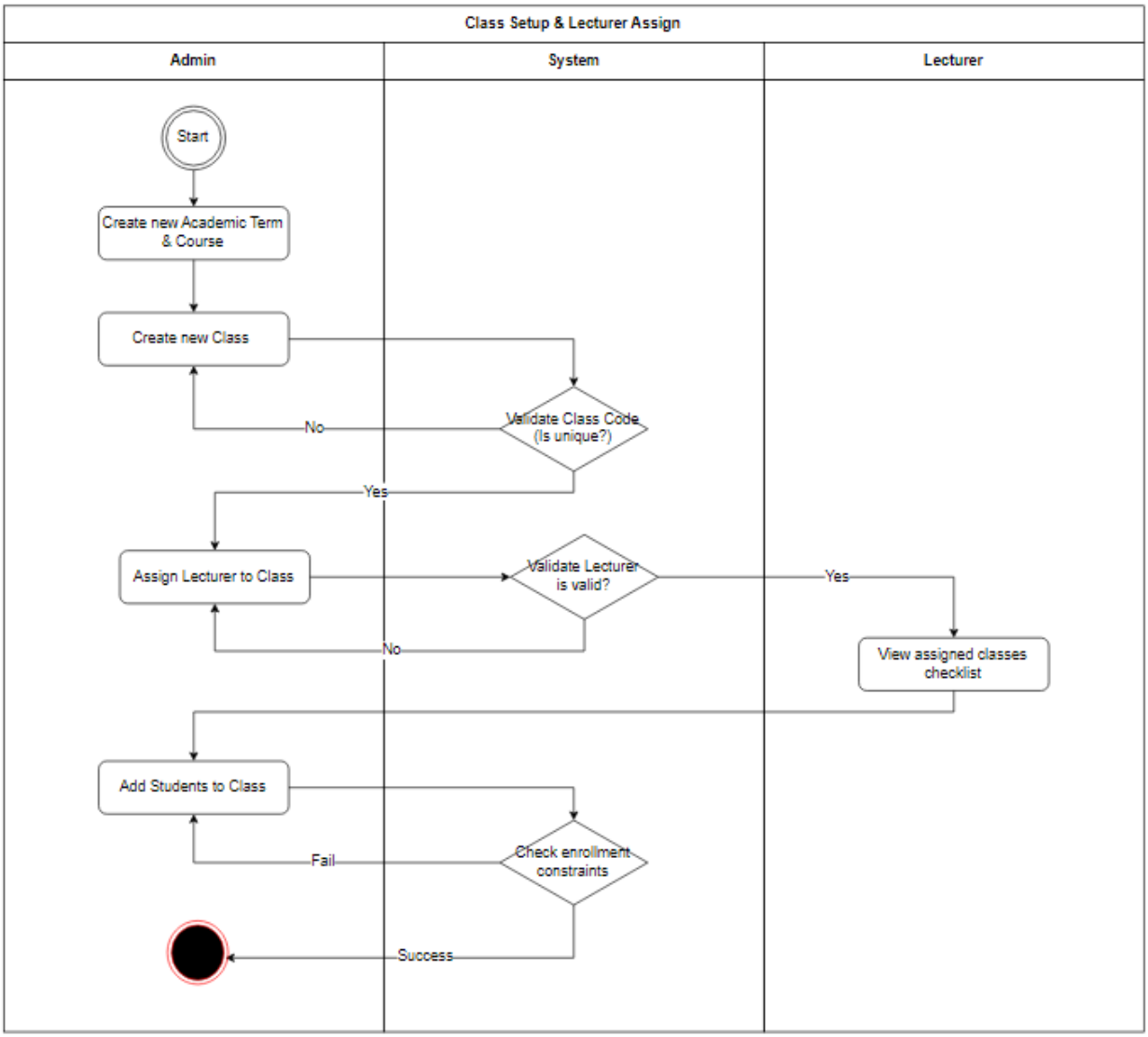
3. Business Processes

3.1 Student Assignment & Grading Process



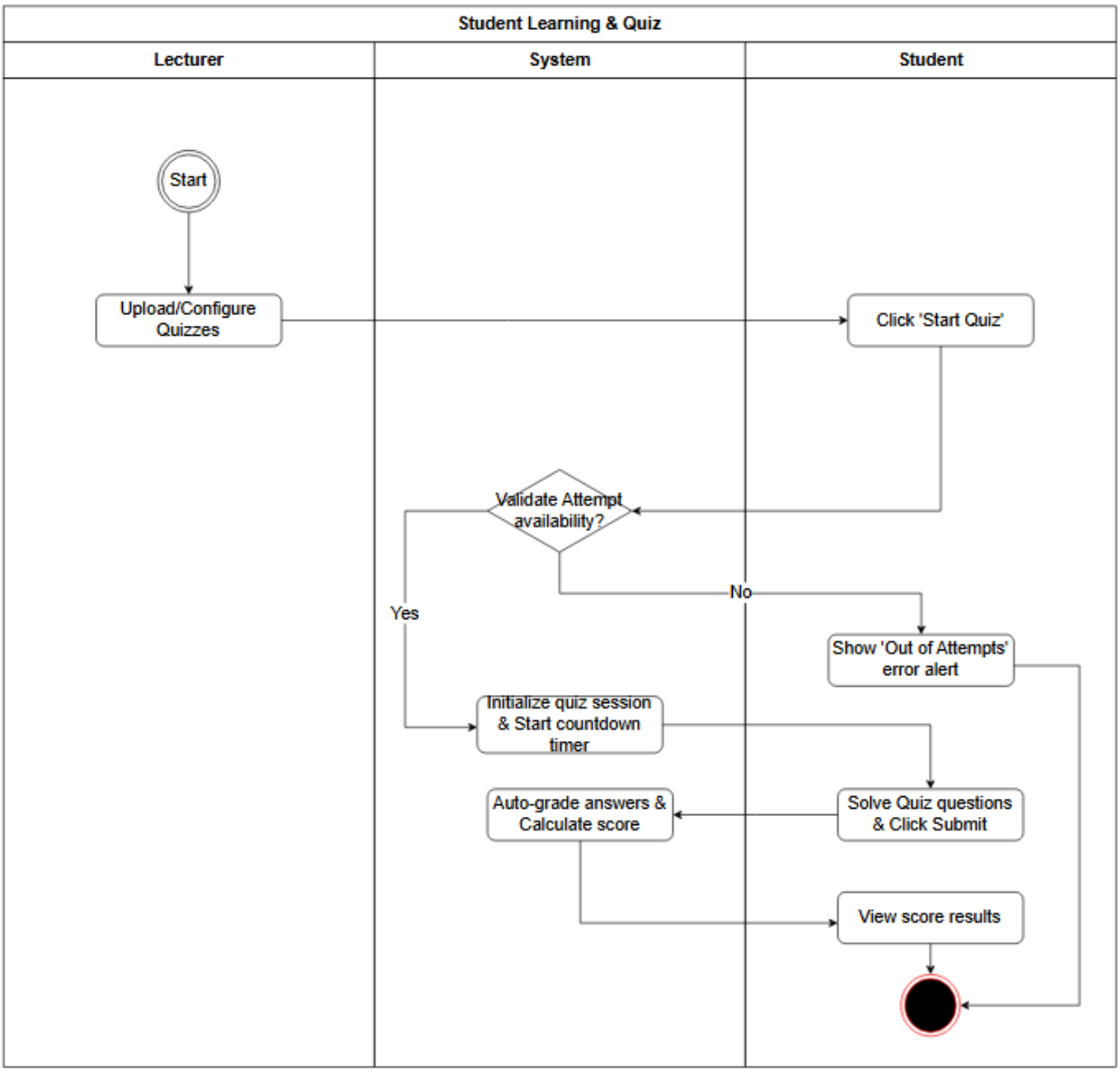
Link Diagram

3.2 Class Setup & Lecturer Assign Process



Link Diagram

3.3 Student Learning & Quiz Process



Link Diagram

4. User Requirements

ID	Use Case	Feature	Main Actor	Use Case Description
01	User Login	Authentication	Admin, Lecturer, Student	Authenticate users (Admin, Lecturer, Student) via Email and Password to access system features.

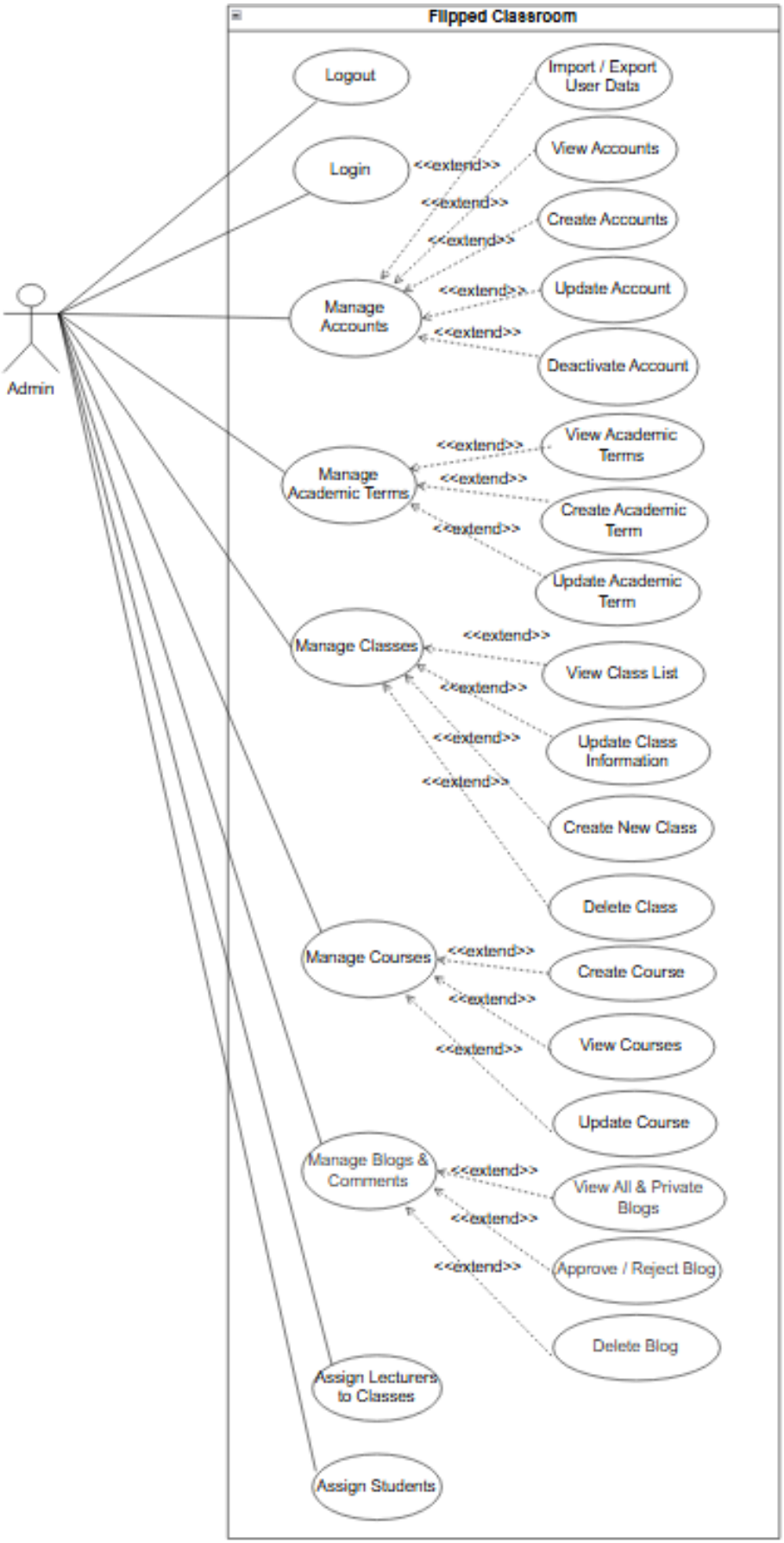
02	User Logout	Authentication	Admin, Lecturer, Student	Safely terminate active user sessions, clear stored tokens, and redirect users to the login page.
03	Manage Accounts	Account Management	Admin	Admin can manage all user accounts in the system including students and lecturers.
04	View Accounts	Account Monitoring	Admin	Admin can view account details, roles, and account status of all users.
05	Create Accounts	Account Creation	Admin	Admin can create new student or lecturer accounts with necessary information and permissions.
06	Import / Export User Data	Account Management	Admin	Admin can import user accounts in batch from Excel/CSV files and export student lists to CSV format.
07	Update Account	Account Maintenance	Admin	Admin can edit user account information such as name, email, role, and password.
08	Deactivate Account	Account Control	Admin	Admin can delete or deactivate accounts that are no longer active or valid.
09	Manage Academic Terms	Academic Term Management	Admin	Admin can manage academic semesters or terms in the system.
10	View Academic Terms	Academic Term Monitoring	Admin	Admin can view existing academic terms and their details.
11	Update Academic Term	Academic Term Maintenance	Admin	Admin can update information related to academic terms when necessary.
12	Manage Classes	Class Management	Admin	Admin can manage all classes available in the system.
13	View Class List	Class Monitoring	Admin	Admin can view the list of classes, schedules, and assigned lecturers.
14	Create New Class	Class Creation	Admin	Admin can create new classes and assign related course information.
15	Update Class Information	Class Maintenance	Admin	Admin can modify class details such as class name, schedule, and capacity.
16	Delete Classes	Class Control	Admin	Admin can remove classes that are canceled or no longer available.
17	Assign Lecturers to Classes	Lecturer Assignment	Admin	Admin can assign lecturers to teach specific classes in the system.
18	Assign Students	Student Assignment	Admin	Admin can assign or enroll students into appropriate classes.

19	Submit Question Feedback	Student Support	Student, Lecturer	Submit private questions or help requests related to lessons.
20	View Personal Learning Roadmap	Learning Roadmap	Student	Students can view personalized learning progress, milestones, and recommended learning paths.
21	View Class	Class Management	Student	Students can view enrolled classes, schedules, lesson materials, and announcements.
22	View Student Information	Profile Management	Student	Students can view personal information such as student ID, email, enrolled courses, and academic status.
23	View Materials	Learning Materials	Student, Lecturer	Students, Lecture can view learning materials organized by chapter and access their content.
24	Download Materials	Learning Materials	Student	Students can download learning materials for offline learning.
25	Mark Material as Completed	Learning Materials	Student	Students can mark learning materials as completed to update their learning progress.
26	View Assignments	Assignment	Student	Students can view assignments, deadlines, instructions, attached files, and submission status.
27	Submit Assignment	Assignment	Student	Students can upload and submit assignment files before the deadline.
28	View Grades & Feedback	Assignment	Student	Students can view grades and feedback after their assignments are evaluated.
29	Resubmit Assignment	Assignment	Student	Students can resubmit their assignments before grading and before the submission deadline.
30	Manage Learning Materials	Course Management	Lecturer	Lecturers can manage course materials and learning resources for assigned classes.
31	Upload Materials	Material Upload	Lecturer	Lecturers can upload lecture slides, documents, videos, and other learning materials for students.
32	Update Materials	Material Maintenance	Lecturer	Lecturers can edit or update uploaded learning materials when necessary.
33	Delete Materials	Material Control	Lecturer	Lecturers can remove outdated or incorrect learning materials from the system.

34	Manage Class	Class Management	Lecturer	Lecturers can manage learning activities and assignments within assigned classes.
35	Create Assignment	Assignment Management	Lecturer	Lecturers can create assignments, quizzes, or tasks for students in a class.
36	Provide Feedback	Feedback Management	Lecturer	Lecturers can provide comments and feedback on student submissions and learning performance.
37	Respond Support Forum Thread	Private Support Forum	Lecturer, Student	Lecturers can respond to students' questions, comments, or discussion feedback in the system.
38	Grade Assignment	Grading System	Lecturer	Lecturers can evaluate and assign grades to submitted assignments and activities.
39	View Student Progress	Progress Monitoring	Lecturer	Lecturers can monitor student learning progress and class performance statistics.
40	Track Student	Student Tracking	Lecturer	Lecturers can track individual student participation, submission status, and academic performance.
41	Manage Courses	Course Management	Admin	Admin can manage the list of courses in the system.
42	Create Course	Course Management	Admin	Admin can create a new course.
43	View Courses	Course Management	Admin	Admin can view detailed information of courses.
44	Update Course	Course Management	Admin	Admin can edit course information.
45	Manage Blogs	Blog Management	Admin, Lecturer, Student	Admin can manage user blog posts, Lecture and Student can manage their blog.
46	All & Private Blogs	Blog Management	Admin	Admin can view all blogs, including private ones.
47	Approve / Reject Blog	Blog Management	Admin	Admin has the right to approve or reject posted blog content.
48	Delete Blog	Blog Management	Admin, Lecturer, Student	Admin can delete violating or inappropriate blogs, Lecturer and Student can delete personal blog.
49	Forgot password	Authentication	Admin, Lecturer, Student	Users (Admin, Lecturer, Student) can request a new password when they forget their login password via email recovery.

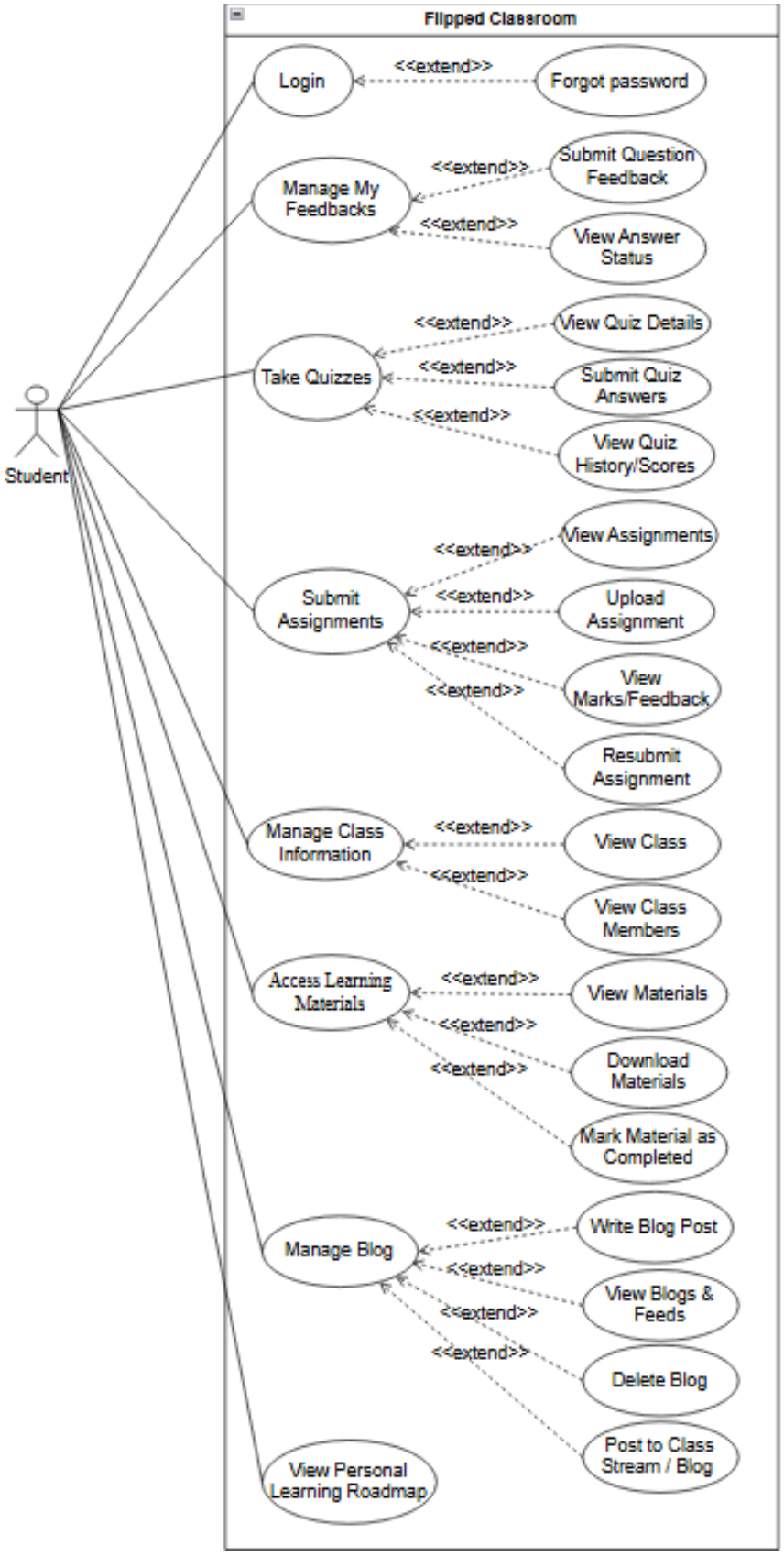
50	Manage Quizzes	Quiz Management	Lecturer	Lecturers can manage quiz lists and overall configurations.
51	Take Quizzes	Quiz Management	Student	Students can take quizzes assigned by lecturers.
52	View Quiz Details	Quiz Management	Student	Students can view the time limit, number of questions, and rules of a quiz.
53	Submit Quiz Answers	Quiz Management	Student	Students can submit their quiz answers to the system.
54	View Quiz History/Scores	Quiz Management	Student	Students can review their quiz history and scores.
55	Write Blog Post	Blog Management	Student, Lecturer, Admin	Students can write and publish personal blog posts to share knowledge.
56	Post to Class Stream / Blog	Class Newsfeed / Blog	Student	Directly post quick updates or private blogs onto the class newsfeed without prior approval.
57	Role Promotion	Class Management	Lecturer	Lecturers can propose role promotions for students (e.g., teaching assistant).
58	Create Quizzes	Quiz Management	Lecturer	Lecturers can create new multiple-choice quizzes with time limits, max attempts, and question sets for assigned classes.
59	View Quizzes	Quiz Management	Lecturer	Lecturers can view the list of created quizzes.
60	Update Quiz	Quiz Management	Lecturer	Lecturers can edit quiz content, questions, or time limits.
61	Delete Quiz	Quiz Management	Lecturer	Lecturers can delete quizzes that have no student attempts from the system.

4.1 UCs for Admin


Flipped Classroom
Admin
Logout
Login
Manage Accounts
Import / Export User Data
View Accounts
Create Accounts
Update Account
Deactivate Account
Manage Academic Terms
View Academic Terms
Create Academic Term
Update Academic Term
Manage Classes
View Class List
Update Class Information
Create New Class
Delete Class
Manage Courses
Create Course
View Courses
Update Course
Manage Blogs & Comments
View All & Private Blogs
Approve / Reject Blog
Delete Blog
Assign Lecturers to Classes
Assign Students
<<extend>>

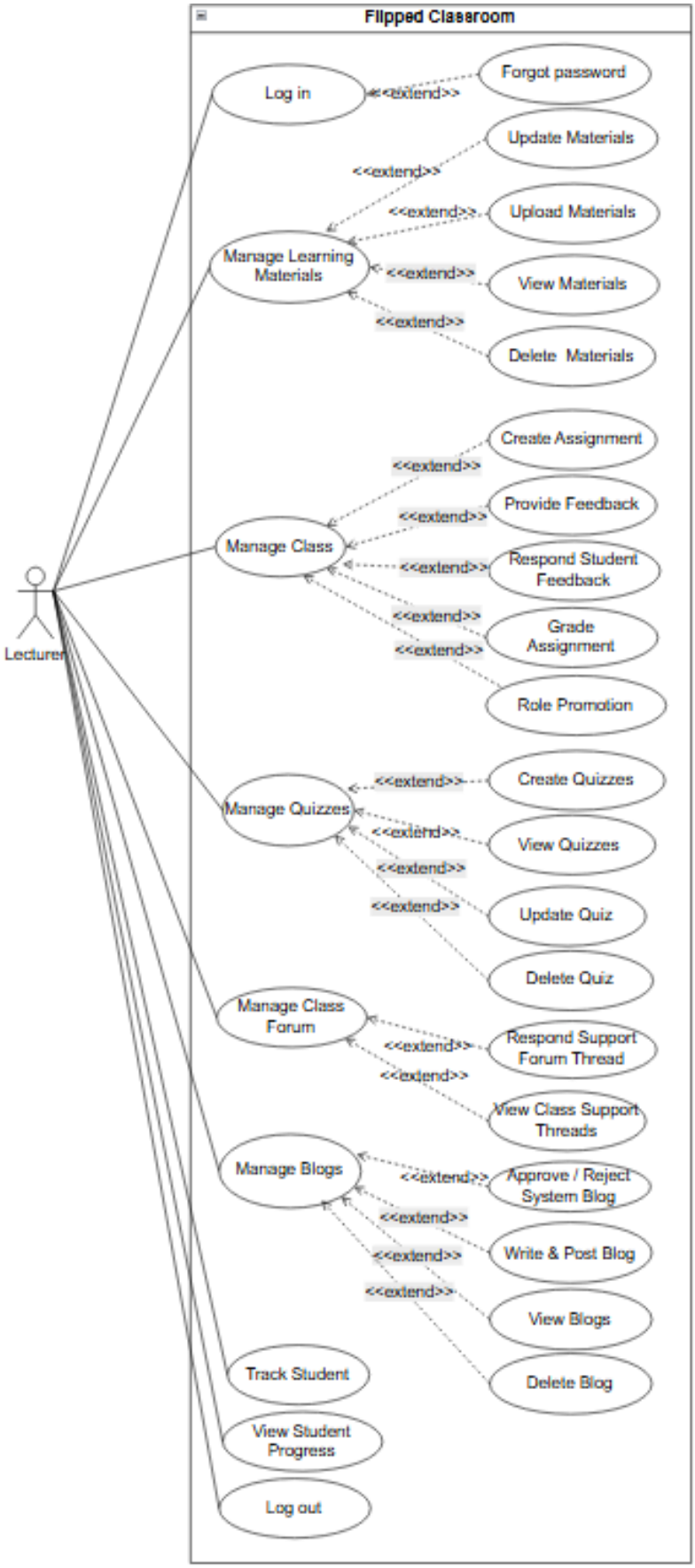
Link UseCase Diagram

4.2 UCs for Student


Flipped Classroom
Student
Login
Forgot password
Manage My Feedbacks
Submit Question Feedback
View Answer Status
Take Quizzes
View Quiz Details
Submit Quiz Answers
View Quiz History/Scores
Submit Assignments
View Assignments
Upload Assignment
View Marks/Feedback
Resubmit Assignment
Manage Class Information
View Class
View Class Members
Access Learning Materials
View Materials
Download Materials
Mark Material as Completed
Manage Blog
Write Blog Post
View Blogs & Feeds
Delete Blog
Post to Class Stream / Blog
View Personal Learning Roadmap
<<extend>>

Link UseCase Diagram

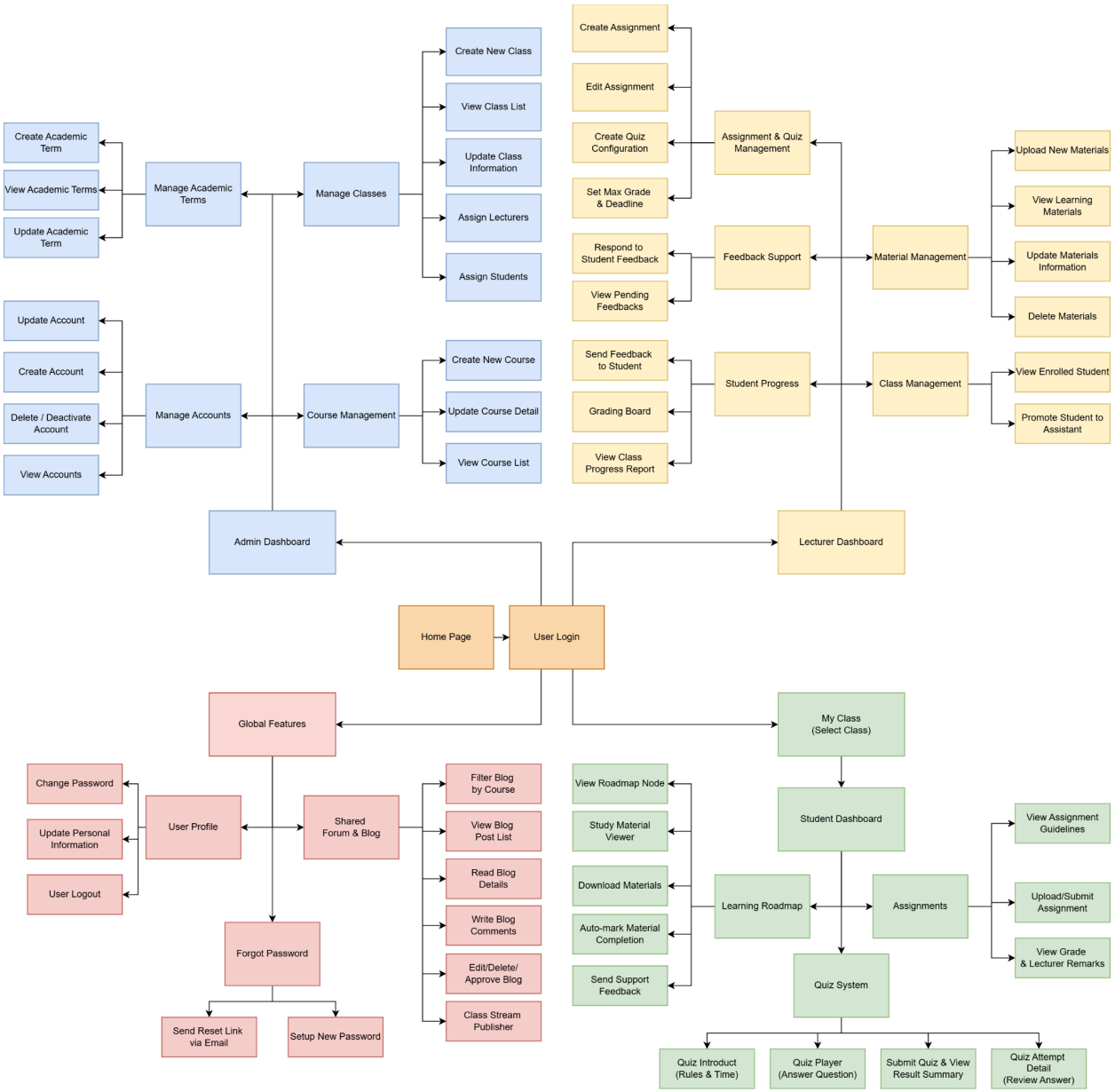
4.3 UCs for Lecturer


Flipped Classroom
Lecturer
Log in
Forgot password
<<extend>>
Update Materials
<<extend>>
Upload Materials
<<extend>>
Manage Learning Materials
<<extend>>
View Materials
<<extend>>
Delete Materials
Create Assignment
<<extend>>
Provide Feedback
<<extend>>
Manage Class
Respond Student Feedback
<<extend>>
<<extend>>
Grade Assignment
<<extend>>
Role Promotion
Create Quizzes
<<extend>>
Manage Quizzes
View Quizzes
<<extend>>
<<extend>>
Update Quiz
<<extend>>
Delete Quiz
Manage Class Forum
Respond Support Forum Thread
<<extend>>
<<extend>>
View Class Support Threads
Manage Blogs
Approve / Reject System Blog
<<extend>>
<<extend>>
Write & Post Blog
<<extend>>
View Blogs
<<extend>>
Delete Blog
Track Student
View Student Progress
Log out

[Link UseCase Diagram]

5. System Functionalities

5.1 Screens Flow



[Link Screen Flow]

5.2 Screen Authorization

Screen	Admin	Lecturer	Student
Home Page / Landing Page			
Query All Data (Xem giao diện chung)	x	x	x
User Login & Authentication			
Query Own Data (Xác thực tài khoản)	x	x	x
Forgot & Reset Password			
Add New Data (Gửi yêu cầu reset)	x	x	x
Update Own Data (Đổi mật khẩu mới)	x	x	x
User Profile			
Query Own Data (Xem thông tin cá nhân)	x	x	x
Update Own Data (Chỉnh sửa hồ sơ)	x	x	x
Admin Dashboard (Overview)			
Query All Data (Xem thống kê tổng quan)	x		
Manage Accounts (Danh sách tài khoản)			
Query All Data (Tìm kiếm, Lọc tài khoản)	x		
Add New Data (Tạo tài khoản mới)	x		
Update All Data (Chỉnh sửa thông tin)	x		
Delete Data (Khóa/Xóa tài khoản)	x		
Course Management (Quản lý Môn học)			
Query All Data (Xem danh sách môn học)	x		
Add New Data (Thêm môn học mới)	x		
Update All Data (Sửa thông tin môn)	x		
Manage Academic Terms (Kỳ học)			
Query All Data (Xem danh sách kỳ học)	x		
Add New Data (Tạo kỳ học mới)	x		
Update All Data (Đóng/Mở kỳ học)	x		
Manage Classes (Quản lý Lớp học)			
Query All Data (Xem danh sách toàn bộ lớp)	x		
Add New Data (Tạo lớp học mới)	x		
Update All Data (Gán Giảng viên/Sinh viên)	x		
Lecturer Dashboard (Thống kê Lớp)			
Query Own Data (Xem thống kê lớp mình)		x	
Class List & Role Promotion			
Query Own Data (Xem sĩ số, ds sinh viên)		x	
Update Own Data (Gán chức trợ giảng)		x	
Material Management (Quản lý Học liệu)			
Query Own Data (Xem kho học liệu)		x	
Add New Data (Tải lên Học liệu)		x	
Update Own Data (Sửa mô tả, giới hạn)		x	
Delete Data (Xóa/Ẩn học liệu)		x	
Assignment & Quiz Management			

Query Own Data (Xem ds bài tập/quiz)		x	
Add New Data (Ra bài tập, Tạo đề Quiz)		x	
Update Own Data (Gia hạn, Đổi đề thi)		x	
Student Progress & Grading Board			
Query Own Data (Xem bài nộp của Sinh)		x	
Update Own Data (Chấm điểm, nhận xét)		x	
Feedback Support (Hỗ trợ câu hỏi)			
Query Own Data (Xem câu hỏi gửi đến)		x	
Update Own Data (Trả lời feedback)		x	
Student Dashboard (My Classes)			
Query Own Data (Xem danh lớp đang học)			x
Learning Roadmap (Lộ trình học)			
Query Own Data (Xem tiến trình Roadmap)			x
Study Material Viewer			
Query All Data (Xem Video, Tải File)			x
Add New Data (Auto-mark completed)			x
Submit Assignments (Nộp Bài tập)			
Add New Data (Upload file bài làm)			x
Query Own Data (Xem điểm, Lời phê)			x
Quiz Player (Thi trắc nghiệm)			
Add New Data (Làm bài & Nộp bài)			x
Query Own Data (Xem kết quả, đáp án)			x
Shared Forum & Blog (Diễn đàn chung)			
Query All Data (Xem, Lọc bài viết)	x	x	x
Add New Data (Đăng Blog, Comment)	x	x	x
Update Own Data (Sửa bài/Comment)	x	x	x
Delete Data (Xóa bài)	x (All)	x (Own)	x (Own)

5.3 Non-UI Functions

#	Feature	System Function	Description
1	Security & Auth	JWT Token Verification & Session Timeout	Generates Access and Refresh tokens upon login. Automatically validates session expiration in the background and triggers an auto-logout process to ensure account security.
2	Password Encryption	BCrypt Hashing Algorithm	A background security mechanism that hashes user passwords into irreversible strings before storing them in the database, preventing internal data breaches.
3	Cloud File Handling	Cloudinary Asset Upload & Optimization	Background API integration with Cloudinary. Automatically converts, compresses, and optimizes image/video files for faster loading, and saves the secure URL to the database.

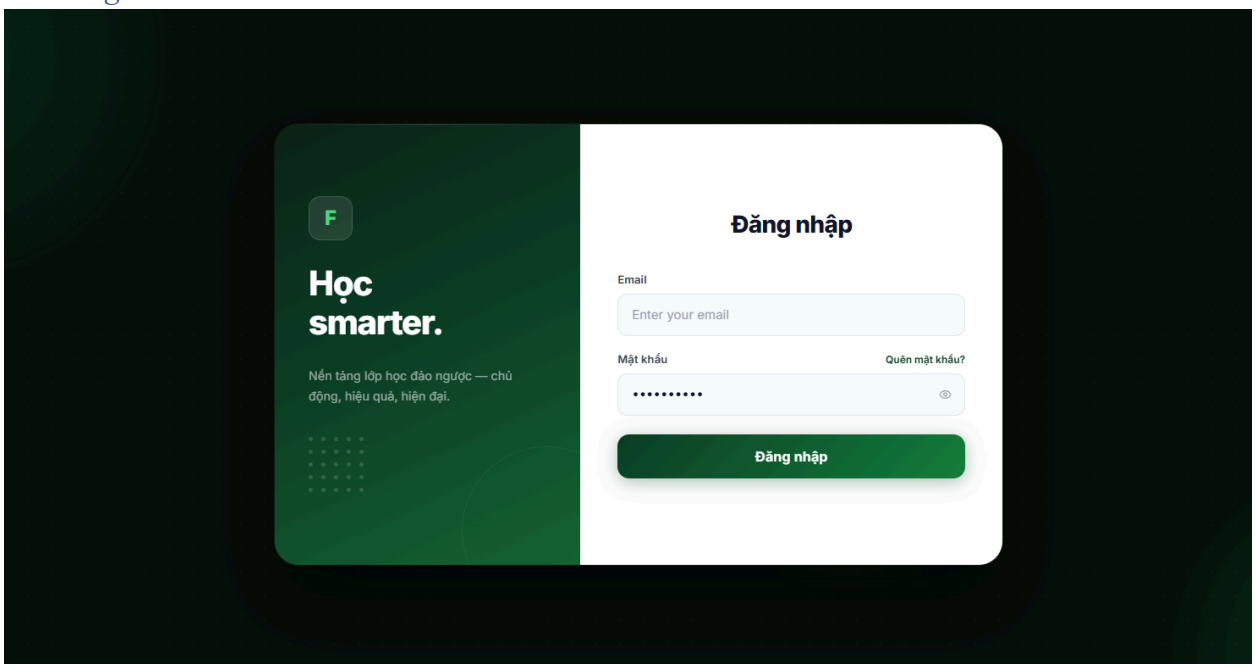
4	Quiz Auto-Grading	Background Assessment Calculator	Automatically cross-references student quiz submissions with the encrypted answer keys. Calculates the final score (Scale 10 or 100) and updates the pass/fail status in milliseconds upon submission.
5	Roadmap Tracker	Material Completion Listener	A background event listener that detects when a student scrolls to the end of a PDF document or reaches the required video completion percentage, subsequently triggering an automatic "Mark as Completed" action.
6	Email Service	SMTP Mail Dispatcher	A background job that connects to the SMTP server (e.g., Gmail/SendGrid) to package and deliver real-time automated emails containing OTPs or password reset links.
7	Term Auto-Scheduler	Academic Term Status Cronjob	A background cronjob that scans the database daily to automatically transition the status of Academic Terms (Upcoming ➔ Active ➔ Completed) based on their predefined Start and End Dates.
8	Assignment Status	Deadline Enforcement Validator	A backend validation mechanism that automatically intercepts and rejects assignment submission API requests if the current timestamp exceeds the lecturer's strict deadline, ensuring absolute fairness.
9	Soft Delete System	Entity IsDeleted Flagging	When a user attempts to delete an entity (Account, Course, Material), the system bypasses hard deletion and instead toggles an IsDeleted = true flag in the background. This hides the record from the UI while preserving data for future recovery.

II. Functional Requirements

1. User Authentication Features

1.1 User Login

1.1.1 Login Screen



This screen allows all users (Admin, Lecturer, Student) to:

- Enter Account Credentials: input registered Email and Password to authenticate with the system.
- Toggle Password Visibility: show or hide plain text password by clicking the eye icon.
- Recover Forgotten Password: initiate password reset workflow if credentials are lost.

On the screen, s/he can also:

- Submit Authentication: click the "Sign In" button to send login request to the server and receive JWT authentication token.
- Choose to go to Password Recovery screen by clicking the "Forgot password?" link.

Mapped Use Case: UC-01: User Login

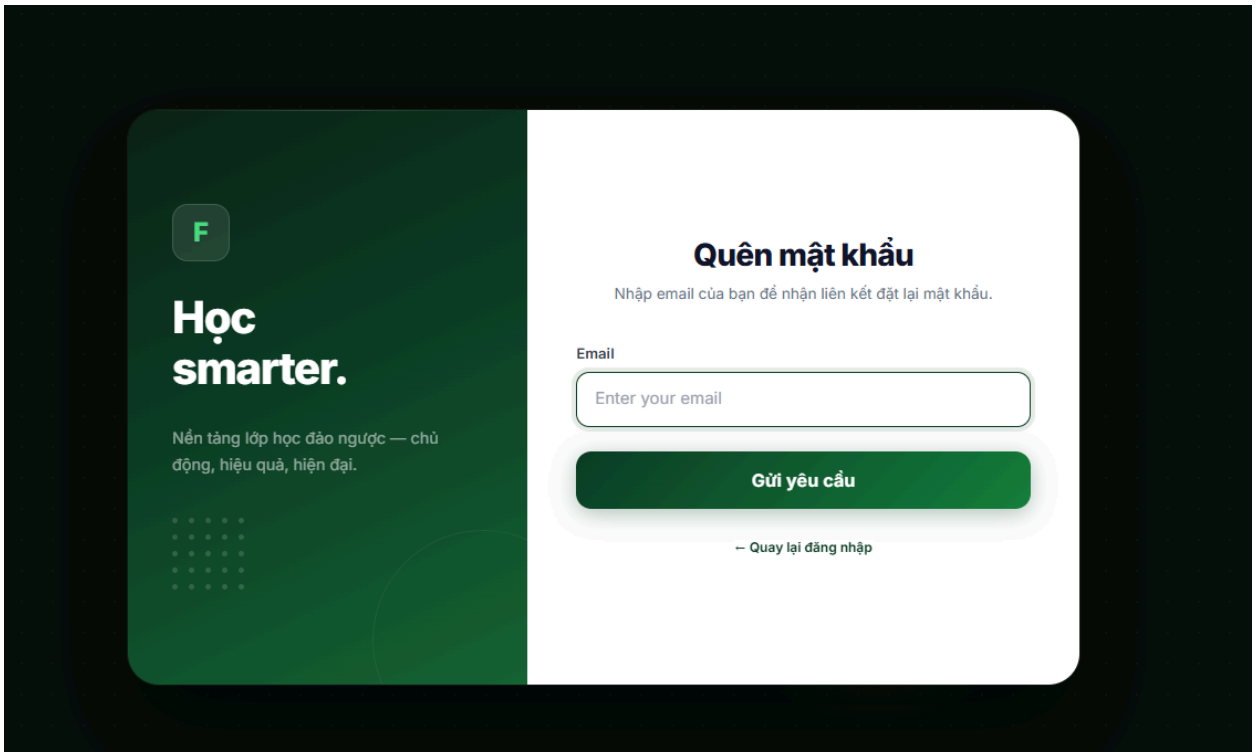
Field Description

Field Name	Description
Email	Textbox. Required. Accepts the user's registered email address (e.g., student@fpt.edu.vn). Placeholder: Enter your email. Validates standard email format.
Password	Password Input. Required. Accepts the user's account password. Placeholder: ••••••••. Includes an interactive eye icon to toggle password visibility.

Forgot password? (Quên mật khẩu)	Hyperlink. Redirects the user to the Password Recovery page (/forgot-password) to request a password reset link.
Sign In (Đăng nhập)	Primary Button. Sends authentication request to POST /api/Auth/login. Displays a loading spinner during API execution and redirects to the user's dashboard upon success.
Error Alert Banner	Dynamic Alert Box. Displays authentication error message (*"Email hoặc mật khẩu không đúng. Vui lòng thử lại"*) when invalid credentials are provided.

1.2 Password Recovery & Reset

1.2.1 Forgot Password Screen



This screen allows all users (Admin, Lecturer, Student) to:

- Enter Registered Email: input their system-registered email address to request a password recovery link.
- Receive Password Recovery Email: receive an automated email containing a password reset token via EmailService.

On the screen, s/he can also:

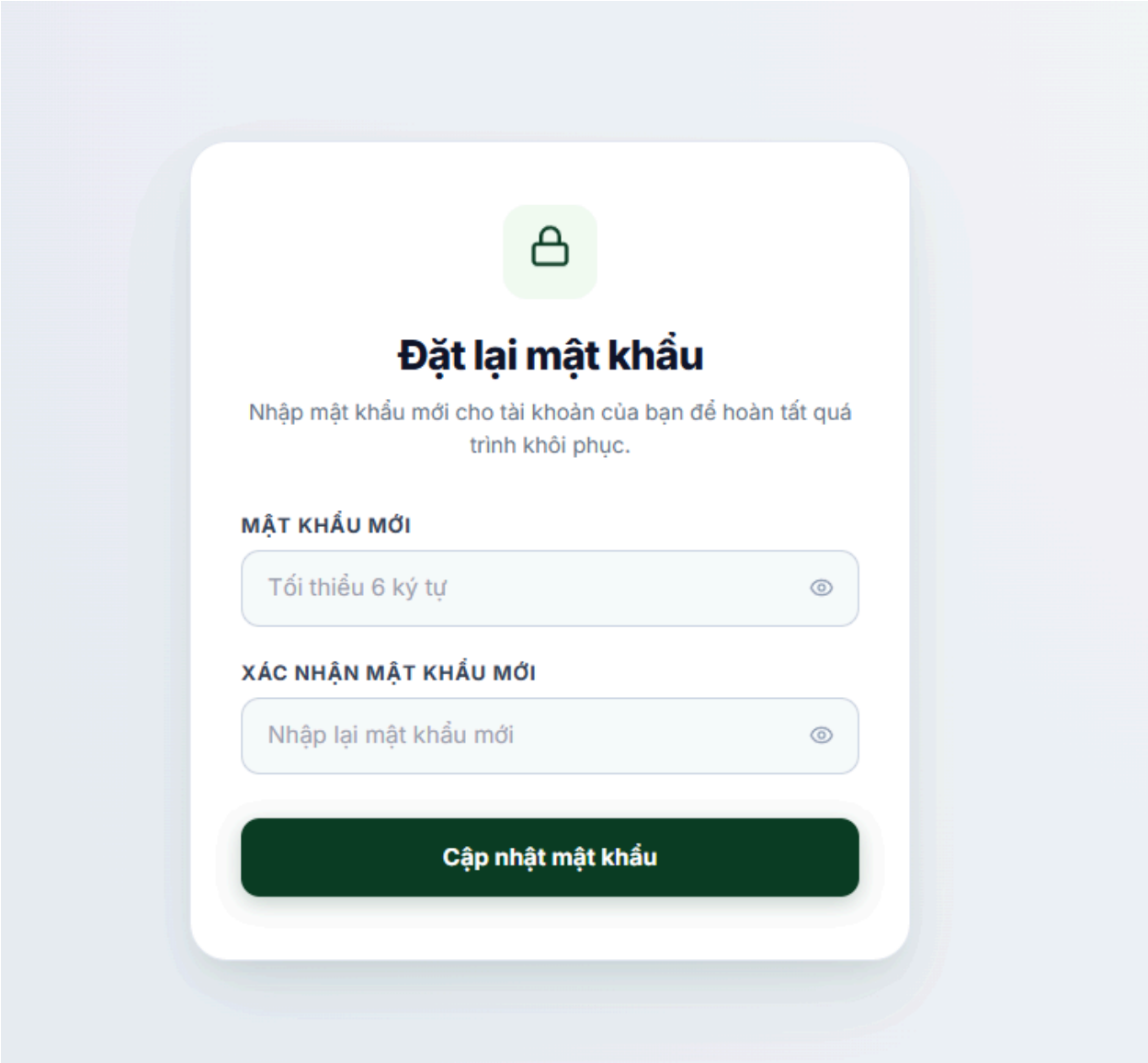
- Send Recovery Request: click the "Gửi yêu cầu" button to send the password reset request to the API.
- Choose to return to the Login screen by clicking the "← Quay lại đăng nhập" link.

Mapped Use Case: UC-49: Forgot password

Field Description

Field Name	Description
Email	Textbox. Required. Accepts the user's registered email address. Placeholder: Enter your email. Validates standard email format.
Gửi yêu cầu	Primary Button. Sends a password recovery request to API POST /api/Auth/forgot-password. Displays a loading spinner during execution.
Quay lại đăng nhập	Link Button. Redirects the user back to the main Login Screen.
Success Alert Banner	Notification Box. Displays success message (*"Nếu email tồn tại trong hệ thống, link khôi phục mật khẩu đã được gửi đi."*) upon valid submission.

1.2.2 Reset Password Screen



This screen allows all users (Admin, Lecturer, Student) to:

- Enter New Password: input a new account password meeting minimum length requirements.
- Confirm New Password: re-enter the new password to ensure exact match and avoid input mistakes.
- Toggle Password Visibility: click the eye icon to view or hide plain text passwords.

On the screen, s/he can also:

- Update Password: click the "Cập nhật mật khẩu" button to submit the new password and complete recovery.

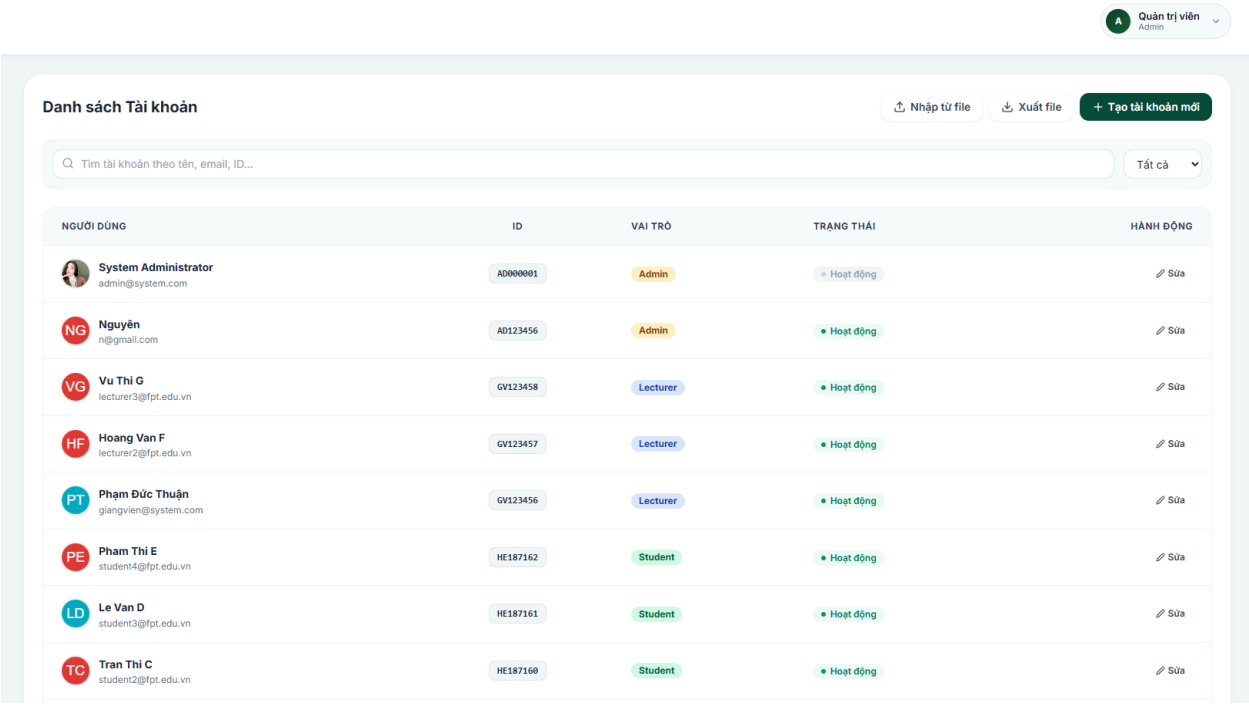
Field Description

Field Name	Description
NEW PASSWORD	Password Input. Required. Accepts the new account password. Placeholder: Tối thiểu 6 ký tự. Features an eye icon toggle to show/hide password.
CONFIRM NEW PASSWORD	Password Input. Required. Re-enters the new password to confirm. Placeholder: Nhập lại mật khẩu mới. Must match NEW PASSWORD exactly.
Update Password	Primary Button. Submits the updated credentials to API POST /api/Auth/reset-password. Upon success, updates user password in database.

2. System Administration Features

2.1 Account Management

2.1.1 Account Management Screen



This screen allows the Administrator to:

- View Account List: view the list of all registered system users including Students, Lecturers, and Admins.
- Filter Account List: filter user accounts by role (ALL, Admin, Lecturer, Student).
- Search Accounts: enter keyword(s) to search user accounts by Full Name, Email, or User ID.
- Sort Account List: sort user accounts (ascending, descending) by clicking column headers.

On the screen, s/he can also:

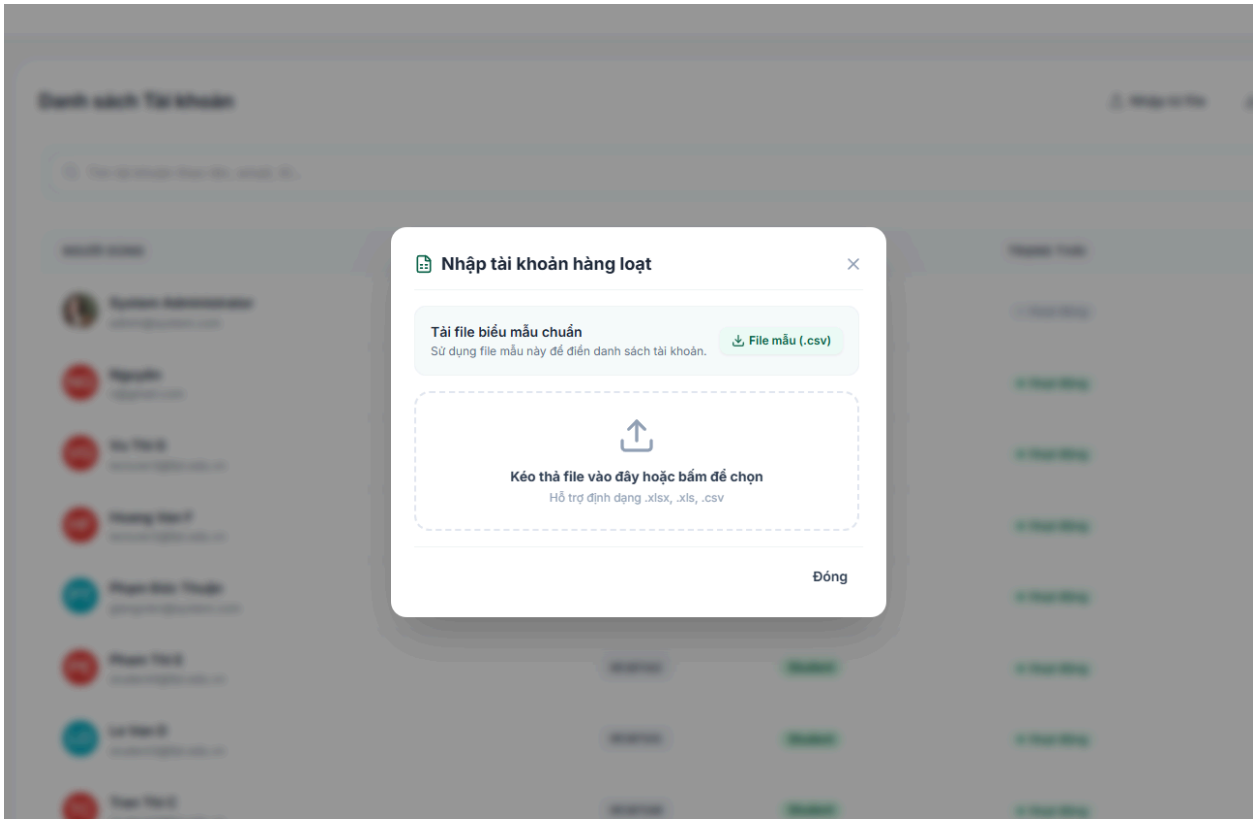
- Activate/Deactivate Account: toggle the active/inactive status switch of a specific user account.
- Export Student List: select student accounts and export their details to a CSV file.

- Choose to go to Create Account modal by clicking the "Create New Account" button, or open Batch Import modal by clicking the "Import File" button.

Field Description

Field Name	Description
Search Input	Textbox. Enter User ID, Full Name, or Email to search accounts. Filters table contents dynamically in real-time
Role Filter Dropdown	Combobox. Filters account list by role. Options: ALL, Admin, Lecturer, Student. Default: ALL.
Import File Button	Secondary Button. Opens the Batch User Import Modal to upload Excel/CSV user account lists.
Export File Button	Secondary Button. Exports selected student user records into a downloadable .csv file.
Create New Account Button	Primary Button. Opens the Create User Account Modal to manually register a new single account.
User ID	Data Column. Displays unique account ID (e.g., HE187159 for Student, GV123456 for Lecturer).
Full Name & Avatar	Data Column. Displays user's profile avatar and full name
Email	Data Column. Displays registered user email address.
Role Badge	Data Badge. Displays user role with distinct color coding (Admin in amber, Lecturer in blue, Student in emerald).

2.1.2 Batch User Import Modal



This modal allows the Administrator to:

- Download Import Template: download a standard .csv template file containing predefined columns (StudentId, FullName, Email, Role).
- Upload Batch File: drag and drop or browse Excel (.xlsx, .xls) or .csv files for batch user creation.
- Review Data Validation Errors: inspect line-by-line validation errors reported by the backend API before executing import.

On the screen, s/he can also:

- Execute Batch Import: click the "Start Import" button to process file data via API POST /api/Users/import.
- Cancel Operation: click the "Cancel" button to close the modal without saving data.

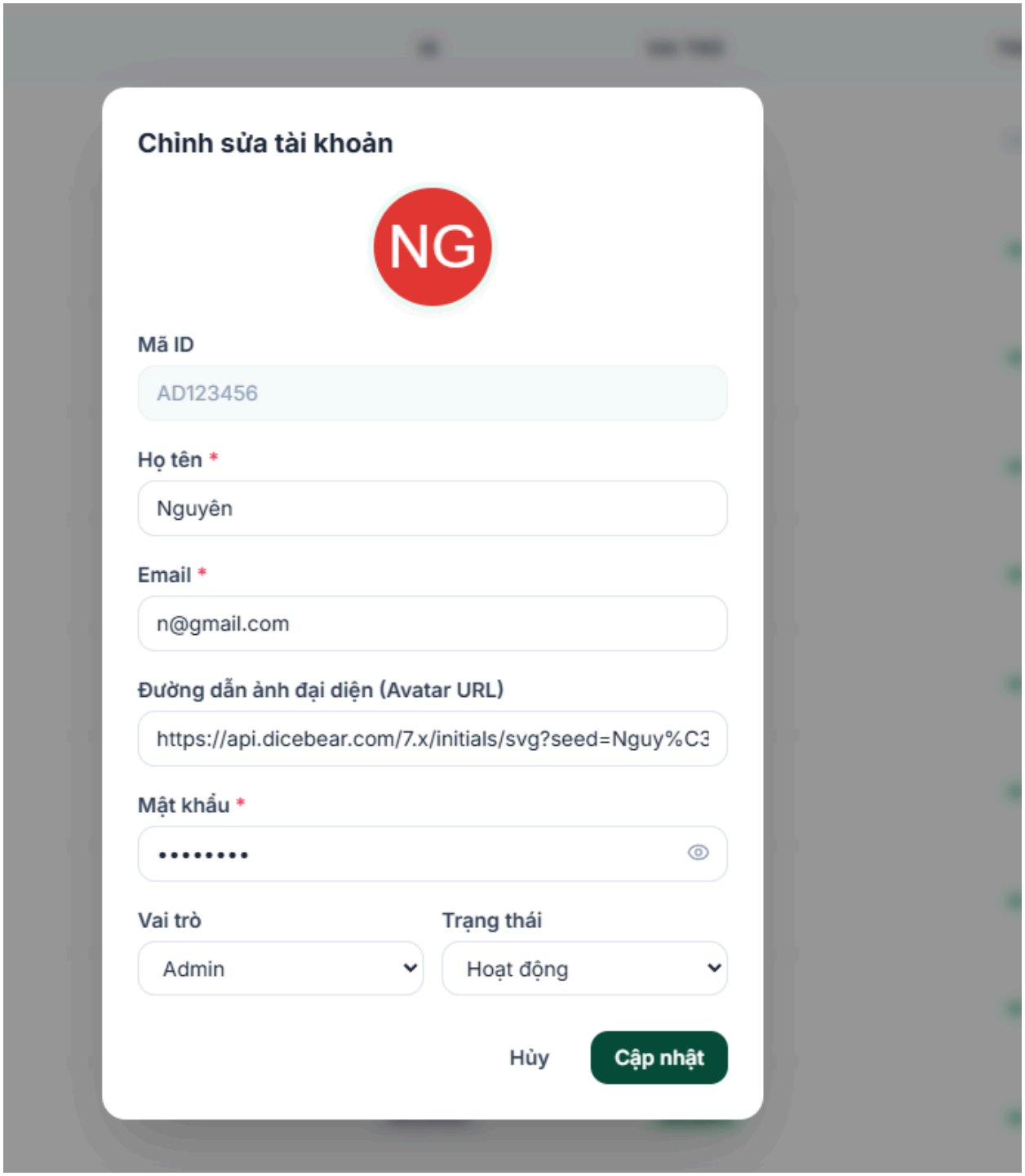
Mapped Use Case: UC-06: Import / Export User Data

Field Description

Field Name	Description
Download Template (.csv)	Secondary Button. Downloads the standardized CSV template file (user_import_template.csv).

Drag & Drop Upload Zone	Interactive Dropzone. Accepts file upload via drag-and-drop or file selector window. Supported formats: .xlsx, .xls, .csv.
File Preview Card	File Card. Displays uploaded file details: file name (e.g., user_list.xlsx), file size (KB), and a remove icon button.
Validation Error List Container	Red Alert Box. Displays row-level validation errors returned by backend. Format: Row [Index]: [Message] (e.g., *Row 3: Invalid Email format*).
Cancel Button	Neutral Button. Closes the import modal and resets selected file state.
Start Import Button	Primary Button. Submits the file payload to POST /api/Users/import. Displays a loading spinner during execution.

2.1.3 Edit User Account Modal



This modal allows the Administrator to:

- View Account Details: inspect current account information including User ID, Full Name, Email, Avatar URL, Role, and Status.
- Update Account Credentials: modify user's Full Name, Email, Avatar URL, new Password, assigned Role, and Account Status.
- Toggle Password Visibility: show or hide plain text password by clicking the eye icon.

On the screen, s/he can also:

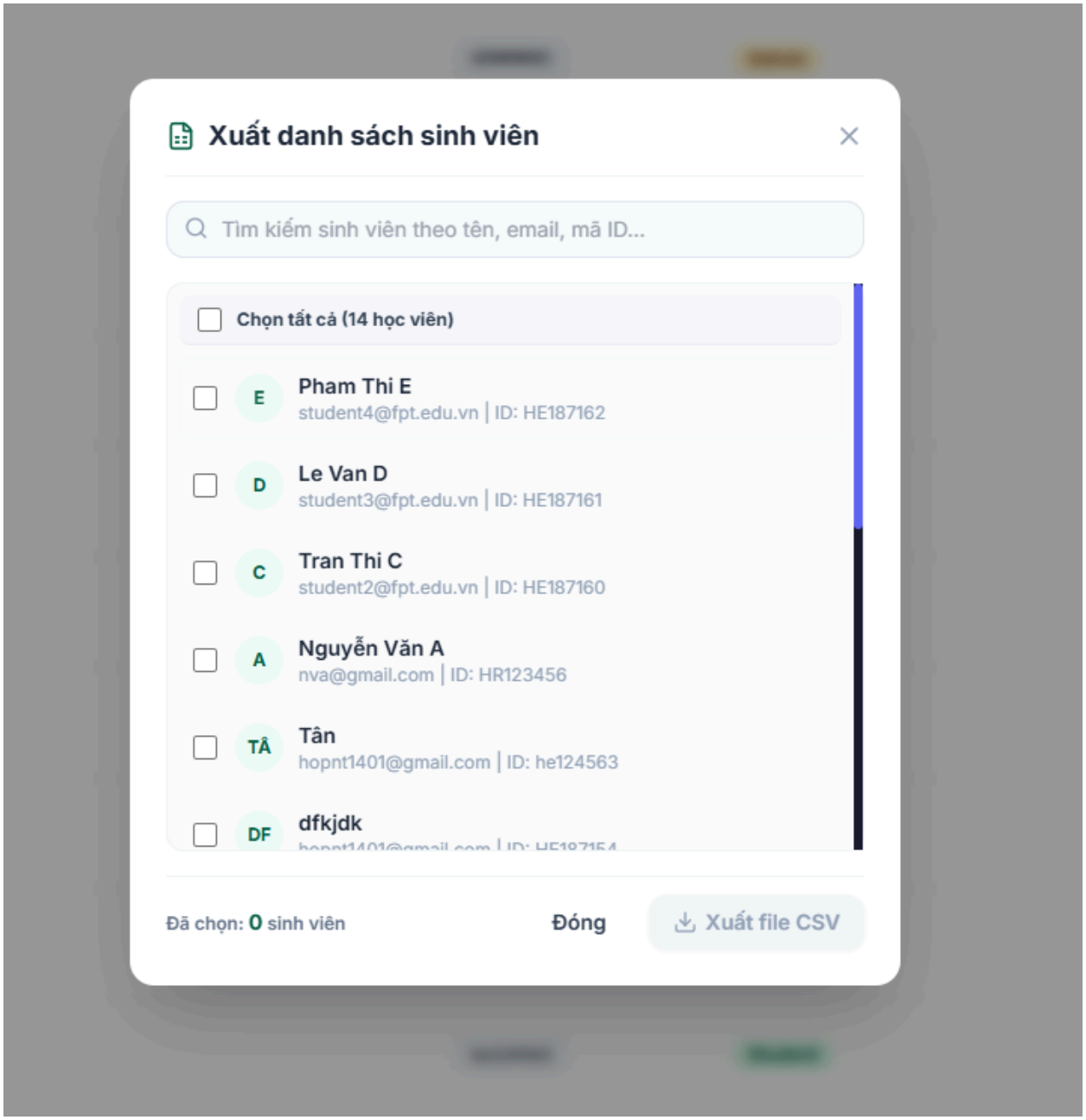
- Submit Account Updates: click the "Cập nhật" button to save changes to CSDL via API PUT /api/Users/{id}.
- Cancel Operation: click the "Hủy" button to close the modal without saving changes.

Mapped Use Case: UC-07: Update Account

Field Description

Field Name	Description
User ID	Textbox. Disabled (Read-only). Displays the unique user ID (e.g., AD123456). Cannot be modified once created.
Full Name	Textbox. Required. Accepts user's full name. Max length: 100 characters.
Email	Textbox. Required. Accepts user's email address. Must be unique in the system and valid email format.
Avatar URL	Textbox. Optional. Accepts image URL string (e.g., Dicebear avatar API link) to set profile avatar.
Password	Password Input. Required. Accepts password update. Features an eye toggle icon (EyeIcon) to show/hide text.
Role	Selectbox. Required. Assigns user system role. Options: Student, Lecturer, Admin.
Status	Selectbox. Required. Sets account status. Options: Active (ACTIVE), Inactive (INACTIVE).
Cancel	Neutral Button. Closes the edit modal and discards uncommitted changes.
Update	Primary Button. Submits form payload to API PUT /api/Users/{id}. Displays success toast upon successful update

2.1.4 Export Student List Modal



This modal allows the Administrator to:

- Search Student Roster: enter keyword(s) to filter student records by Full Name, Email, or Student ID within the modal.
- Select Students: select individual students using checkboxes or select all enrolled students at once using the "Chọn tất cả" option.
- Track Selected Count: monitor real-time counter displaying the number of currently selected student records.

On the screen, s/he can also:

- Export to CSV: click the "Xuất file CSV" button to generate and download the .csv file containing the selected student records.
- Close Modal: click the "Đóng" button or the top-right "X" icon to dismiss the modal.

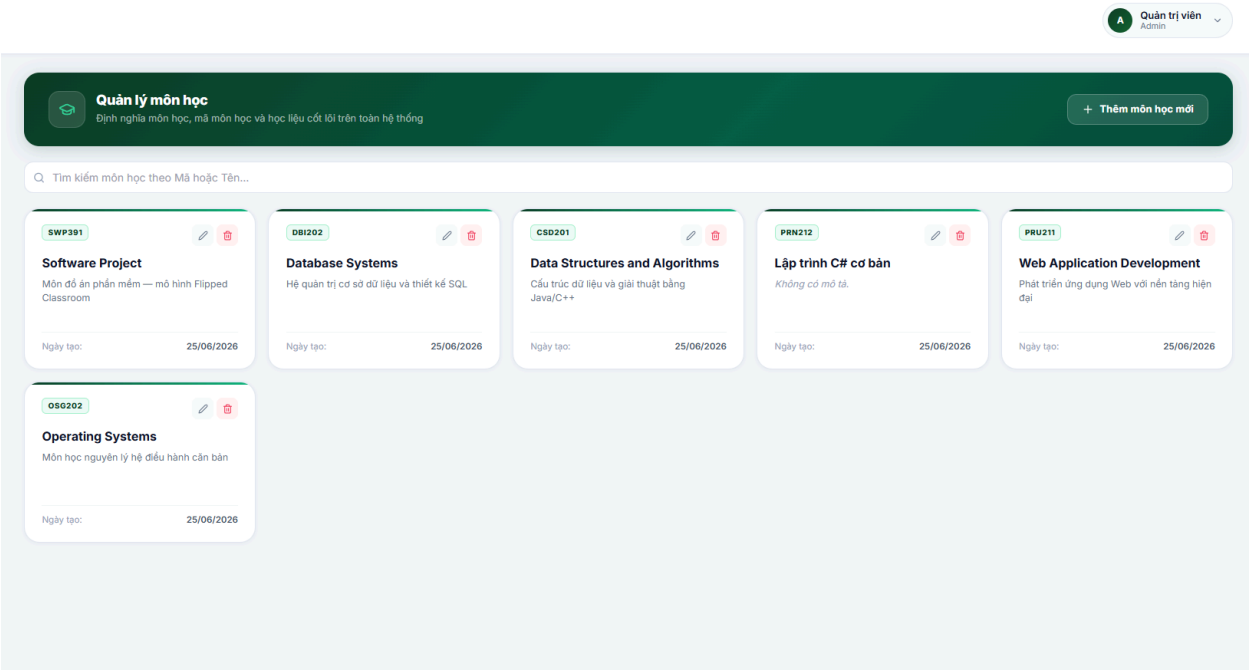
Mapped Use Case: UC-06: Import / Export User Data

Field Description

Field Name	Description
Search Input	Textbox. Search students by Name, Email, or Student ID. Placeholder: Tìm kiếm sinh viên theo tên, email, mã ID....
Select All Checkbox	Checkbox. Toggles selection for all listed students at once (e.g., Chọn tất cả (14 học viên)).
Student Record Checkbox List	Checkbox List. Displays student avatar, full name, email address, and student ID. Allows multi-selection.
Selected Counter	Dynamic Counter Label. Displays real-time count of currently selected student items (e.g., Selected: X students).
Close	Neutral Button. Closes the export modal.
Export CSV File	Primary Button with Download Icon. Generates and downloads danh_sach_sinh_vien_export.csv. Disabled when 0 students are selected.

2.2 Course Management

2.2.1 Course Management Screen



This screen allows the Administrator to:

- View Course List: view all registered courses in the curriculum displayed in a clean grid card layout containing Course Badge, Title, Description, and Creation Date.
- Search Courses: enter keyword(s) to search courses dynamically by Course Code or Course Title.

On the screen, s/he can also:

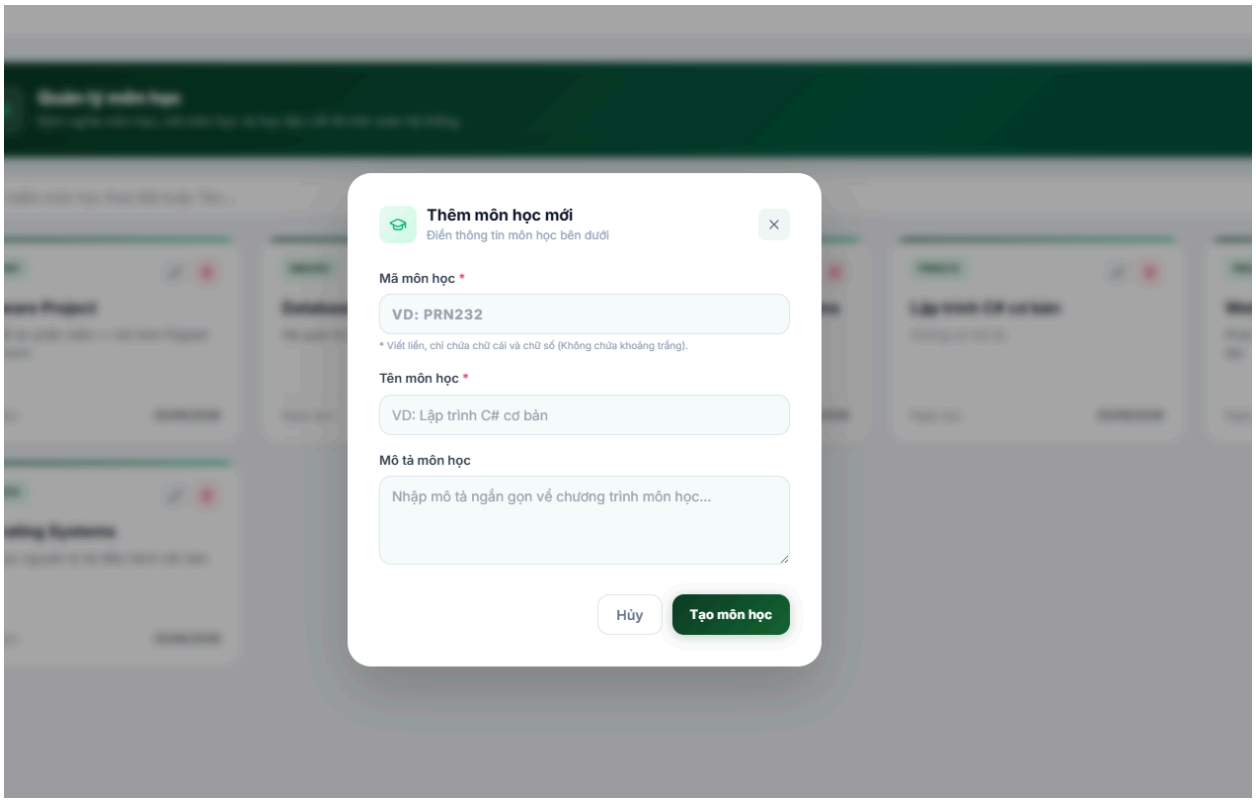
- Edit/Delete Course: click the pencil icon to modify course details or click the red trash icon to delete a course.
- Choose to go to Create Course modal by clicking the "+ Thêm môn học mới" button.

Mapped Use Case: UC-41: Manage Courses,UC-43: View Courses.

Field Description

Field Name	Description
Search Input	Textbox. Enter Course Code or Title to filter course cards. Placeholder: Search courses by Code or Title.... Filters list dynamically in real-time.
Add New Course	Primary Button. Opens the Create Course Modal to define a new course code, title, and description.
Course Code Badge	Data Badge. Displays the unique course code tag (e.g., SWP391, DBI202, CSD201).
Course Title	Card Heading. Displays the full title of the course (e.g., *Software Project*, *Database Systems*).
Course Description	Card Paragraph. Displays a summary description of the course scope and curriculum details.
Created Date	Card Metadata. Displays the creation timestamp date of the course record (e.g., 25/06/2026).
Edit Icon Button (Pencil)	Action Button. Opens the Edit Course Modal with pre-filled course data.
Delete Icon Button (Trash)	Action Button. Prompts confirmation modal to delete/remove the target course from the Database.

2.2.2 Create / Edit Course Modal



This modal allows the Administrator to:

- Enter Course Code: input or inspect the unique course identifier code.
- Enter Course Title & Description: input full course title and detailed curriculum description.

On the screen, s/he can also:

- Submit Course Changes: click the "Lưu thay đổi" button to save updates to CSDL via API POST /api/Courses (or PUT /api/Courses/{code}).
- Cancel Operation: click the "Hủy" button or top-right "X" icon to close the modal without saving changes.

Mapped Use Case: UC-42: Create Course, UC-44: Update Course

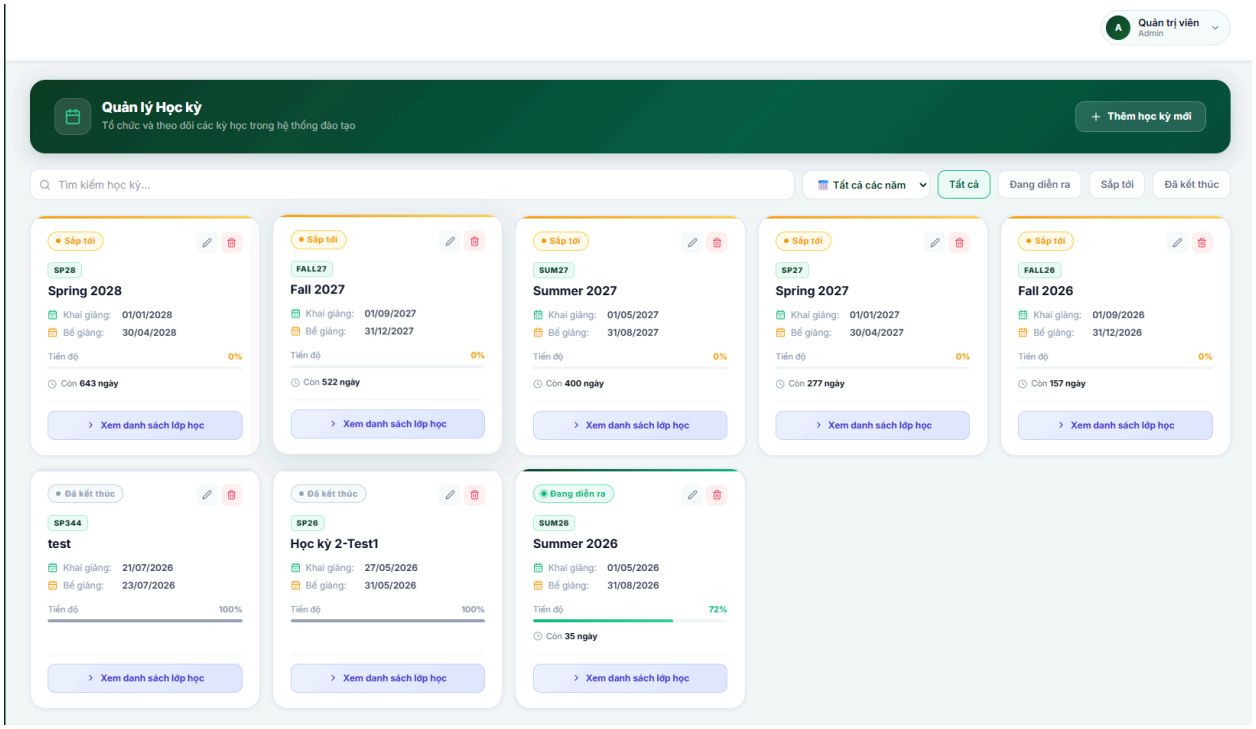
Field Description

Field Name	Description
Course Code	Textbox. Required. Accepts unique course code (e.g., DBI202). Validation note: * Single word, letters and numbers only (No spaces allowed). Max length: 20 characters.
Course Title	Textbox. Required. Accepts full course title (e.g., *Database Systems*). Max length: 150 characters.

Course Description	Textarea. Optional. Accepts detailed course overview description (e.g., *Database management systems and SQL design*). Max length: 500 characters.
Cancel	Neutral Button. Closes the course form modal and discards uncommitted inputs.
Save Changes	Primary Button. Submits form payload to backend API (POST/PUT /api/Courses). Reloads course grid upon success.

2.3 Academic Terms Management

2.3.1 Academic Term Management Screen



This screen allows the Administrator to:

- View Academic Term List: view all academic semesters displayed in a card grid featuring Status Badge, Term Code Tag, Term Name, Start/End Dates, Semester Progress bar, and Countdown days.
- Filter Academic Terms: filter terms by Year dropdown (e.g., Tất cả các năm) and Status chips (Tất cả, Đang diễn ra, Sắp tới, Đã kết thúc).
- Search Academic Terms: enter keyword(s) to search terms dynamically by Term Name or Term Code.

On the screen, s/he can also:

- Edit/Delete Academic Term: click the pencil icon to modify term details or click the red trash icon to delete a term.

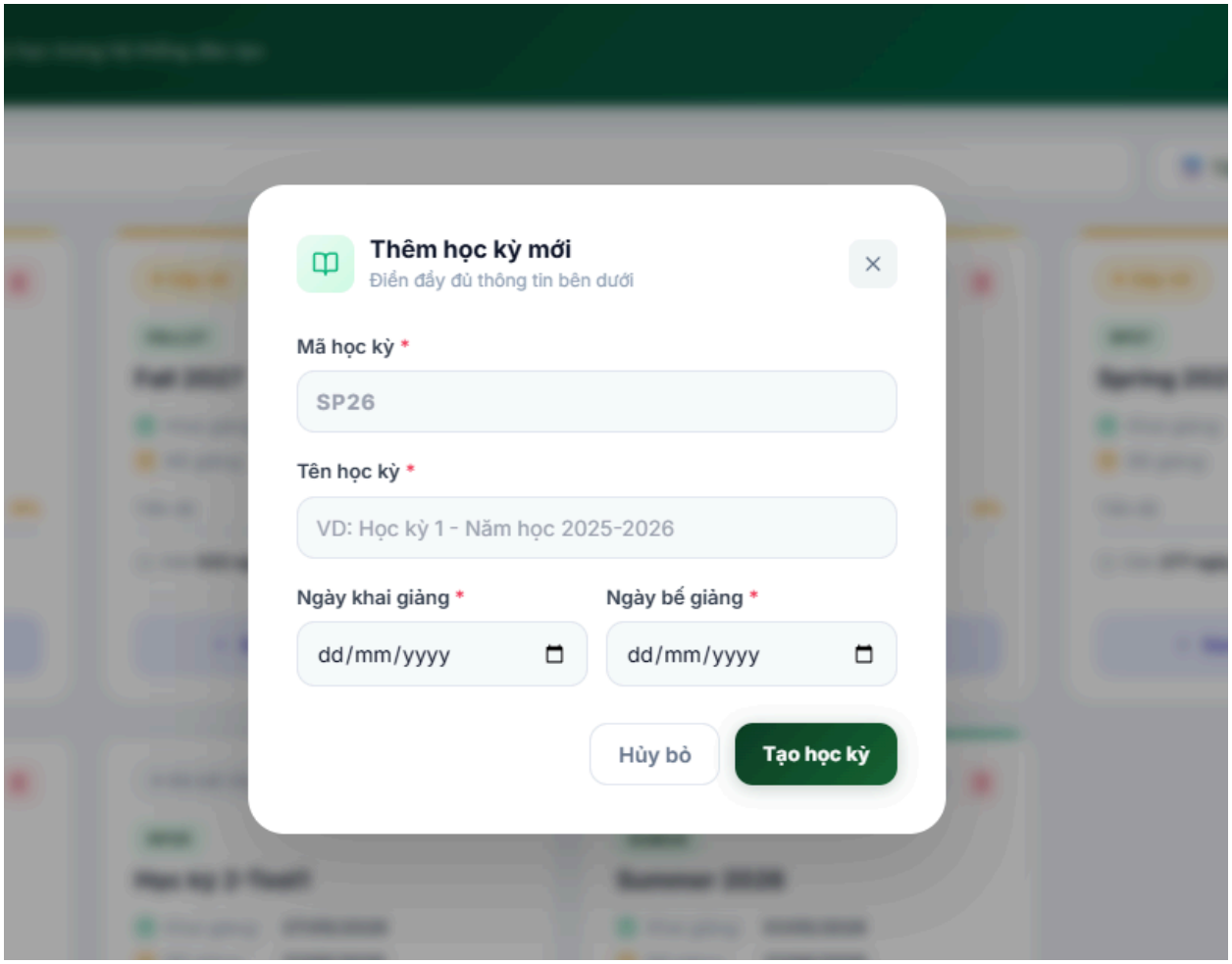
- Navigate to Filtered Classes: click the "> Xem danh sách lớp học" button to open the Class Management screen pre-filtered by that specific academic term.
- Choose to go to Create Term modal by clicking the "+ Thêm học kỳ mới" button.

Mapped Use Case: UC-09: Manage Academic Terms, UC-10: View Academic Terms.

Field Description

Field Name	Description
Search Input	Textbox. Search terms by code or name. Placeholder: Search academic terms.... Filters list dynamically.
Year Filter Dropdown	Combobox. Filters terms by calendar year (e.g., All years, 2026, 2027).
Status Filter Chips	Filter Buttons. Filters terms by state: All, Active, Upcoming, Completed.
+ Add New Term	Primary Button. Opens the Create Academic Term Modal to set up a new semester.
Status Badge	Data Badge. Displays status: • Active (green), • Upcoming (amber), • Completed (gray).
Term Code Tag	Data Tag. Displays short term code (e.g., SUM26, FALL27, SP28).
Term Title	Card Heading. Displays full term title (e.g., *Summer 2026*, *Spring 2028*).
Start Date / End Date	Date Metadata. Displays start date (Start Date) and end date (End Date) of the term.
Progress Bar & Percentage	Progress Bar. Displays percentage completion of the semester based on current date.
Countdown Label	Time Metadata. Displays remaining days in term or days until term starts (e.g., 35 days remaining).
Edit Icon Button (Pencil)	Action Button. Opens the Edit Academic Term Modal with pre-filled term dates.
Delete Icon Button (Trash)	Action Button. Prompts confirmation modal to delete the academic term.
View Class List	Action Button. Redirects to Class Management page filtered by this specific Academic Term.

2.3.2 Create / Edit Course Modal



This modal allows the Administrator to:

- Enter Term Code & Title: input unique term code and semester title.
- Select Term Dates: select official Start Date (Ngày khai giảng) and End Date (Ngày bế giảng) using date picker inputs.

On the screen, s/he can also:

- Create Academic Term: click the "Tạo học kỳ" button to save the new term record via API POST /api/AcademicTerms.
- Cancel Operation: click the "Hủy bỏ" button or top-right "X" icon to close the modal without saving.

Mapped Use Case: UC-09: Manage Academic Terms, UC-11: Update Academic Term.

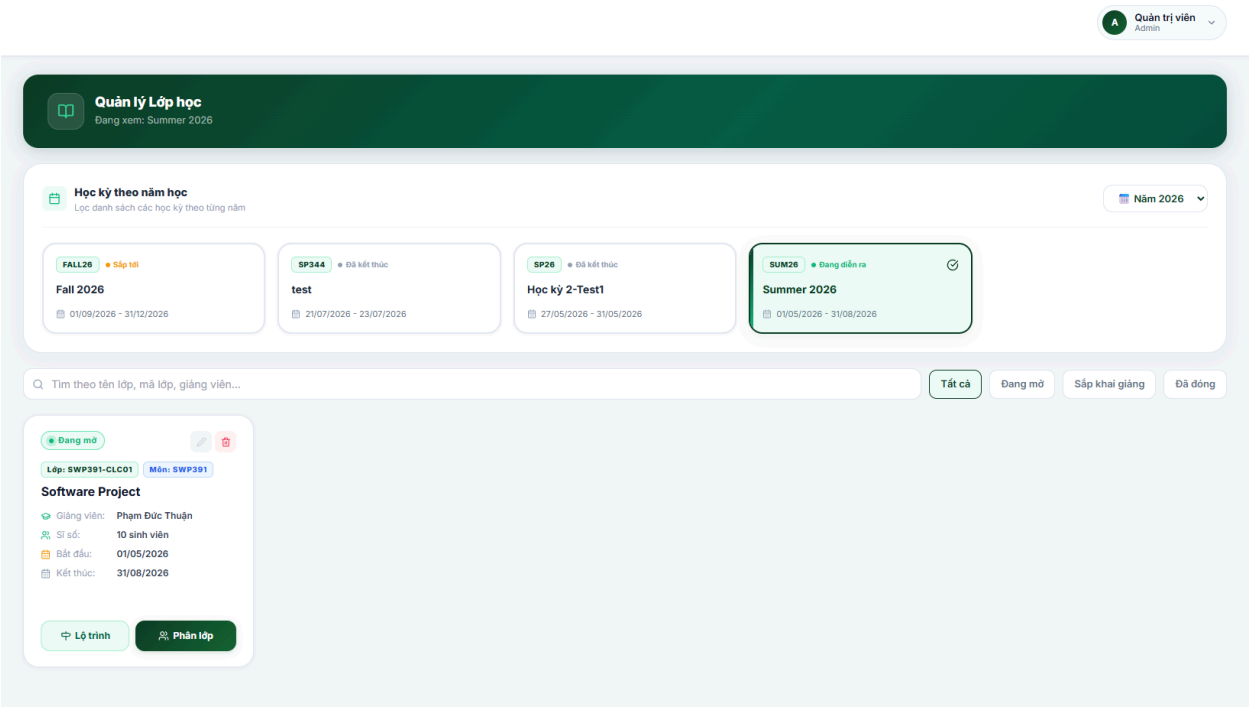
Field Description

Field Name	Description
Term Code	Textbox. Required. Accepts unique term code. Placeholder: SP26. Max length: 30 characters.
Term Name	Textbox. Required. Accepts full semester title. Placeholder: e.g. Semester 1 - Academic Year 2025-2026. Max length: 100 character

Start Date	Date Picker. Required. Accepts official term start date. Placeholder: dd/mm/yyyy.
End Date	Date Picker. Required. Accepts official term end date. Placeholder: dd/mm/yyyy. Validation note: Must be later than Start Date.
Cancel	Neutral Button. Closes the term creation modal and discards uncommitted inputs.
Create Term	Primary Button. Submits form payload to backend API POST /api/AcademicTerms. Reloads academic term grid upon success.

2.4 Class Management

2.4.1 Class Management Screen



This screen allows the Administrator to:

- Filter Classes by Year and Term: select a calendar year using the dropdown (e.g., Năm 2026) and click an Academic Term card (e.g., Summer 2026) to filter the classes roster.
- Filter Classes by State: filter class cards by status using the chips (Tất cả, Đang mở, Sắp khai giảng, Đã đóng).
- Search Classes: enter keyword(s) to search classes dynamically by Class ID, Course Code, or Lecturer Name.
- View Class Details: inspect class name, course tag, class code, assigned lecturer, student count, start date, and end date in a card layout.

On the screen, s/he can also:

- Edit/Delete Class: click the pencil icon to modify class details or click the red trash icon to delete a class.

- Allocate Class Members: click the "Phân lớp" button to manage lecturer assignments and student enrollments.
- View Class Roadmap: click the "Lộ trình" button to navigate to the class curriculum roadmap workspace.

Mapped Use Case: UC-12: Manage Classes, UC-13: View Class List, UC-15: Update Class Information, UC-16: Delete Classes.

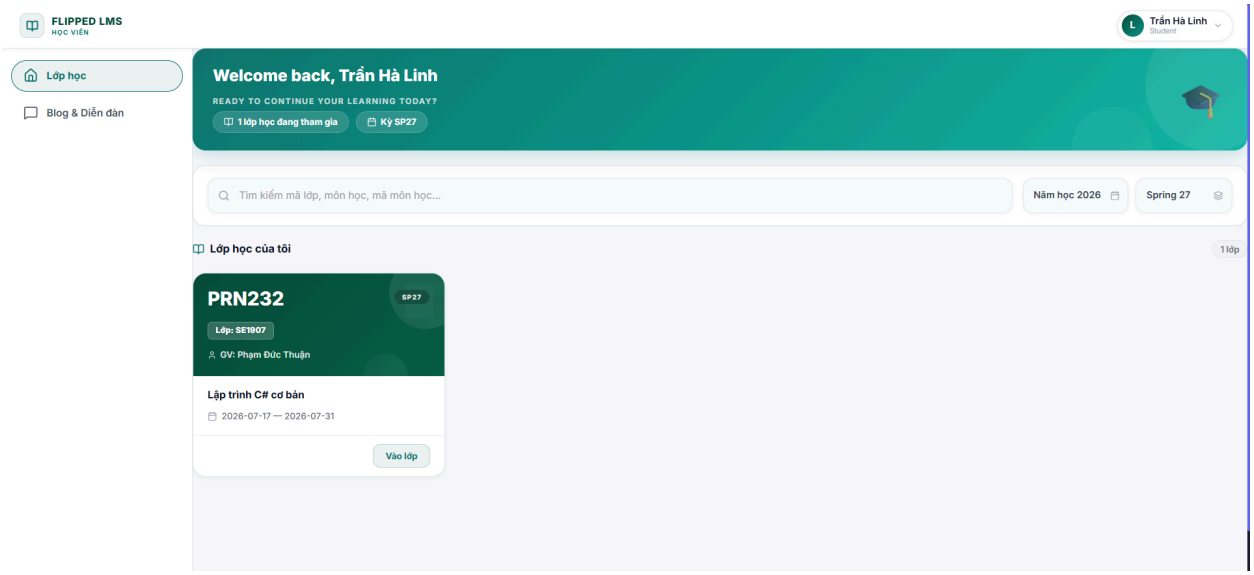
Field Description

Field Name	Description
Year Selector Dropdown	Combobox. Filters academic terms in the panel by year (e.g., Year 2026).
Academic Term Card Panel	Interactive Cards. Clicking a card selects a term (e.g., SUM26 Summer 2026) to filter active classes. Displays term status (e.g., • Active).
Search Input	Textbox. Search classes by class code, course title, or lecturer name. Placeholder: Search by class name, class code, lecturer....
Status Filter Chips	Filter Buttons. Filters active classes by status: All, Open, Upcoming, Closed.
Status Badge	Data Badge. Displays class status (e.g., • Open).
Class & Course Tags	Data Tags. Displays Class Code (e.g., Class: SWP391-CLC01) and Course Code (e.g., Course: SWP391).
Class Title	Card Heading. Displays the course name of the class (e.g., *Software Project*).
Lecturer	Metadata. Displays the assigned Lecturer's full name.
Student Count	Metadata. Displays current student count enrolled in the class (e.g., 10 students).
Start Date / End Date	Metadata. Displays Start Date and End Date of the class.
Edit Icon Button (Pencil)	Action Button. Opens the Edit Class Modal to modify class information.
Delete Icon Button (Trash)	Action Button. Prompts confirmation modal to delete the class.
Learning Roadmap	Action Button. Navigates to the Class Learning Roadmap.
Class Allocation	Action Button. Opens the Member Allocation panel to assign lecturers and manage student rosters.

3. Student Learning Management System

3.1. Course Management & Overview

3.1.1. My Classes



This screen allows the Student to:

- View My Classes List: view the list of currently enrolled classes .
- Filter My Classes List: filter enrolled classes by academic year and semester.
- Search My Classes: enter keyword to search enrolled classes by class code, course name, or course code.

On the screen, s/he can also

- View quick summary tags of their learning status, such as the number of participating classes and the current semester .
- Choose to go to the Class Details/Learning screen for a specific course by clicking the Enter class button on the class card.

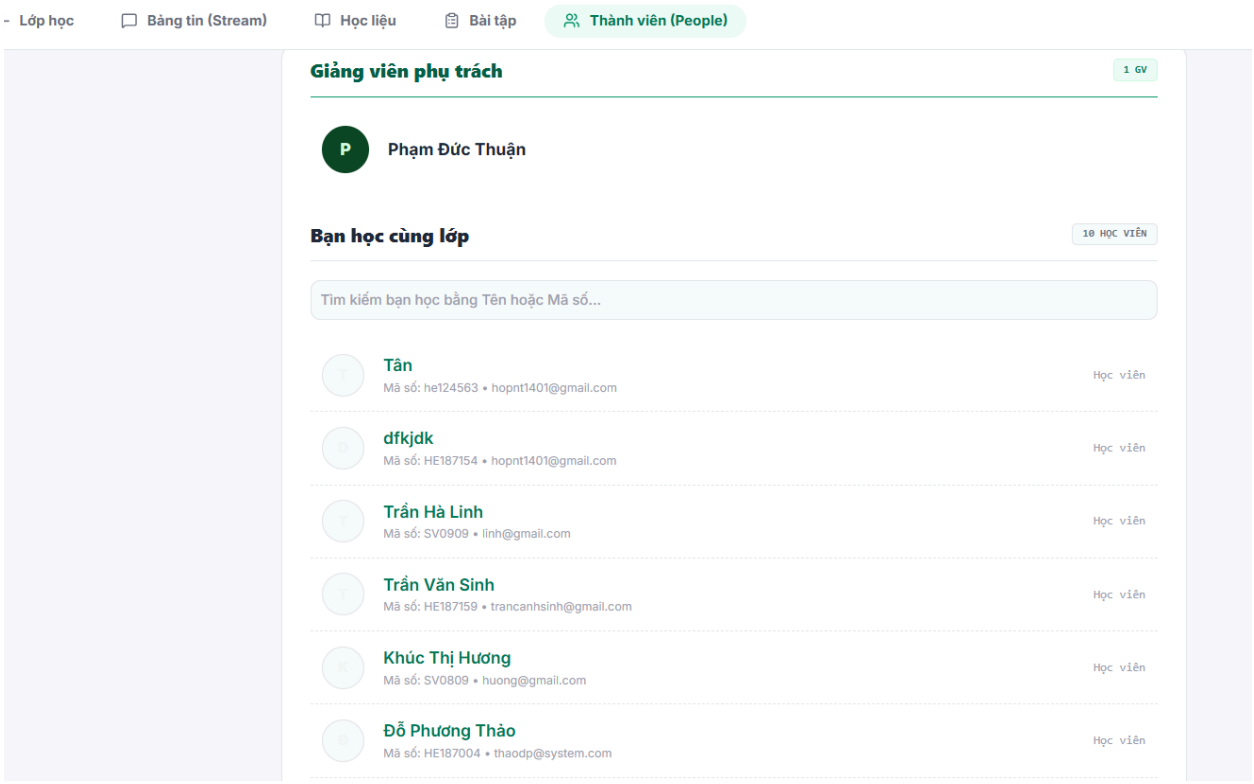
Mapped Use Case: UC-21: View Class

Field Description

Field Name	Description
Search Box	Initial values: null or blank. Placeholder text displays: “Search class code, course name, course code...”
Academic Year	Dropdown lists to filter the displayed classes by a specific Academic Year and/or Semester.
Search	Button to execute the search and filter action based on the inputted criteria. .

Class Card Details	Displays information for each enrolled class, including: Course Code, Course ID, Class Name, Instructor Name, Course Name, and Start Date — End Date.
Enter Class	Action button to navigate the student to the detailed workspace of that specific class.

3.1.2.Class Members



This screen allows the user (Student / Learner) to:

- View Instructor Information: View the instructor(s) assigned to the current class.
- View Classmates List: View the list of enrolled peers/classmates in the class.
- Search Classmates: Enter keywords to search classmates by Student Name or Student ID.
- View Summary Indicators: View quick badges displaying the total count of instructors and enrolled classmates.

Mapped Use Case: UC-26: Class Members

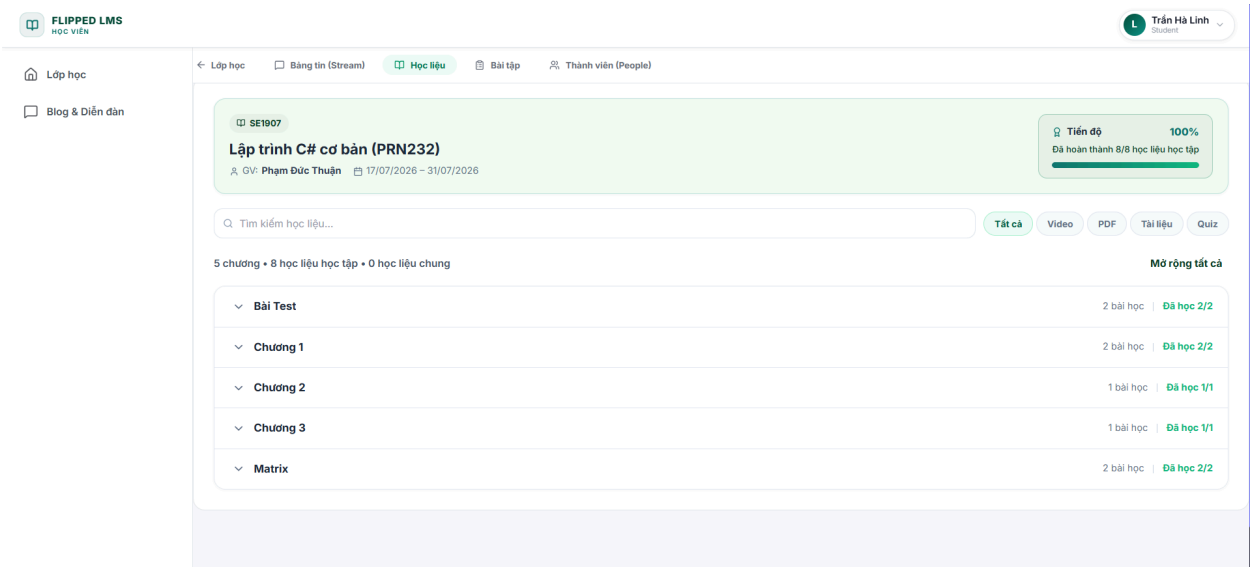
Field Description

Field Name	Description

Class Navigation Bar	Navigation bar for the class workspace containing tabs: Class (Lớp học), Stream (Bảng tin), Learning Materials (Học liệu), Assignments (Bài tập), and People (Thành viên). The active tab is currently People.
Instructor Section Header	Header displaying the label "Giảng viên phụ trách" along with a tag/badge showing the total number of assigned instructors (e.g., 1 GV).
Instructor Card / Details	Displays information for each assigned instructor, including: Avatar (initials or profile image) and Full Name (e.g., Phạm Đức Thuận).
Classmates Section Header	Header displaying the label "Bạn học cùng lớp" along with a tag/badge showing the total count of enrolled classmates (e.g., 10 HỌC VIÊN).
Search Box	Search input field to filter classmates. Initial values: null or blank. Placeholder text displays: "Tìm kiếm bạn học bằng Tên hoặc Mã số..."
Classmate List Item	Displays information for each classmate,

3.2. Learning Roadmap & Materials

3.2.1. Student Roadmap



This screen allows the user to:

- View Class Materials: view the list of chapters, learning materials, and general materials for a specific class.

- Filter Materials: filter the material list by specific types.
- Search Materials: enter keyword(s) to search for specific materials within the class.

On the screen, student can also

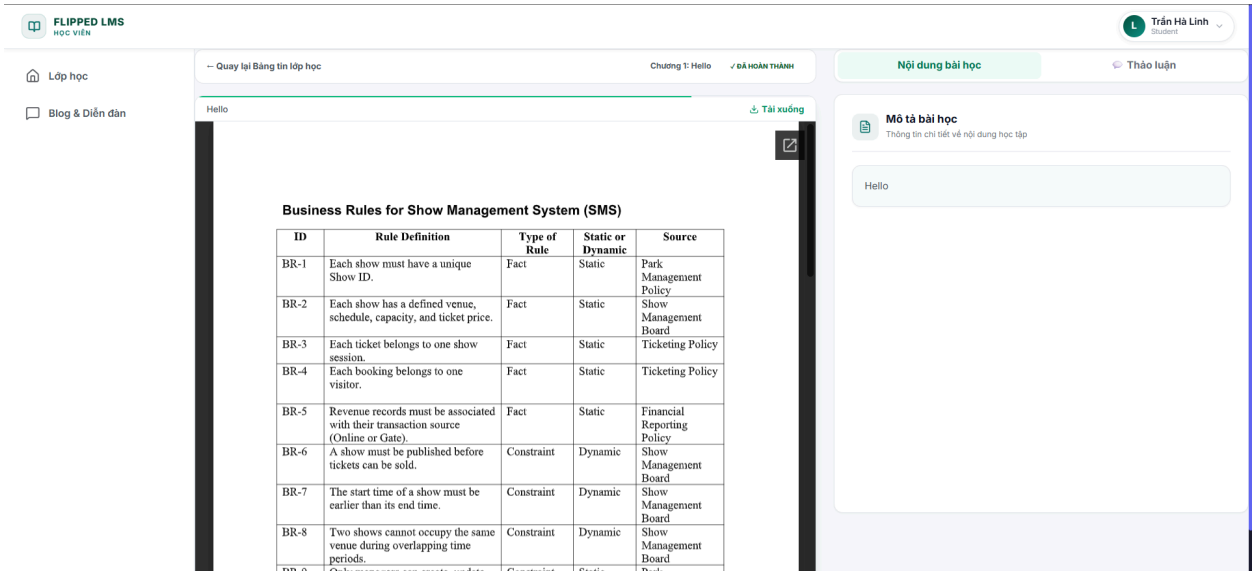
- View overall class details, including the class name, instructor, and duration (start/end dates).
- Track learning progress via a completion status progress bar.
- Expand or collapse chapter lists to view the specific number of lessons and their individual learning statuses.
- Navigate to other areas of the class workspace, such as the Stream, Assignments, or People tabs.

Mapped Use Case: UC-20: View personal learning roadmap

Field Description

Field Name	Description
Class Navigation Tabs	Links to navigate back to the main "Classes" list or switch between current class views (Stream, Materials, Assignments, People).
Class Information	Displays the master data for the selected class, including: Class Code, Class Name, Instructor Name, and Start Date — End Date.
Progress Box	Displays the user's overall material completion status as a numerical percentage (e.g., 100%) and a visual progress bar.
Material Type	Dropdown list to filter the displayed materials by specific categories (Default value: "All types").
Search Field	Initial values: null or blank. Placeholder text displays: “Search materials...”
Search	Button to execute the search and filter action based on the inputted criteria.
Materials Summary	Text summary displaying the "Total chapters • Total learning materials • Total general materials" within the class.
Expand all	Action button/link to simultaneously expand (or collapse) all chapter and material accordions below.
Chapter / Section Accordion	Expandable rows displaying the "Chapter Name" or "General materials section". It also indicates the number of lessons/materials contained within and the user's specific learning status for that chapter.

3.2.2. Learning Materials



This screen allows the Student to:

- View the learning material content within the system.
- Read lesson descriptions and related learning information.
- Download learning materials for offline study.
- Mark the lesson as completed after finishing the content.
- Navigate back to the Class Dashboard.
- Switch between the Lesson Content and Discussion tabs.
- Participate in discussions related to the current lesson.

On this screen, the student can also:

- View the lesson title and chapter information.
- Open the learning material in full-screen mode for easier reading.
- Track the completion status of the lesson.

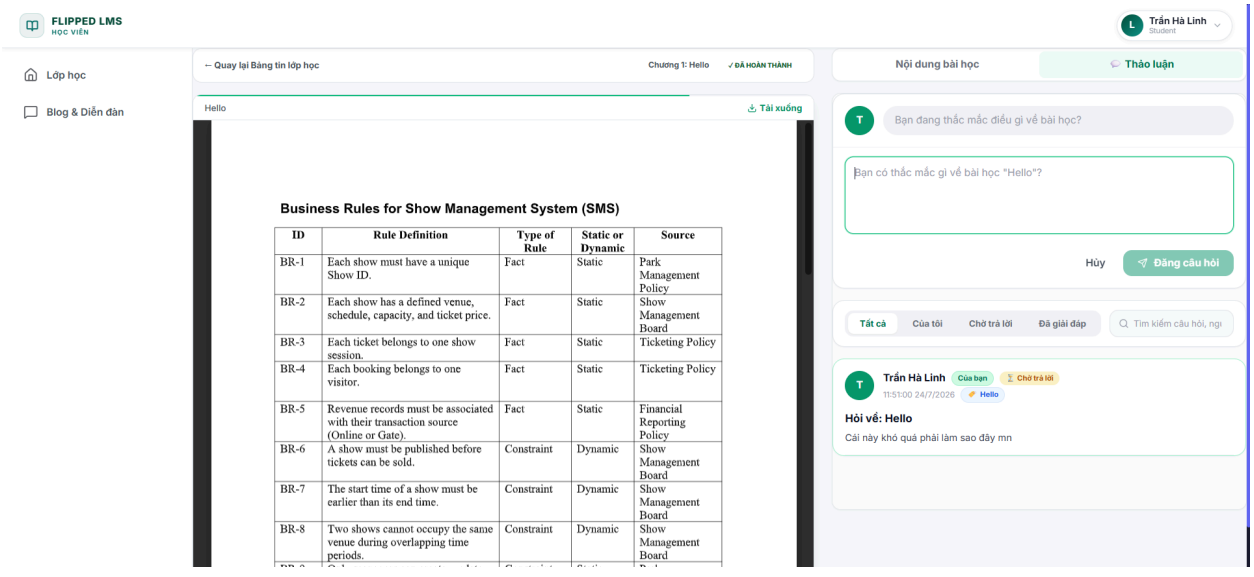
Mapped Use Case: UC-23: View Materials, UC-24: Download Materials, UC-25: Mark Material as Completed

Field Description

Field Name	Description

Back to Class Dashboard	Navigates the student back to the selected class dashboard without affecting the current learning progress.
Chapter Title	Displays the current chapter or lesson name (e.g., *Chapter 1: Hello*).
Completion Status	Displays whether the lesson has been completed (e.g., Completed) based on the student's learning progress.
Lesson Title	Displays the title of the currently selected learning material.
Download	Downloads the attached learning material (PDF, document, slide, etc.) to the student's device for offline access.
Learning Material Viewer	Displays the content of the learning material directly within the system. Supports scrolling through the lesson content without downloading the file.
Open in New Tab	Opens the current learning material in a new browser tab for a larger viewing experience.
Lesson Content Tab	Displays the lesson description, attached learning material, and additional information related to the lesson.
Discussion Tab	Switches to the discussion area where students can view or participate in lesson discussions.
Lesson Description	Displays the description or overview of the current learning material provided by the instructor.

3.2.3. Lesson Discussion



This screen allows the Student to:

- View discussion threads related to the current learning material.
- Create a new question or discussion topic about the lesson.
- Search for discussion topics using keywords.
- Filter discussion threads by category (All, My Questions, Waiting for Response, Answered).
- Monitor the response status of submitted questions.

On this screen, the student can also:

- Enter a question title and detailed content.
- Submit a new discussion topic.
- Cancel the current discussion draft before submission.
- View previously submitted questions and their current status.

Mapped Use Case: UC-19: Submit Question Feedback

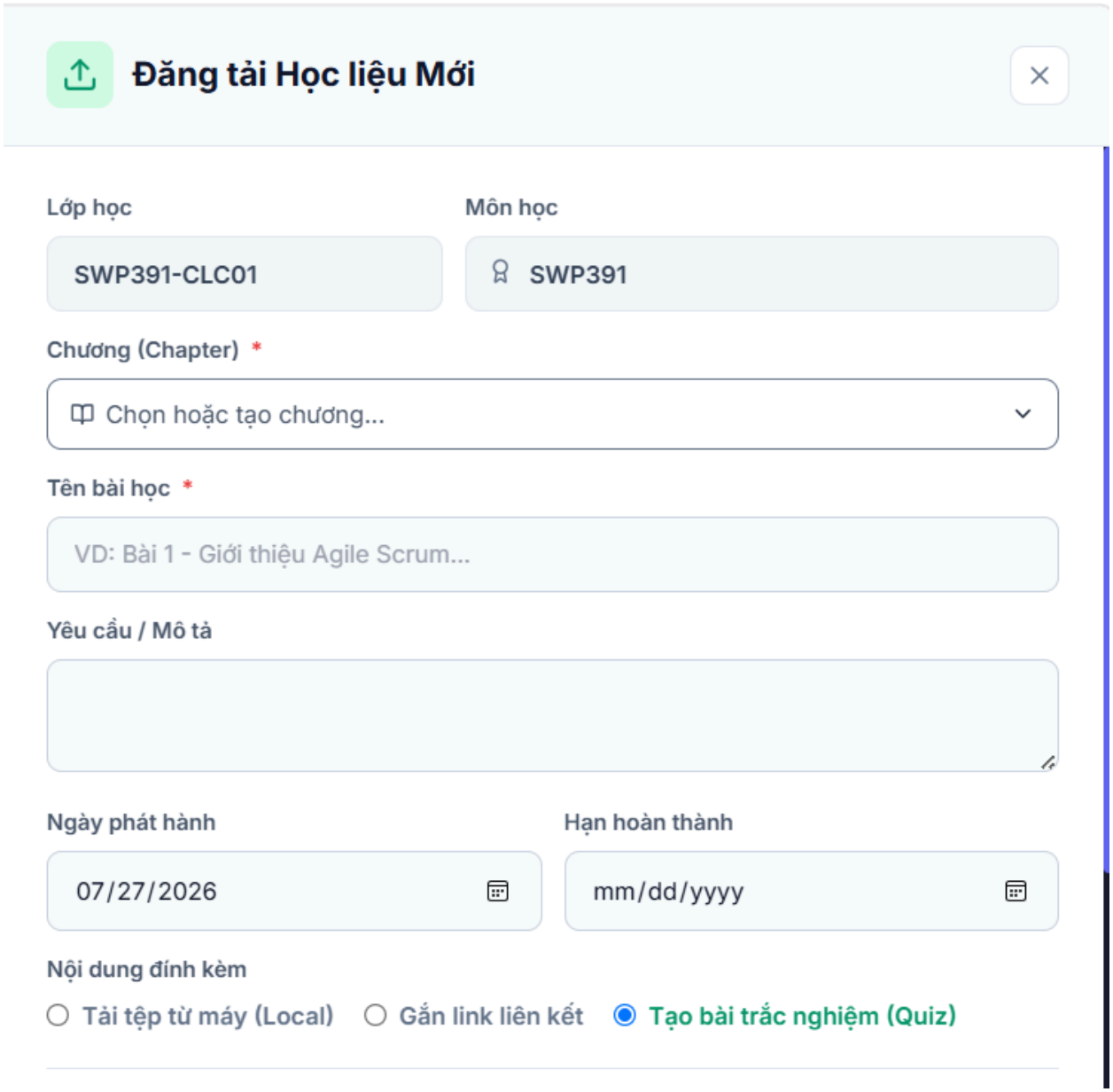
Field Description

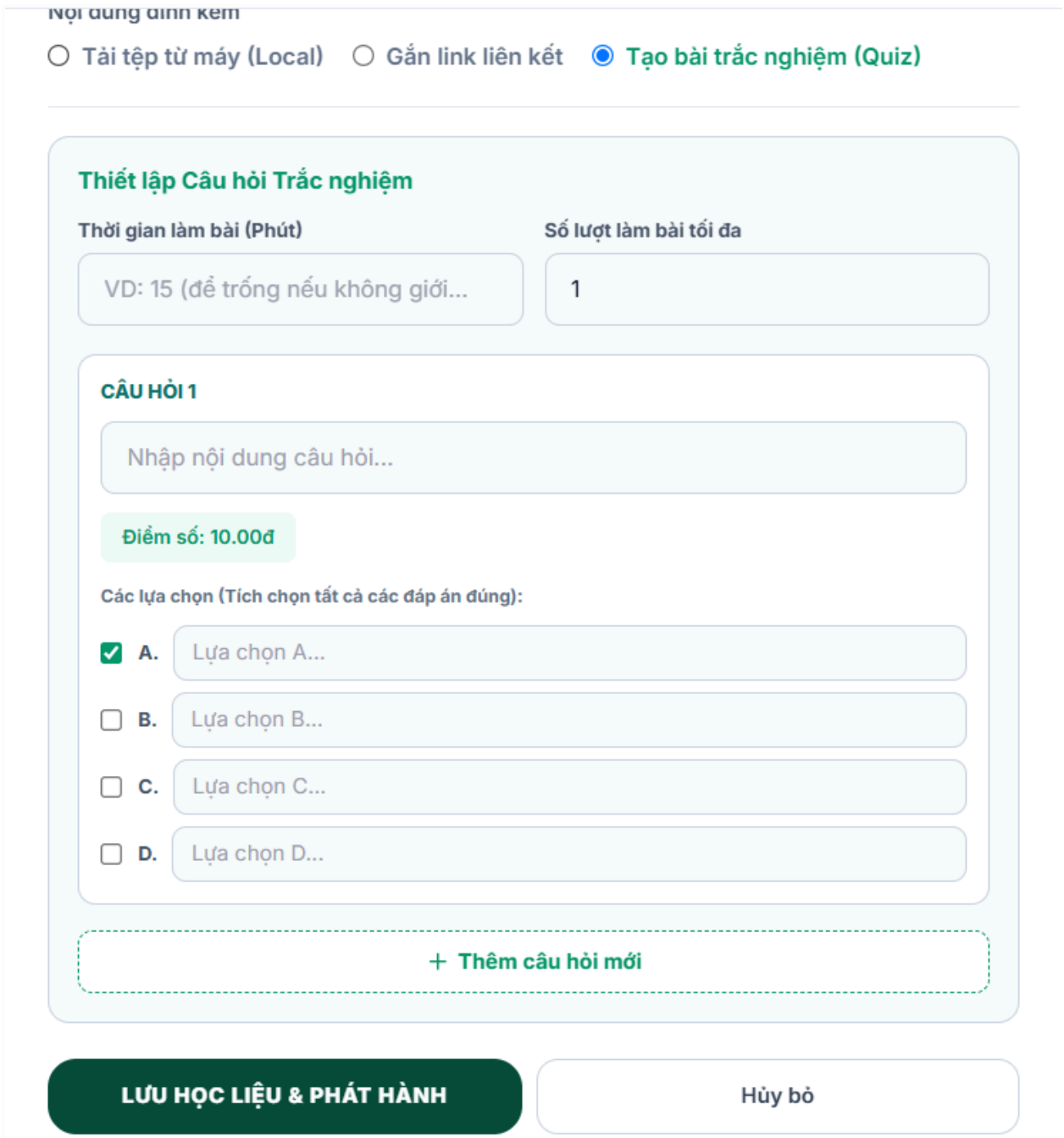
Field Name	Description
Lesson Content Tab	Switches back to the lesson content screen.
Discussion Tab	Displays the discussion area for the current learning material.
Question Title	Input field for entering the title or subject of the discussion.
Question Content	Multi-line text area for entering the detailed question or discussion content.
Cancel	Clears the current input or cancels the discussion creation process.
Post Question	Submits the discussion topic to the system for review by instructors and other participants.
All	Displays all discussion topics related to the current lesson.
My Questions	Displays only the discussion topics created by the current student.
Waiting for Response	Displays discussion topics that have not yet received a reply.
Answered	Displays discussion topics that have already received at least one response.
Search Box	Searches discussion topics by keyword.

Discussion List	Displays all discussion topics matching the selected filter or search keyword.
Discussion Card	Displays summary information of a discussion, including author, lesson, title, content preview, posting time, and discussion status.
Discussion Status	Indicates the current status of the discussion (e.g., Waiting for Response, Answered).
Lesson Tag	Displays the learning material associated with the discussion topic.

3.3. Quizzes System & Assignments

3.3.1. Create Quiz





This screen allows the Instructor/Teacher to:

- Upload new learning materials (Đăng tải Học liệu Mới) for a specific enrolled class and subject.
- Categorize the lesson by selecting or creating a chapter.
- Define the lesson name and provide a detailed description or requirements.
- Set a specific release date and due date for the material.
- Choose the type of attached content (Local file, URL link, or Quiz).

On the screen, if "Create Quiz" (Tạo bài trắc nghiệm) is selected, s/he can also:

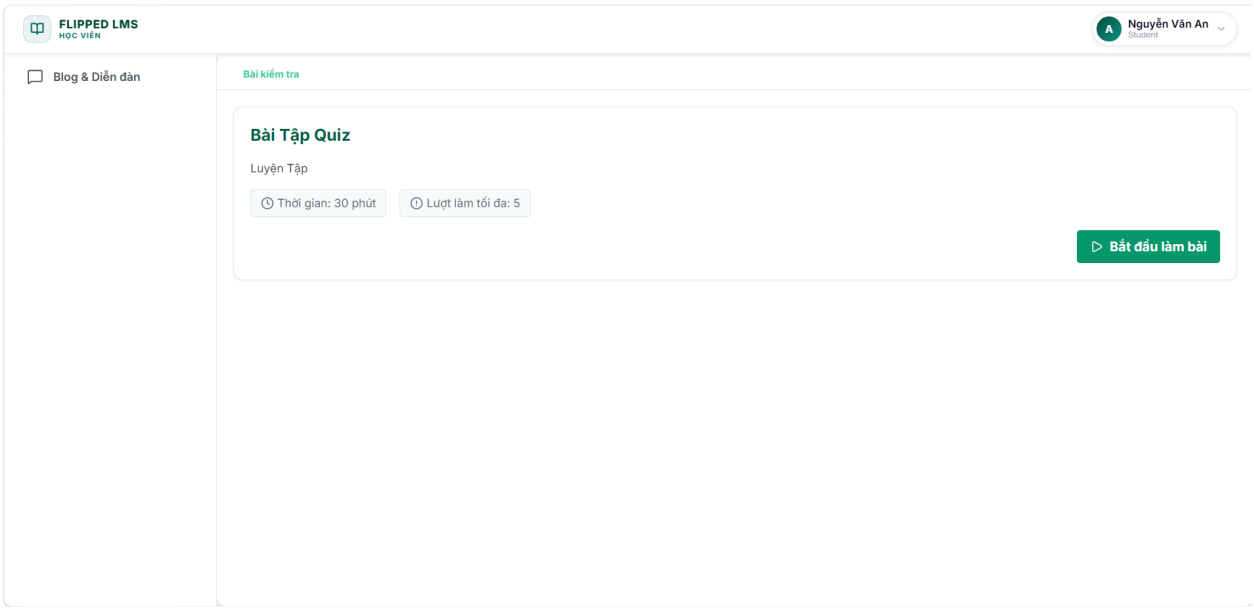
- Configure the overall quiz settings, including the time limit and maximum number of attempts.
- Create multiple-choice questions, set point values, and indicate the correct answer(s).
- Add multiple questions to build a complete quiz.
- Choose to save and publish the material or cancel the creation process.

Mapped Use Case: UC-58: Create Quizzes

Field Description

Field Name	Description
Class	Read-only field displaying the specific class code (e.g., SWP391-CLC01).
Subject	Read-only field displaying the subject code (e.g., SWP391).
Chapter	Dropdown list to select an existing chapter or create a new one. Required field (*).
Lesson Name	Text input field to enter the title of the lesson. Required field (*). Placeholder text displays: "VD: Bài 1 - Giới thiệu Agile Scrum..."
Requirement / Description	Text area to provide additional instructions, requirements, or details about the lesson.
Release Date	Date picker to select when the material becomes visible/available to students.
Due Date	Date picker to select the deadline for students to complete the lesson or quiz.
Attached Content	Radio buttons to select the attachment type: Local File, Link, or Create Quiz.
Time limit (Minutes)	Text input to define the quiz duration. Placeholder indicates it can be left blank for unlimited time.
Maximum attempts	Text input to define how many times a student is allowed to take the quiz.
Question content	Text input to type the multiple-choice question.
Score	Displays or sets the point value awarded for correctly answering the question (e.g., 10.00đ).
Options (A, B, C, D)	Text input fields for answer choices. Includes checkboxes next to each option to allow the instructor to mark all the correct answers.
Add new question	Button to append another question block to the current quiz.
SAVE MATERIAL & PUBLISH	Action button to save the inputted data, create the learning material, and publish it to the students.
Cancel	Action button to discard all changes and exit the upload screen.

3.3.2 Quiz Overview



This screen allows the Student to:

- View Quiz Information: view the details of a specific quiz before beginning.
- Start Quiz: begin the quiz attempt.

On the screen, s/he can also

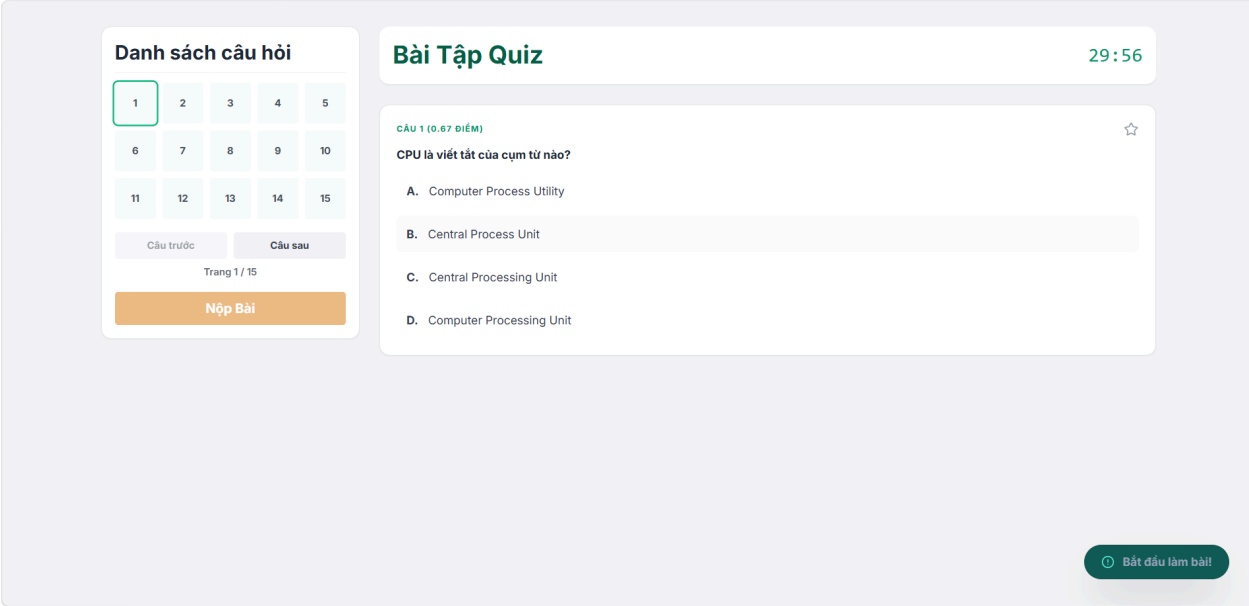
- Review the specific constraints of the quiz, such as the total time allotted and the maximum number of attempts permitted.

Mapped Use Case: UC-51: Take Quizzes

Field Description

Field Name	Description
Quiz Name	Displays the title or name of the quiz.
Quiz description	Displays additional context, instructions, or rules for the quiz.
Time limit tag	An information tag indicating the total duration allowed to complete the quiz (e.g., "Time: 30 minutes").
Max attempts tag	An information tag indicating the maximum number of times the student is permitted to take this specific quiz (e.g., "Max attempts: 3").
Start Quiz	Action button to initiate the quiz attempt and start the countdown timer.

3.3.3. Quiz Player (Quiz Execution)



This screen allows the Student to:

- Take the Quiz: view and select answers for the quiz questions.
- Navigate Questions: jump to specific questions using the Question List grid or the Previous/Next pagination buttons.
- Submit Quiz: finalize and submit their answers for grading.

On the screen, s/he can also

- Track Remaining Time: monitor the countdown timer to manage their time during the attempt.
- Flag Questions: mark specific questions for later review by clicking the star icon.
- Track Progress: identify which question they are currently viewing (highlighted in the Question List).

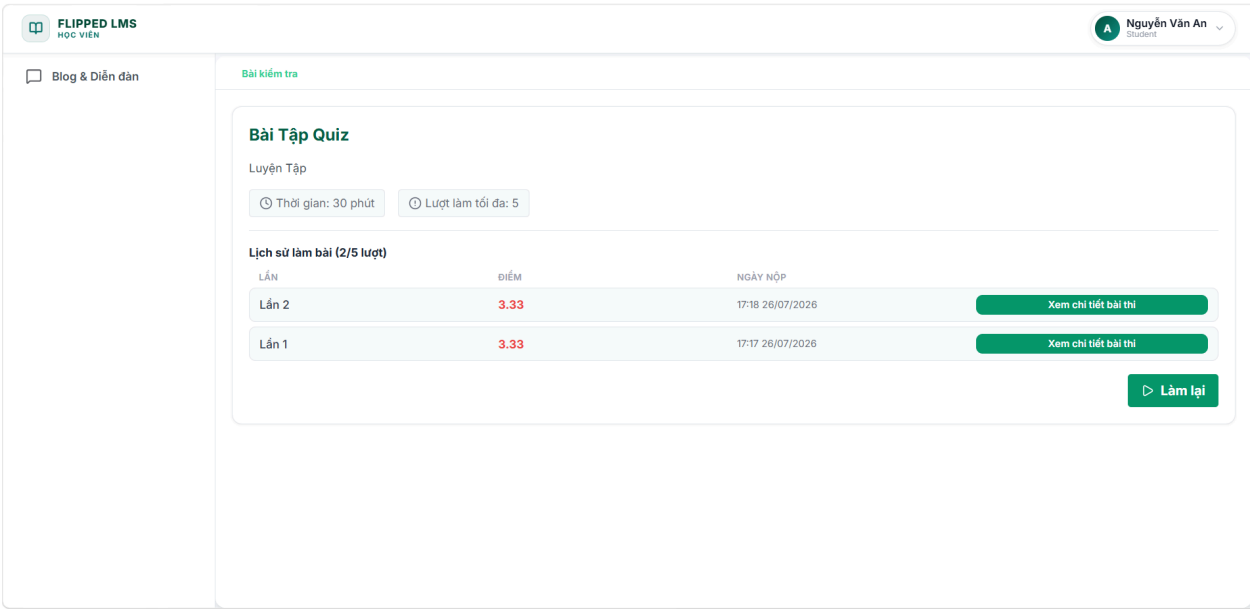
Mapped Use Case: UC-42: View Quiz Details, UC-53: Submit Quizzes

Field Description

Field Name	Description
Question List	A grid displaying all question numbers. Students can click a specific number to navigate directly to that question. The currently viewed question is visually highlighted (e.g., Question 1).
Previous / Next	Action buttons to navigate between pages within the Question List panel.
Page Indicator	Text showing the current page of the Question List out of the total pages (e.g., "Page 1 / 2").
SUBMIT	Action button to complete the quiz attempt and submit all answers.

Quiz Name	Displays the title of the active quiz.
Countdown Timer	A dynamic timer displaying the remaining time for the quiz attempt (e.g., "29:57").
Question Header	Displays the current question number and its allocated point value (e.g., "QUESTION 1 (5 POINTS)").
Star Icon	A toggle button to flag or bookmark the current question, allowing the student to easily find and review it later.
Question Content	The main text body where the question prompt is displayed.
Answer Options	The list of available choices (A, B, C, D) for multiple-choice questions. Clicking an option selects it as the answer, visually indicating the selection with a shaded background.

3.3.4. Quiz History Table



This screen allows the Student to:

- View Quiz Information: view the general details and constraints of a specific quiz.
- View Attempt History: review the log of their past attempts for this quiz.
- View Attempt Details: navigate to a detailed review of a specific past attempt to see the questions and selected answers.

On the screen, s/he can also

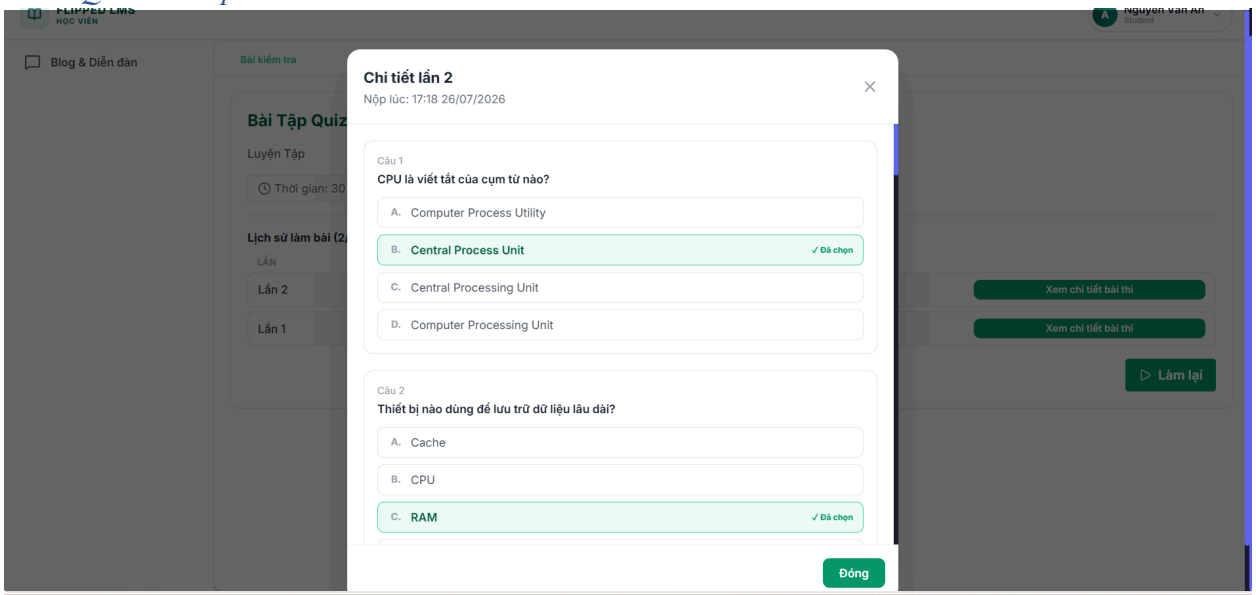
- Check how many attempts they have remaining out of the maximum allowed (e.g., 3/12 attempts).

Mapped Use Case: UC-54: View Quiz History/Scores

Field Description

Field Name	Description
Quiz Name	Displays the title or name of the quiz.
Quiz description	Displays additional context, instructions, or rules for the quiz.
Time limit tag	An information tag indicating the total duration allowed to complete the quiz attempt (e.g., "Time: 15 minutes").
Max attempts tag	An information tag indicating the maximum number of times the student is permitted to take this specific quiz (e.g., "Max attempts: 12").
Attempt history	Section header that also displays the number of completed attempts versus the total allowed (e.g., "(3/12 attempts)").
Countdown Timer	A dynamic timer displaying the remaining time for the quiz attempt (e.g., "29:57").
Attempt History Table	A table listing all previous submissions by the student. It includes the following columns: • ATTEMPT: The sequential number of the attempt (e.g., Attempt 3). • SCORE: The final grade or score achieved on that specific attempt. • SUBMITTED DATE: The exact time and date when the attempt was submitted (e.g., 15:15 15/07/2026).
View attempt details	Action button to navigate the student to a detailed result screen for that specific past attempt.

3.3.5. Quiz Attempt Detail



This screen allows the Student to:

- View Submission Details: review the specific answers they chose during a past quiz attempt.

On the screen, s/he can also

- Check the exact date and time the attempt was submitted.
- Scroll through the list of questions to see which specific options were marked as selected.
- Close the details modal to return to the previous screen.

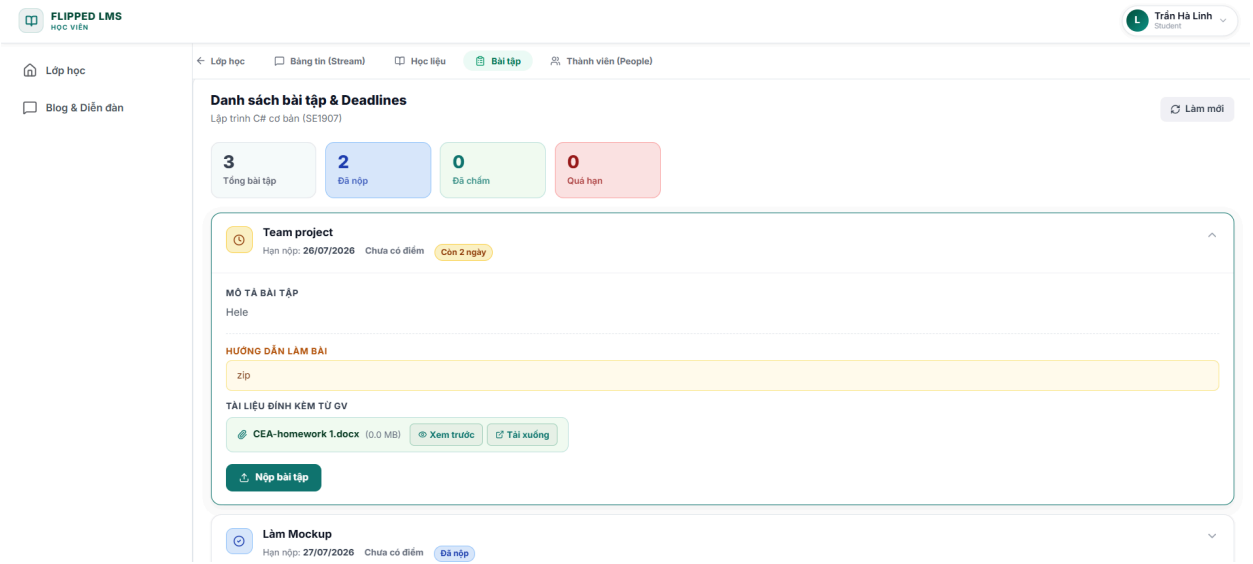
Mapped Use Case: UC-53: View Quiz History/Scores

Field Description

Field Name	Description
Submission time	Displays the timestamp indicating exactly when this specific quiz attempt was finalized and submitted (e.g., "[Submission Time]").
Question Header	Displays the sequential question number (e.g., "Question 1") for reference.
Question content	Displays the main text body or prompt of the question.
Answer Options	Lists all the available choices for that question (e.g., A, B, C, D) along with their corresponding text content.

Selected Indicator	Visually highlights the specific answer option that the student chose during the attempt, indicated by a shaded background, a checkmark icon, and the text "Selected".
Close / 'X' icon	Action buttons located at the bottom and top right (respectively) to dismiss the details modal.

3.3.6. Assignment List and Submission



This screen allows the Student to:

- View all assignments of the selected class.
- Monitor assignment statistics, including the total number of assignments, submitted assignments, graded assignments, and overdue assignments.
- View assignment details, including the description, submission instructions, instructor attachments, and reference links.
- Preview or download files attached by the instructor.
- Submit an assignment before the submission deadline.
- View the latest submitted assignment, submission time, and submission status.
- Resubmit an assignment before the deadline if necessary.
- Monitor assignment deadlines and grading status.

On this screen, the student can also:

- Expand or collapse an assignment card to show or hide its detailed information.
- Identify assignment status through visual indicators such as Submitted, Overdue, or Remaining Days.

- Refresh the assignment list to retrieve the latest assignment information.

Mapped Use Case: UC-26: View Assignments , UC-27: Submit Assignment, UC-29: Resubmit Assignment , UC-28: View Grades & Feedback

Field Description

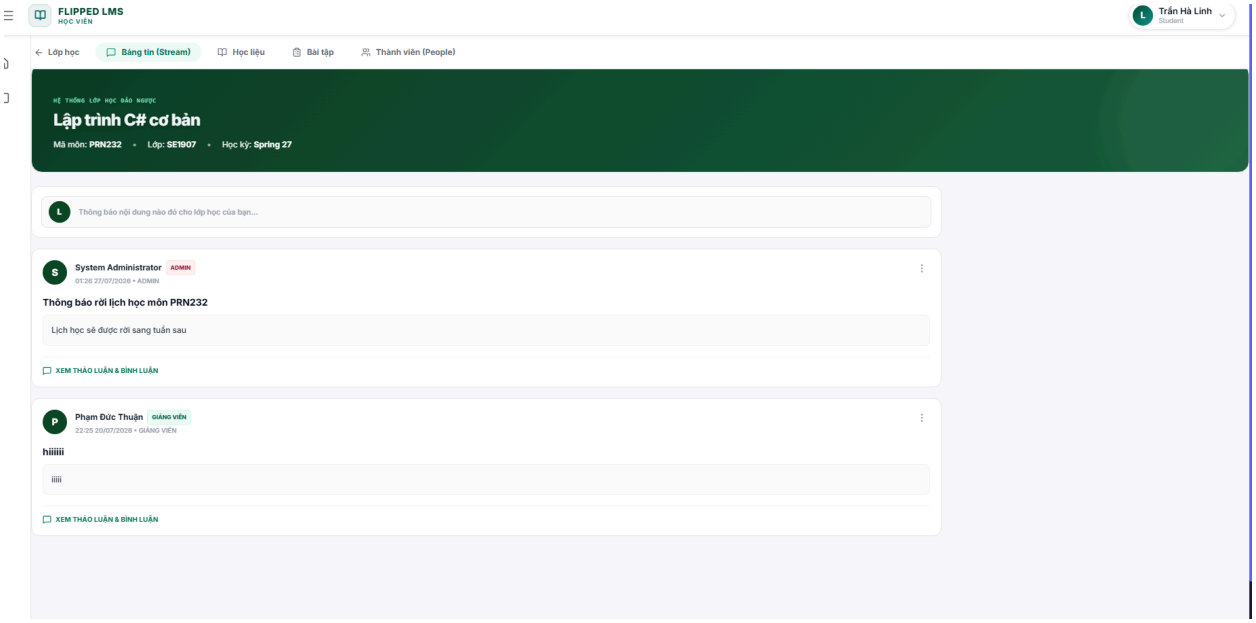
Field Name	Description
Back to Class	Returns the student to the selected class page.
Assignment Tab	Displays the assignment management page of the selected class.
Refresh	Reloads the assignment list and updates the latest information.
Total Assignments	Displays the total number of assignments in the selected class.
Submitted	Displays the number of assignments submitted by the student.
Graded	Displays the number of assignments that have been graded by the instructor.
Overdue	Displays the number of overdue assignments.
Assignment Card	Displays summary information of an assignment, including the title, due date, score, and submission status.
Expand / Collapse	Shows or hides the detailed information of the selected assignment.

Assignment Description	Displays the detailed description of the assignment.
Submission Instructions	Displays instructions provided by the instructor for completing the assignment.
Instructor Attachments	Displays files attached by the instructor.
Preview	Opens the selected attachment for online viewing.
Download	Downloads the selected attachment to the student's device.
Reference Links	Displays external resources or hyperlinks provided by the instructor.
Submit Assignment	Allows the student to upload and submit an assignment before the deadline.
Submitted Assignment	Displays the student's latest submitted file.
Preview Submitted File	Opens the submitted assignment for online viewing.
Submission Time	Displays the date and time of the latest submission.
Resubmit Assignment	Allows the student to replace the previous submission before the deadline.

Submission Status	Displays the current assignment status (e.g., Submitted, Not Submitted, Overdue).
Score	Displays the assignment score after grading. If the assignment has not been graded, the field displays No Score.

3.4. Class Stream

3.4.1 Class Stream Overview



This screen allows the Student to:

- View Class Announcements & Posts: view all internal announcements, discussions, and updates shared within the specific class.
- Create Class Posts: share information, ask questions, or post announcements to the class.

On the screen, s/he can also:

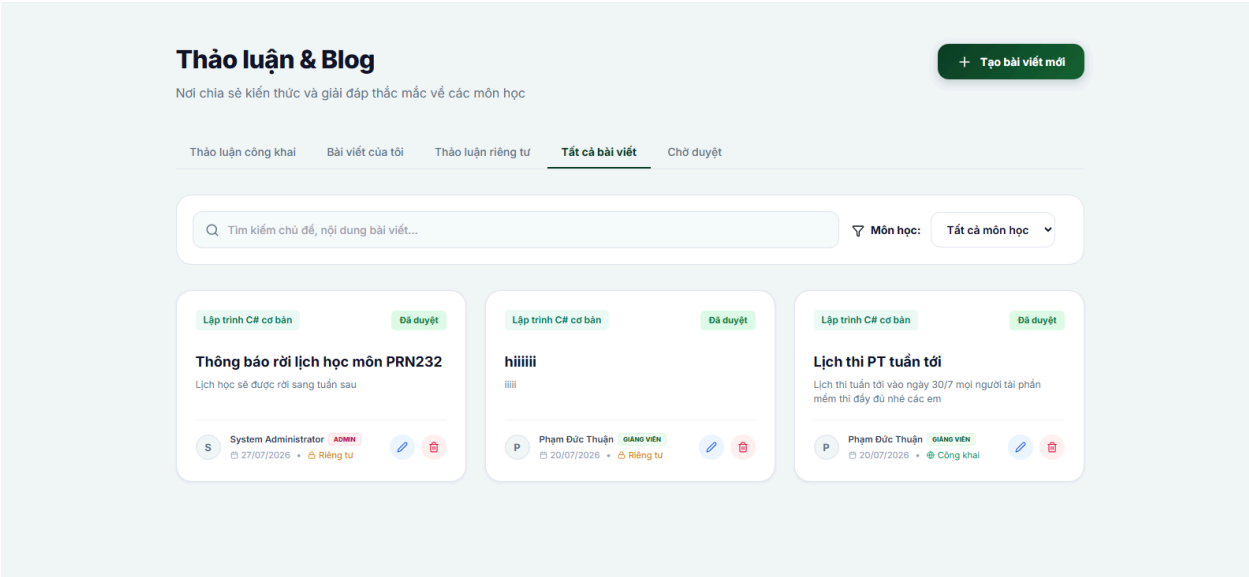
- View & Post Comments: expand discussion threads on individual posts to view existing comments or submit new replies.
- Navigate Class Tabs: switch between Stream, Learning Materials, Assignments, and Class Members.

Field Description

Field Name	Description
Course & Class Banner	Displays the course name, course code, class code, and academic term information (e.g., "Lập trình C# cơ bản", "PRN232", "SE1907", "Spring 27").
Create Post Input	Text input field allowing the student to enter and publish a new announcement or post for the class.
Author Info & Role Badge	Displays the author's full name, creation timestamp, and role badge indicating their system role (e.g., "ADMIN", "GIẢNG VIÊN", "HỌC VIÊN").
Post Title & Content	Displays the title and detailed text content of the class post or announcement.
View Discussion & Comments	Action button to expand the comment section, view existing replies, and post new comments for a specific post.
Navigation Tabs	Navigation bar allowing users to switch between Class Stream, Learning Materials, Assignments, and Class Members.

4. Blog Management

4.1.1 Blog list



This screen allows the User to:

- View and manage blog posts and discussion topics within the system.

On the screen, the person can also:

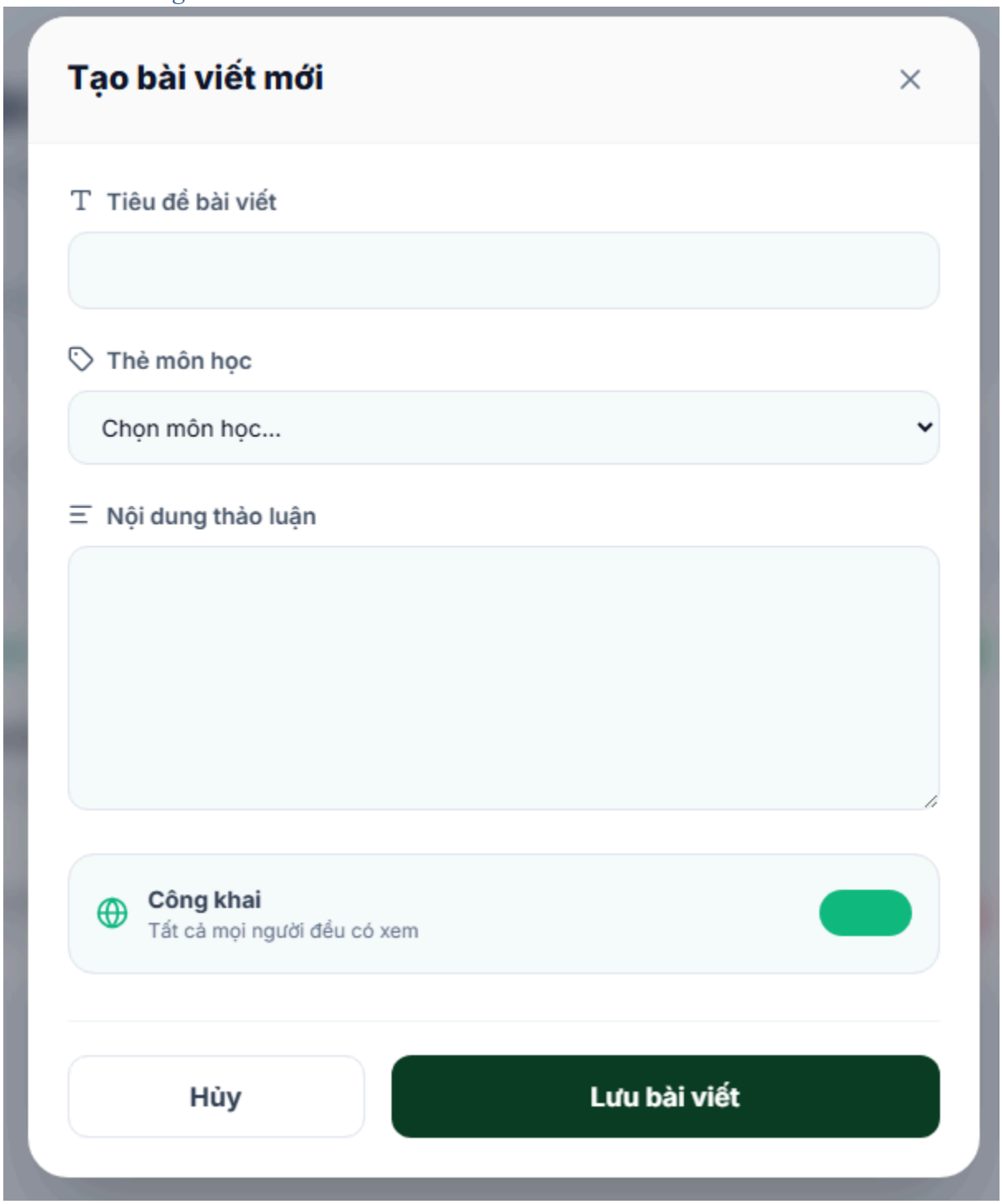
- Filter posts by category tabs (Public Discussions, My Posts, Private Discussions, All Posts, Pending).
- Search posts by topic title and filter them by subject.
- Create a new discussion or blog post by clicking the "Create New Post" button.
- Edit or delete existing posts (for authorized users/authors).

Field Description

Field Name	Description
Create New Post	Button to navigate to the post creation screen.
Tabs (Public Discussions, My Posts, Private Discussions, All Posts, Pending)	Navigation tabs to filter posts based on discussion scope, ownership, and approval status.
Search topic	Input search bar to find specific posts by topic keywords or title.
All Subject	Dropdown menu to filter posts by specific subjects/courses.
Tag Subject	Label tag indicating the subject associated with the post.
Title	The title of the blog post or discussion topic.

Content	Preview summary or snippet content of the post.
Author's name	Display name of the post creator.
Role	The user role of the author (e.g., Student, Lecturer).
Date	The publication date of the post.
Edit	Action button to modify the existing post.
Delete	Action button to remove the post from the system.

4.1.2 Create Blog



This screen allows the User to:

- Create and publish a new blog post or discussion topic within the system.

On the screen, the person can also:

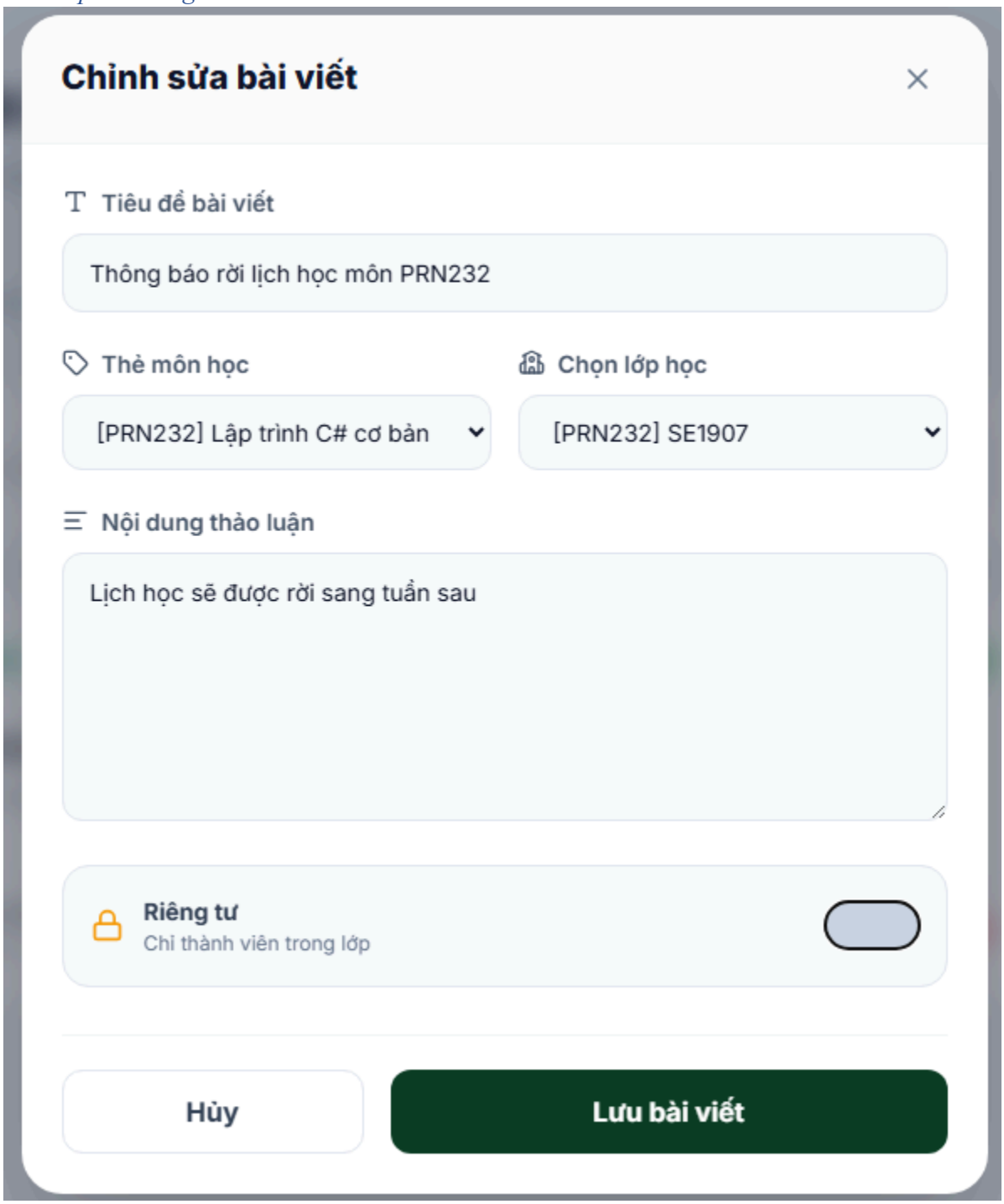
- Input the post title, select/tag an associated subject, and write the main content.
- Set the post visibility scope using the Public/Private toggle switch.

- Submit the post for creation or cancel the action to return to the Blog List screen.

Field Description

Field Name	Description
Title	Text input field to enter or modify the title of the post (Required).
Subject	Input/dropdown field to select or update the subject/course tag associated with the post.
Content	Multi-line text area to edit the detailed body text and description of the post.
Public/Private	Toggle switch to change the visibility scope of the post (Public for general access, or Private for restricted access).
Cancel	Button to discard all changes and return to the Blog List screen without saving.
Submit	Button to save and publish the newly created post into the system.

4.1.3 Update Blog



This screen allows the User to:

- Update and modify the information of an existing blog post or discussion topic.

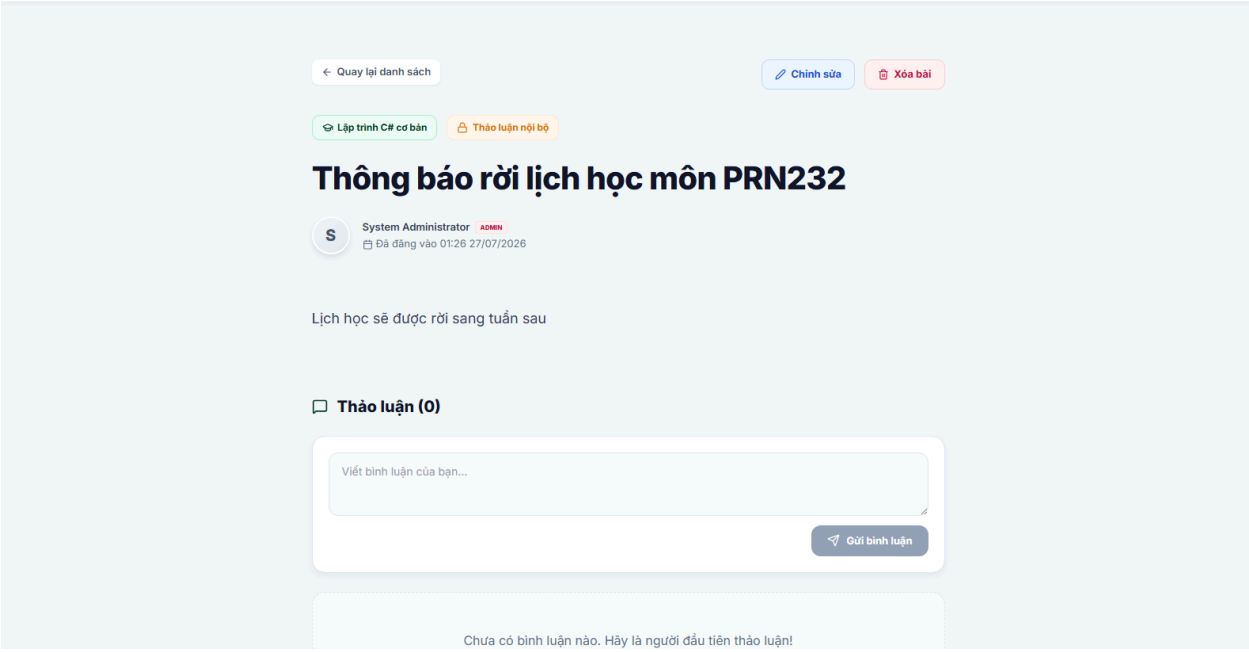
On the screen, the person can also:

- Modify the title, associated subject tag, and body content of the post.
- Toggle the visibility status between Public and Private.
- Save the updated post details or cancel the modification to return to the Blog List screen.

Field Description

Field Name	Description
Title	Text input field to enter or modify the title of the post (Required).
Subject	Input/dropdown field to select or update the subject/course tag associated with the post.
Content	Multi-line text area to edit the detailed body text and description of the post.
Public/Private	Toggle switch to change the visibility scope of the post (Public for general access, or Private for restricted access).
Cancel	Button to discard all changes and return to the Blog List screen without saving.
Save Post	Button to submit and save the updated post information into the system database.

4.1.4 Blog Detail



This screen allows the User (Student / Lecturer / Admin) to:

- View the full details of a blog post or discussion topic and participate in the discussion.

On the screen, the person can also:

- Navigate back to the Blog List screen.

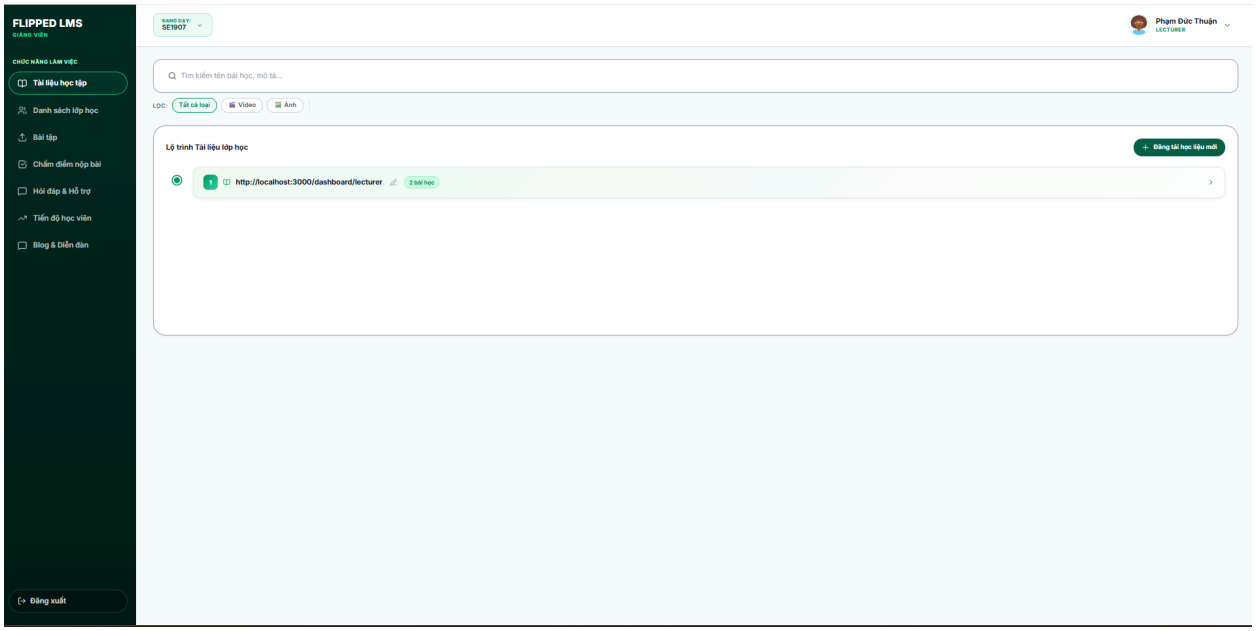
- View subject tags, visibility scope, post title, author information, publication timestamp, and main content.
- Edit or delete the post (for post author or Admin).
- Write and submit new comments/replies to the discussion thread.

Field Description

Field Name	Description
Back to list	Button to return to the Blog List screen.
Edit	Button to navigate to the Edit Post screen (visible to author or Admin).
Delete	Button to remove the post from the system (visible to author or Admin).
Subject Tag	Badge indicating the specific course/subject associated with the post.
Visibility Tag	Badge displaying the post scope (Public or Internal/Private discussion).
Title	The main title header of the post.
Author Info	Displays author's avatar, name, role badge (e.g., ADMIN, LECTURER, STUDENT), and publication date/time.

5. Lecturer Workspace Management

5.1.1 Lecturer Dashboard



This screen allows the User to:

- View a summary of class statistics and navigate to different lecturer features.

On the screen, the person can also:

- Select an assigned class to view its workspace.
- Navigate to Materials, Class List, Assignments, Grading, Feedback, Progress, and Blog using the sidebar.
- Access profile settings or sign out of the system.

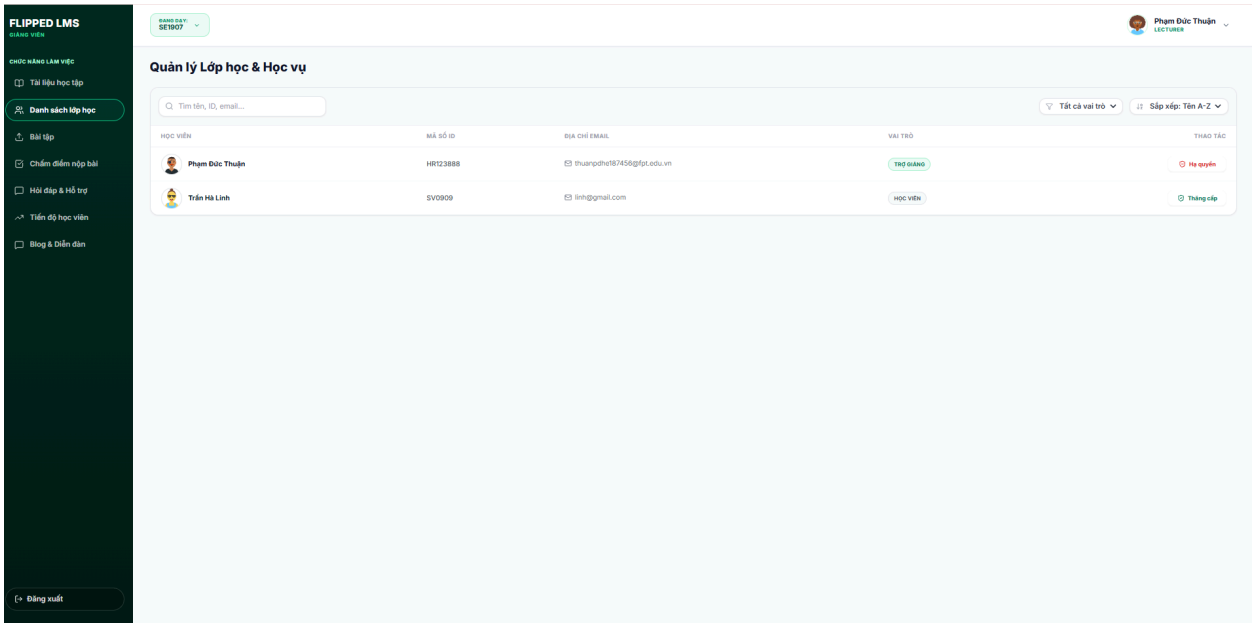
Field Description

Field Name	Description
Class Selector	Dropdown menu to choose from the list of classes assigned to the lecturer.
Sidebar Navigation	Menu panel containing navigation links to Materials, Class List, Assignments, Grading, Feedback, Progress, and Blog.
User Profile	Button to open the profile details and settings modal.
Logout	Button to safely terminate the session and return to the login screen.

Mapped Use Case: UC-01: User Login, UC-02: User Logout

5.2. Class List & Role Management

5.2.1. Student Roster



This screen allows the User to:

- View the student roster of the selected class.

On the screen, the person can also:

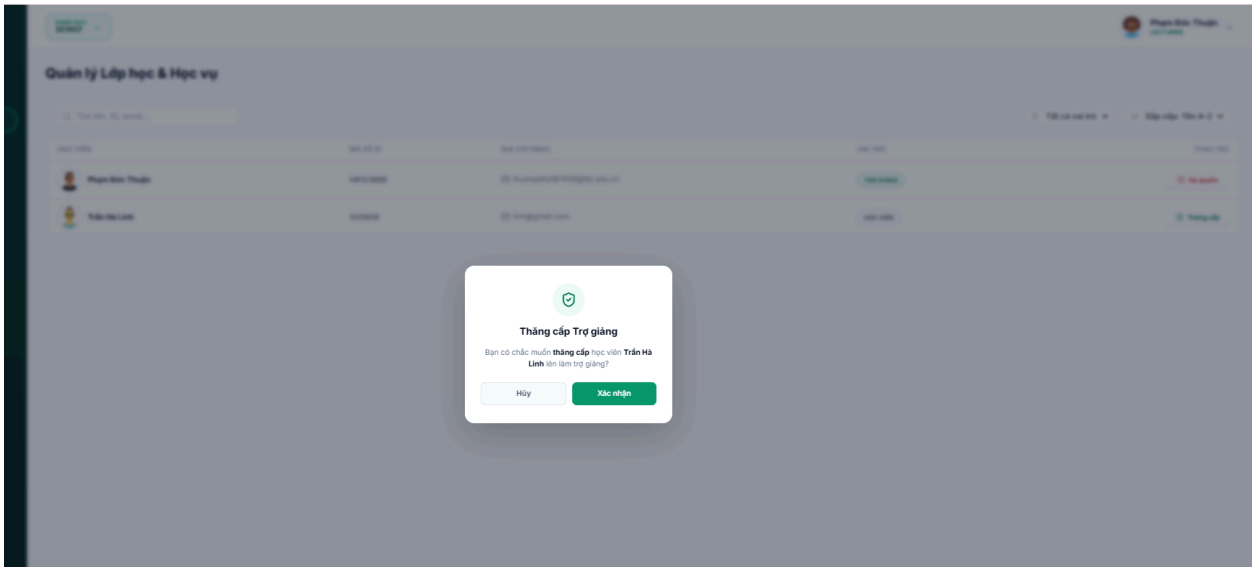
- Search students by name or email.
- Filter the roster by classroom role.
- Open the promotion confirmation dialog for any student.

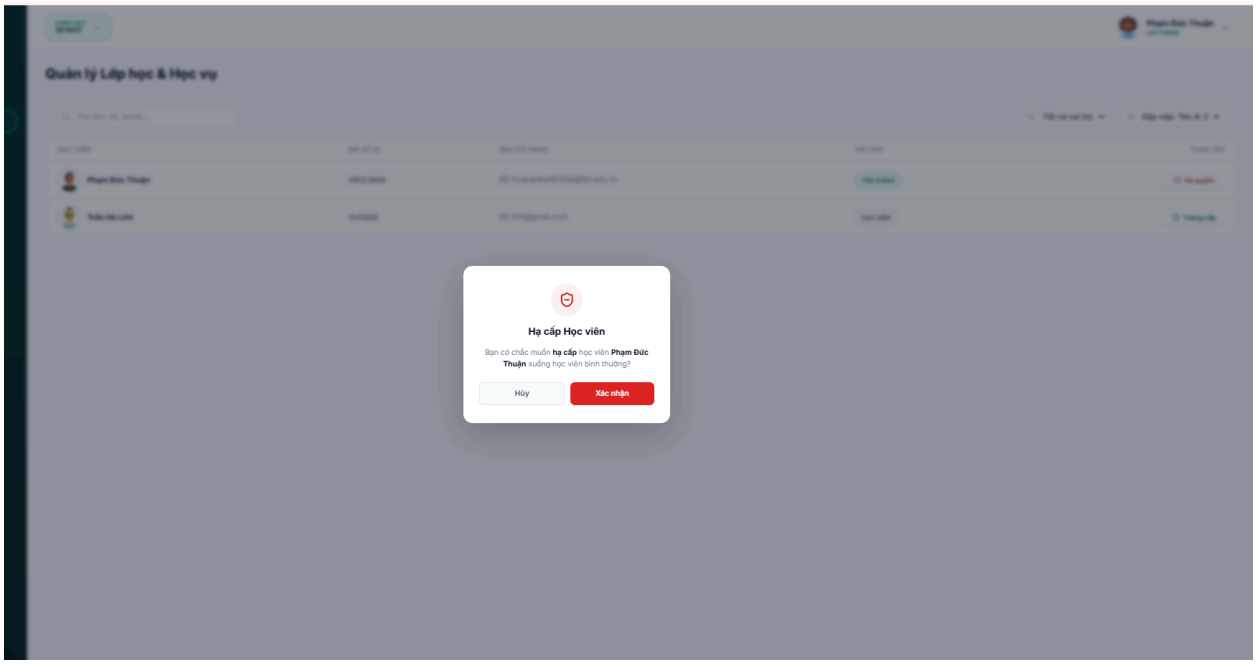
Field Description

Field Name	Description
Student Search	Text input field to search students by name or email.
Role Filter	Dropdown filter to view All, Teaching Assistants, or regular Students.
Sort Option	Dropdown to sort the student roster alphabetically.
Role Status Icon	Visual indicator of the student's current role (Student or Assistant).
Manage Role	Button next to each student to trigger the Role Promotion process.
Student Search	Text input field to search students by name or email.

Mapped Use Case: UC-34: Manage Class, UC-40: Track Student

5.2.2. Role Promotion





This screen allows the User to:

- Confirm or cancel role transitions for a student.

On the screen, the person can also:

- View the student name and the target role (Promote to Teaching Assistant or Demote to Student).
- Confirm the change to apply privileges, or cancel to abort.

Field Description

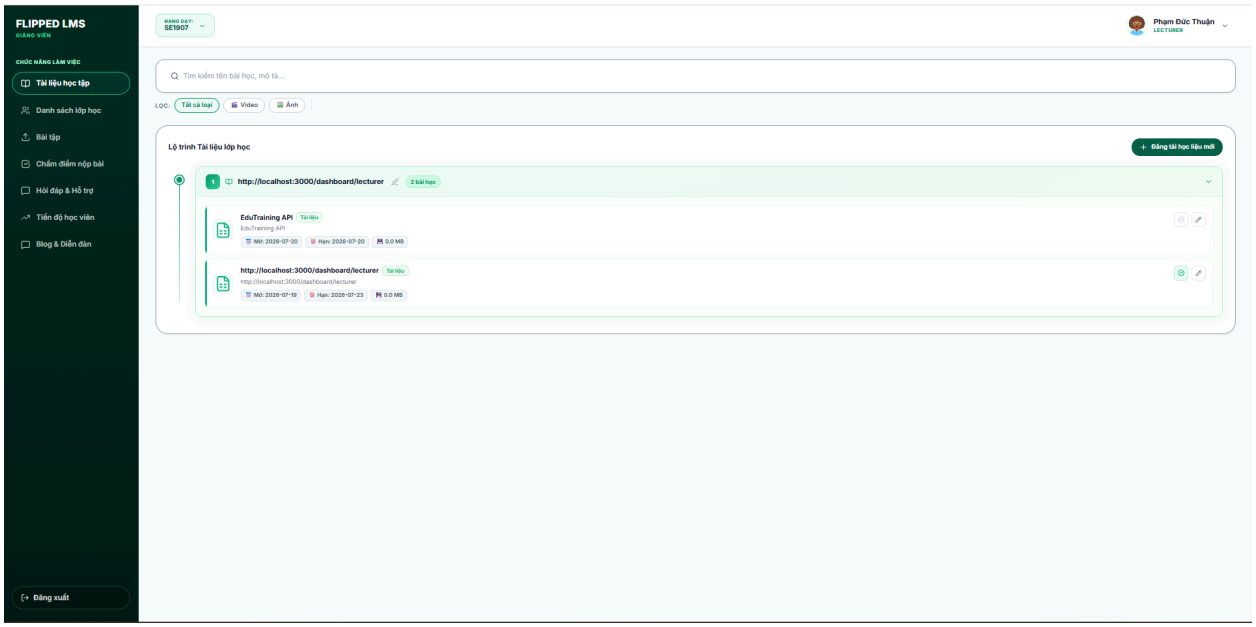
Field Name	Description

Student Name Label	Text showing the name of the student being managed (Read-only).
Target Role Label	Text showing the new role to be assigned (Read-only).
Confirm Button	Button to execute the API call and update the student's role.
Cancel Button	Button to close the modal without making changes.

Mapped Use Case: UC-57: Role Promotion

5.3. Learning Materials Management

5.3.1. Material List



This screen allows the User to:

- View learning resources grouped by class chapters.

On the screen, the person can also:

- Search materials by keyword and filter by file type.
- Trigger chapter renaming or delete specific materials.
- Open modals to upload materials or build quizzes.

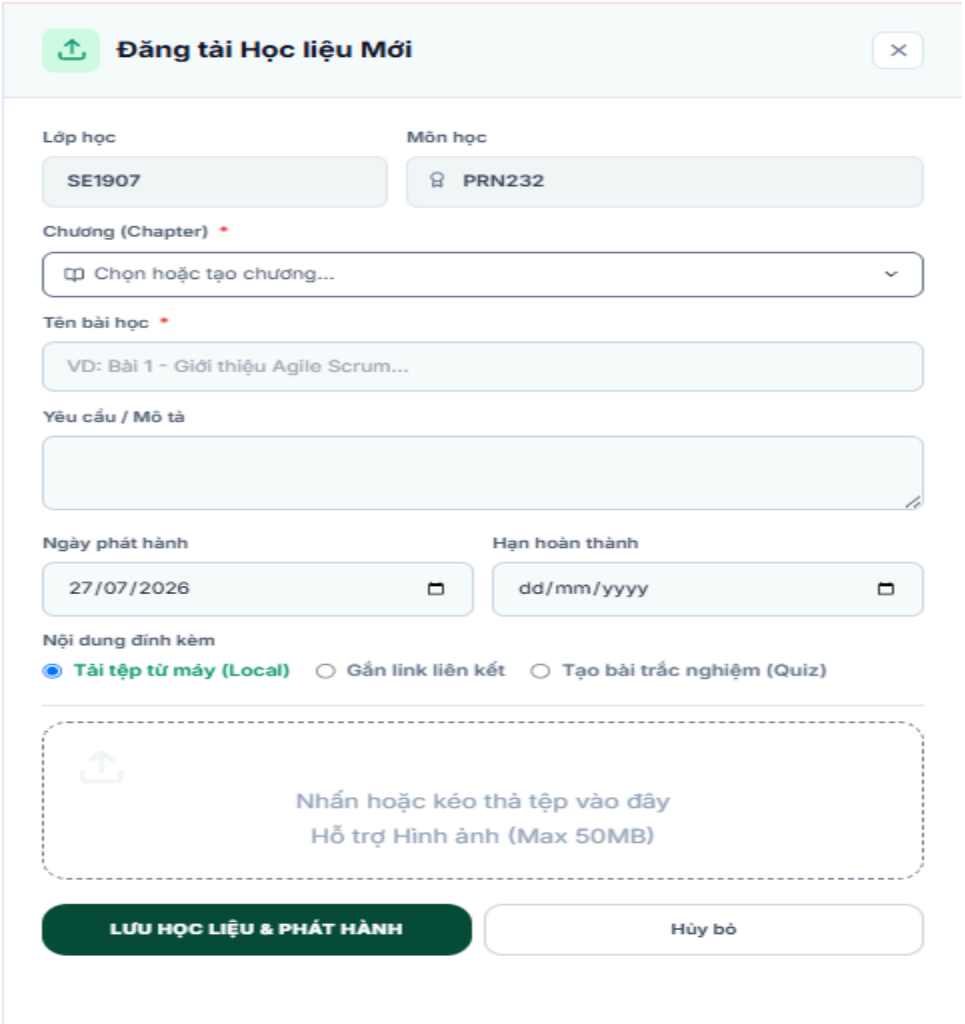
Field Description

Field Name	Description
Materials Search	Text input field to search materials by title.
Type Filter	Tab buttons to filter materials by type: All, Video, PDF, Document, or Quiz.
Add Material Button	Button to trigger the upload modal.

Rename Chapter Link	Clickable option next to a chapter title to edit its name.
Edit Material Button	Button next to a material to open the editing form.
Delete Material Button	Button to remove the material from the class.

Mapped Use Case: UC-30: Manage Learning Materials, UC-23: View Materials

5.3.2. Upload Material



This screen allows the User to:

- Add a new learning material to a class chapter.

On the screen, the person can also:

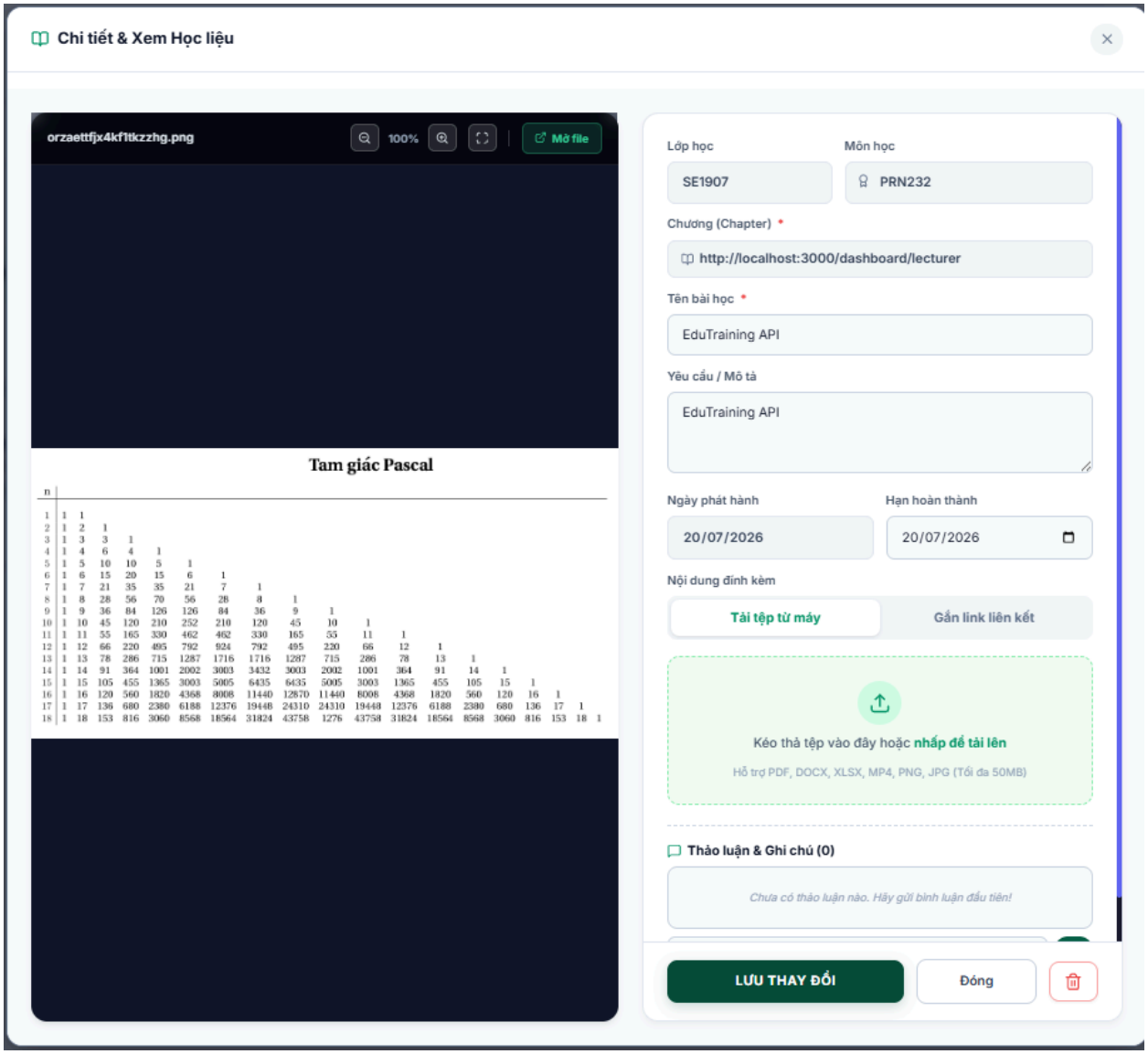
- Enter title, description, chapter, and select type.
- Pick a local file (Video, PDF, Document) or paste an external URL.

Field Description

Field Name	Description
Title	Text input field to enter the title of the material (Required).
Chapter Name	Dropdown selection or text input to assign the material to a chapter (Required).
Type Selector	Dropdown selection to define the material type (Video, PDF, Document, Link, Quiz).
File Picker	Upload area to drag and drop or select files under 50MB (Only for file-based types).
Link URL	Text input field for YouTube/external web URL (Only for Link type).
Cancel Button	Button to discard all inputs and close the modal.
Save Button	Button to upload the file to Cloudinary and save metadata.

Mapped Use Case: UC-31: Upload Materials

5.3.3. Edit Material



This screen allows the User to:

- Edit an existing material's metadata and replace its file.

On the screen, the person can also:

- Modify the title, description, and chapter.
- Upload a new file to replace the old one, and preview PDF, image, or video files.
- Zoom in/out of image previews.

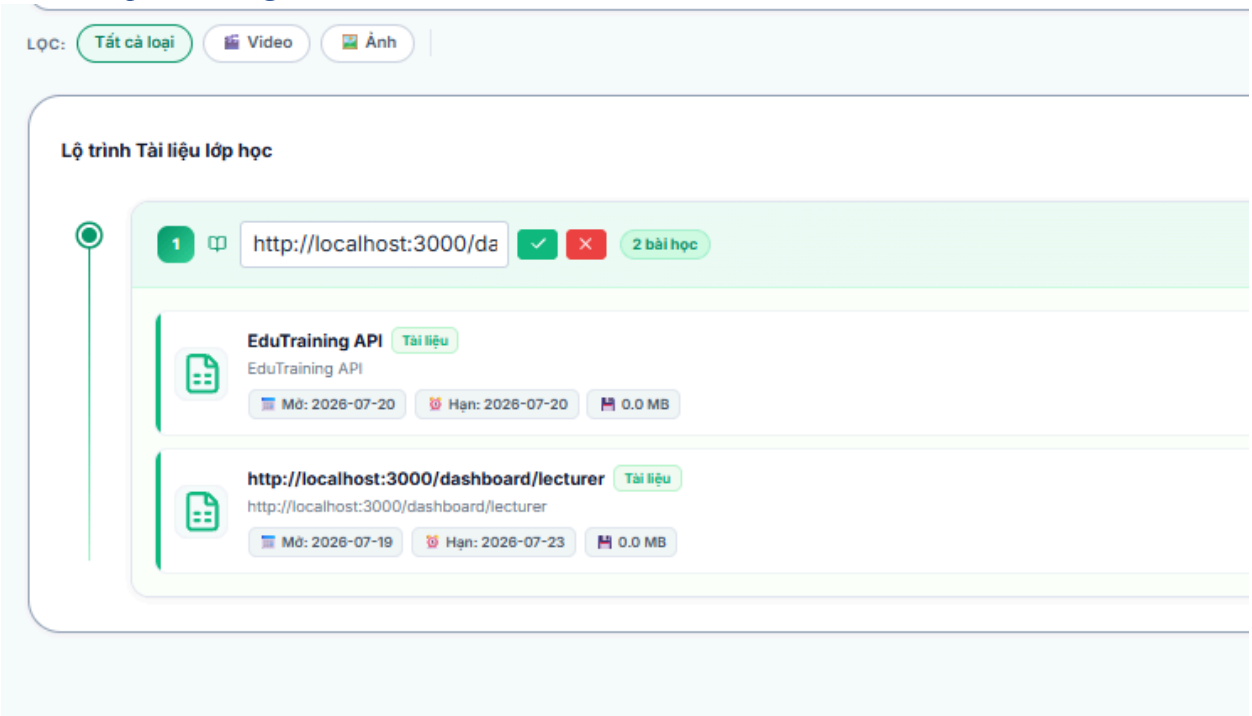
Field Description

Field Name	Description
Edit Title	Text input field to modify the material's title.
Edit Description	Text area to modify the material's description.

Replace File Input	File picker to upload a new file to overwrite the existing asset.
Zoom In/Out Buttons	Control buttons to adjust the scale of the image preview.
Cancel Button	Button to close the modal without saving changes.
Update Button	Button to save updated info and files.

Mapped Use Case: UC-32: Update Materials

5.3.4. Chapter Management



This screen allows the User to:

- Rename an existing chapter group within the materials list.

On the screen, the person can also:

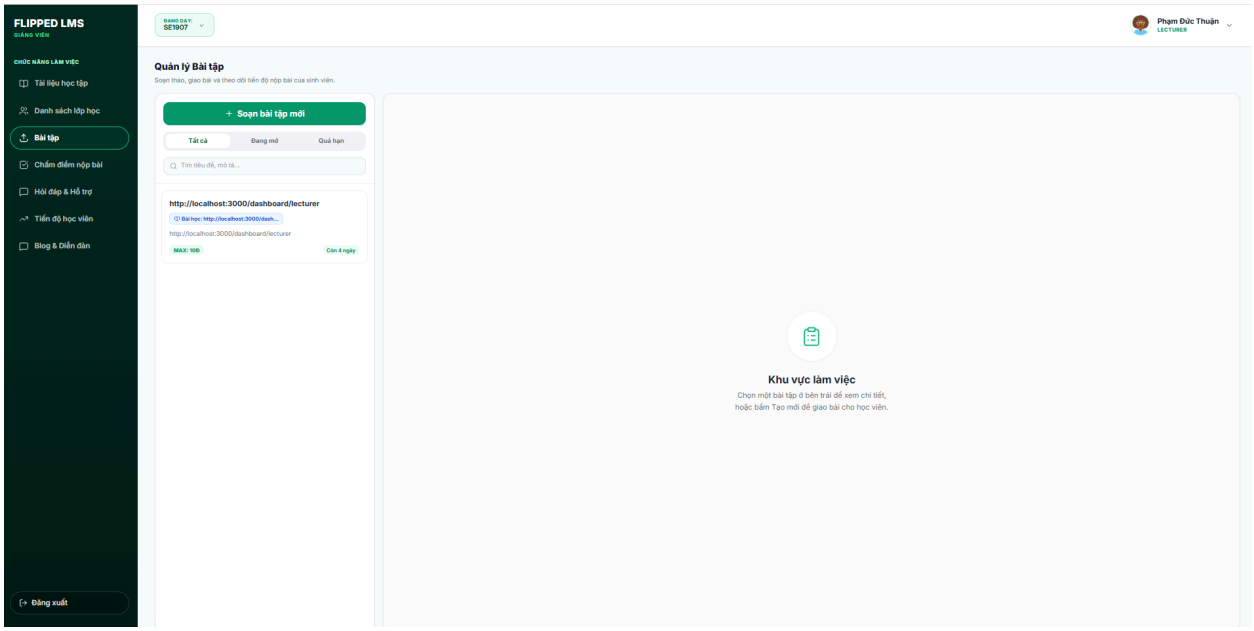
- Enter a new name for the chapter, update it across all linked materials, or cancel.

Field Description

Field Name	Description
Old Chapter Name	Display label showing the current chapter name (Read-only).
New Chapter Name	Text input field to enter the new name of the chapter (Required).
Cancel Button	Button to close the dialog without saving.
Confirm Button	Button to update the chapter name across the database.

5.4. Assignment Management

5.4.1. Assignment List



This screen allows the User to:

- Browse, filter, and search assignments created for the class.

On the screen, the person can also:

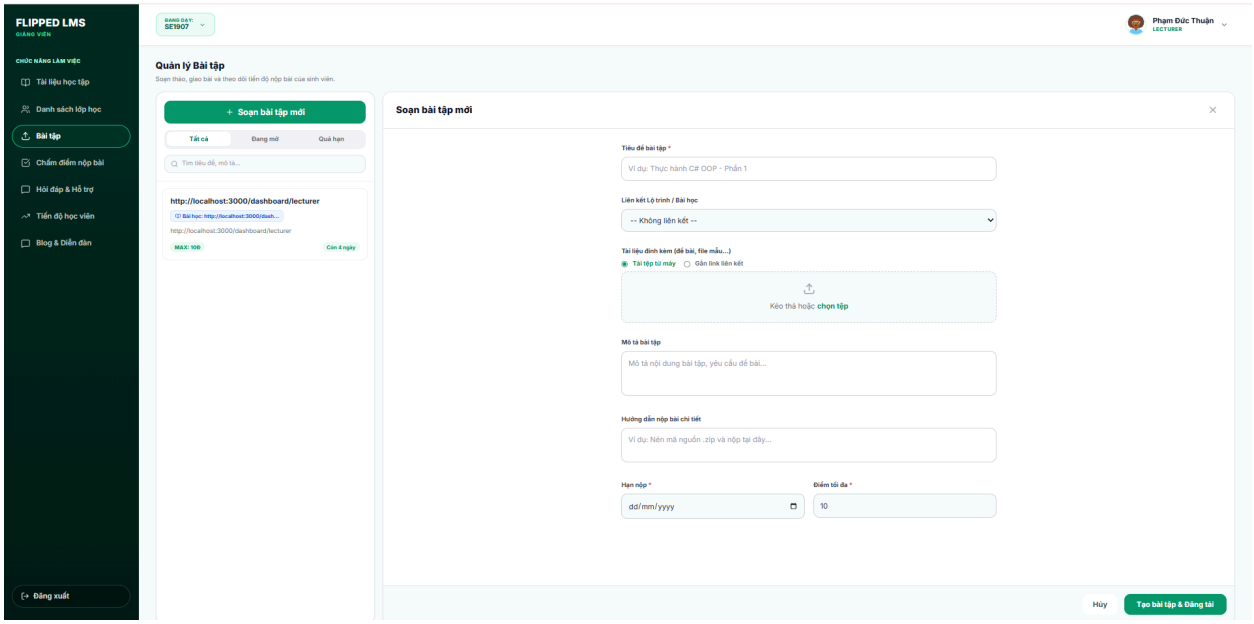
- Filter assignments by their due status, or search them by title.
- Open creation/editing screens or delete assignments.

Field Description

Field Name	Description
Assignment Search	Text input field to search assignments by title.
Status Filter	Filter tabs to display All, Active, or Overdue assignments.
Create Assignment Button	Button to open the assignment creation view.
Edit Assignment Button	Button next to an assignment to open the edit form.
Delete Assignment Button	Button to delete a specific assignment from the class.

Mapped Use Case: UC-34: Manage Class

5.4.2. Create Assignment



This screen allows the User to:

- Create a new classroom assignment task.

On the screen, the person can also:

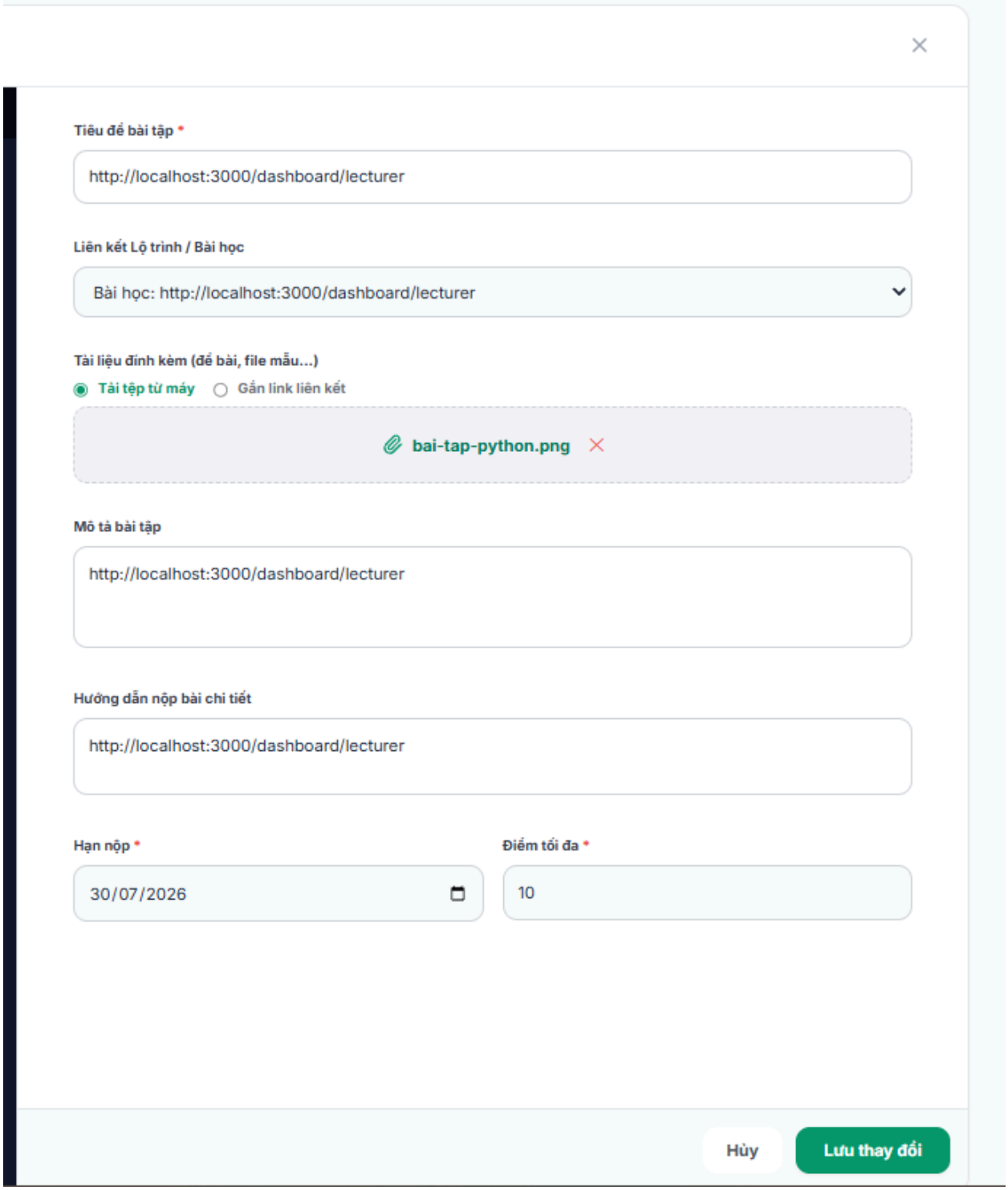
- Input assignment title, instructions, maximum points, and due date.
- Upload reference files or link assignments to study materials.

Field Description

Field Name	Description
Title	Text input field to enter the assignment title (Required).
Instructions	Multi-line text area to write down assignment guidelines.
Max Points	Numeric input field to set maximum grade score (Required).
Due Date	Date-time picker to set the submission deadline (Required).
Attach File Input	File picker to upload guideline sheets or reference docs.
Linked Material Dropdown	Dropdown selection to link the assignment to a specific material node.
Cancel Button	Button to discard input and return to list.
Save Button	Button to publish the assignment.

Mapped Use Case: UC-35: Create Assignment

5.4.3. Edit Assignment



This screen allows the User to:

- Modify an existing assignment's configuration.

On the screen, the person can also:

- Update title, instructions, max points, and deadline dates.

- Upload a new attachment file or change the linked material.

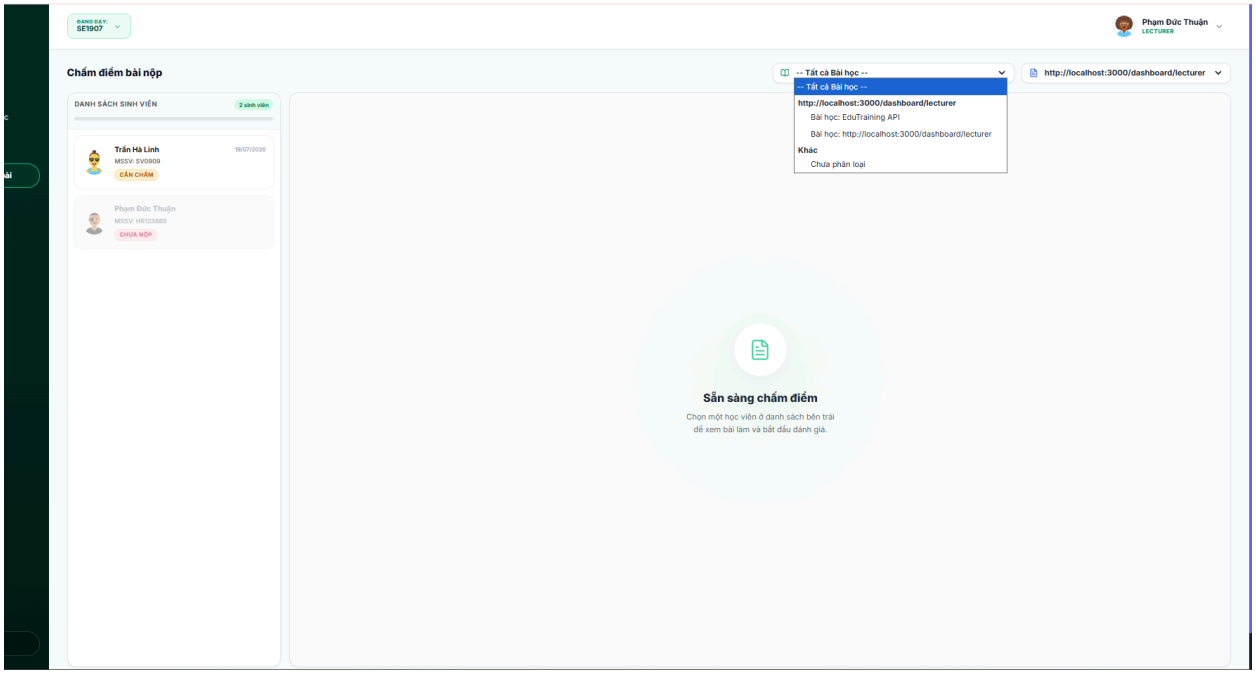
Field Description

Field Name	Description
Edit Title	Text input field to update assignment title.
Edit Instructions	Multi-line text area to modify instructions.
Edit Max Points	Numeric input field to change maximum grade score.
Edit Due Date	Date-time picker to set the new deadline.
Change Attachment	File picker to upload a different guideline file.
Change Linked Material	Dropdown selection to modify the linked material.
Update Button	Button to save changes.

Mapped Use Case: UC-35: Create Assignment

5.5. Grading Board

5.5.1. Submission Overview



This screen allows the User to:

- View a summary of student submissions for a selected assignment.

On the screen, the person can also:

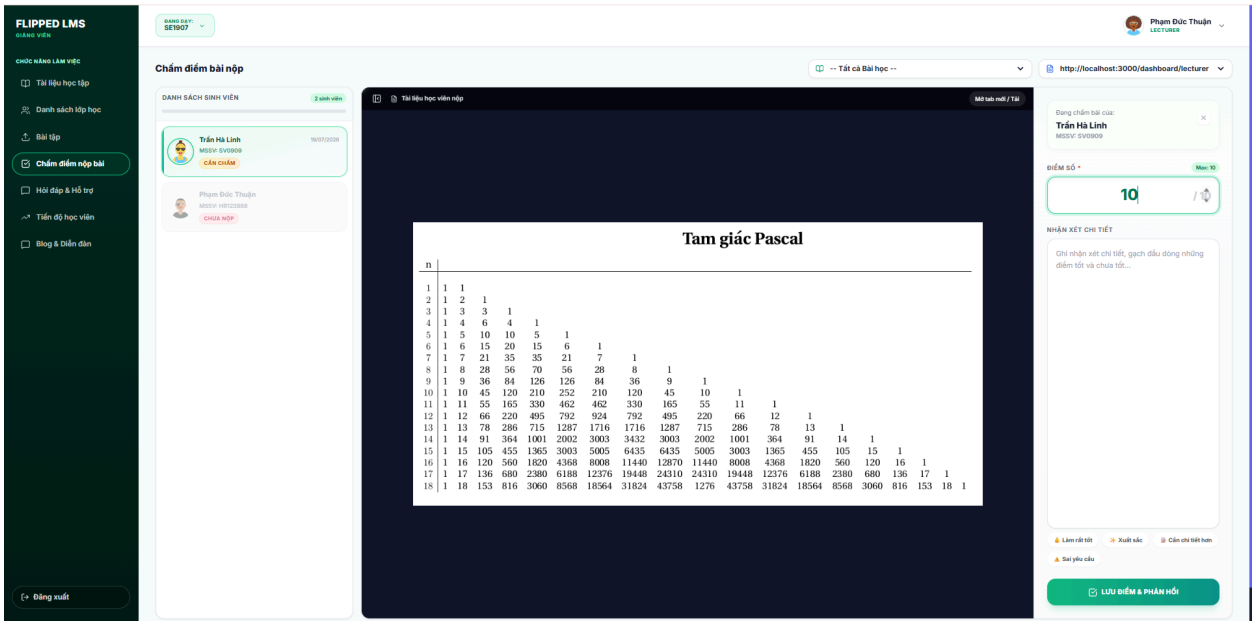
- Filter submissions by linked materials and student submission status.
- Select a student to open the grading details panel.

Field Description

Field Name	Description
Assignment Selector	Dropdown list to choose which assignment to review.
Material Filter	Dropdown list to filter assignments linked to a specific material.
Status Filter	Tab buttons to display Ungraded, Graded, or Not Submitted students.
Student Row	Clickable row in the list to select a student for grading.

Mapped Use Case: UC-38: Grade Assignment

5.5.2. Grade Submission



This screen allows the User to:

- Evaluate student work, assign scores, and write remarks.

On the screen, the person can also:

- Download and preview the submitted file.
- Input a score, write feedback remarks, and enable auto-advance to navigate between students.

Field Description

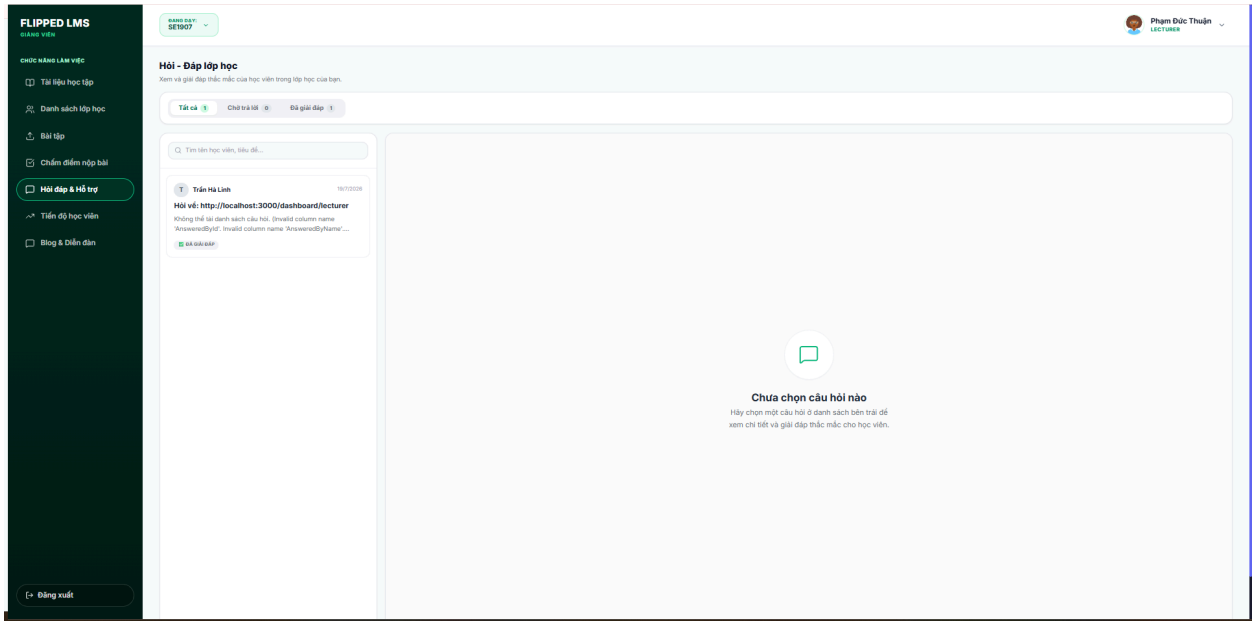
Field Name	Description
File Download Link	Button to download the student's submitted file.
Grade Score Input	Numeric input field to enter the grade score (Must be between 0 and Max Points).
Remarks Textarea	Multi-line text field to write feedback remarks for the student.

Auto-advance Checkbox	Toggle checkbox to automatically load the next ungraded student upon saving.
Previous/Next Buttons	Navigation buttons to cycle through student submissions manually.
Save Grade Button	Button to submit the score and remarks to the database.

Mapped Use Case: UC-38: Grade Assignment, UC-36: Provide Feedback

5.6. Feedback Support

5.6.1. Feedback List



This screen allows the User to:

- Manage Q&A tickets and feedbacks sent by students.

On the screen, the person can also:

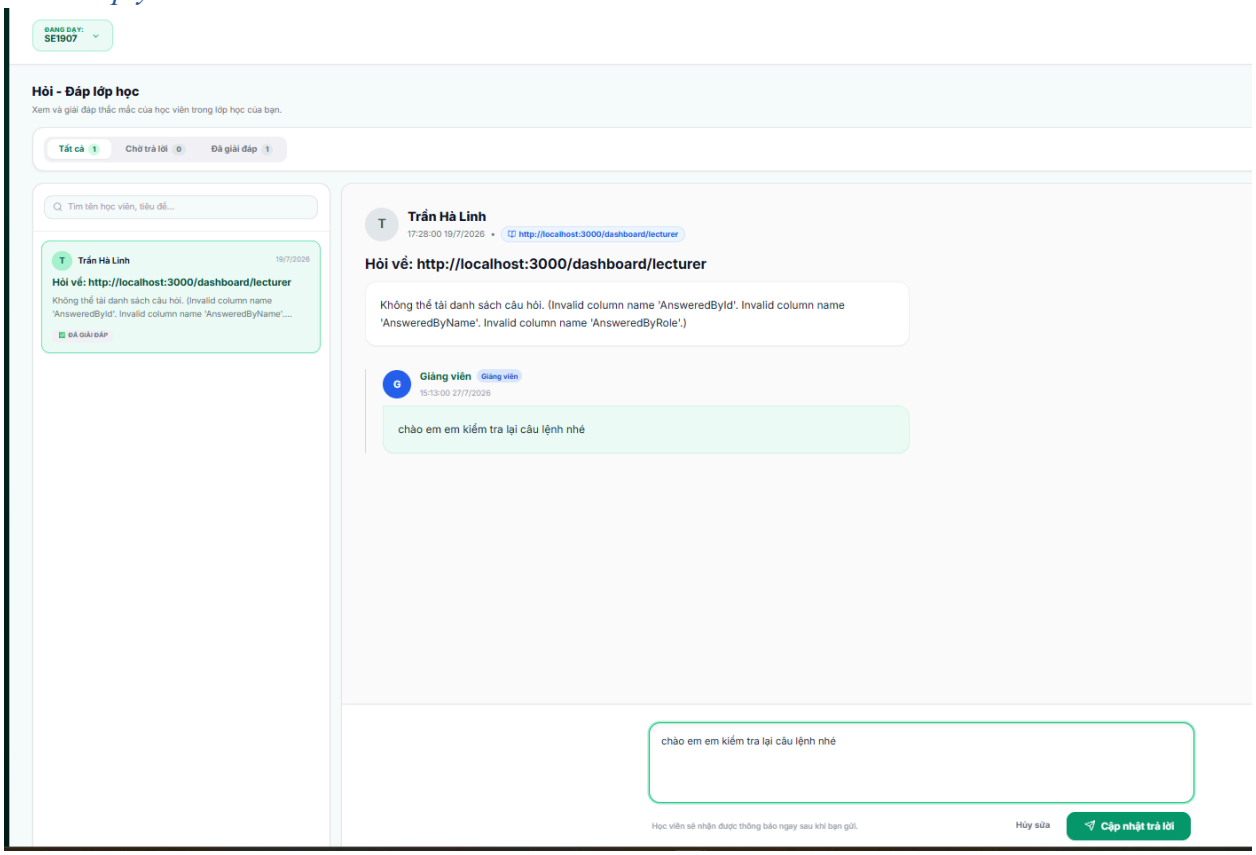
- Filter tickets by their status (Open/Resolved) and search by keywords or student names.
- Select a ticket to open the response editor.

Field Description

Field Name	Description
Status Filter	Tab buttons to display All, Open, or Resolved feedbacks.
Search Feedback	Text input field to search feedback by student name or keywords.
Feedback Card	Clickable list card that selects a feedback ticket for review.

Mapped Use Case: UC-37: Respond Support Forum Thread

5.6.2. Reply Feedback



This screen allows the User to:

- Write, send, or edit responses to student feedback.

On the screen, the person can also:

- View the student's question and any associated material link.
- Write a response, edit a previously sent response, or cancel.

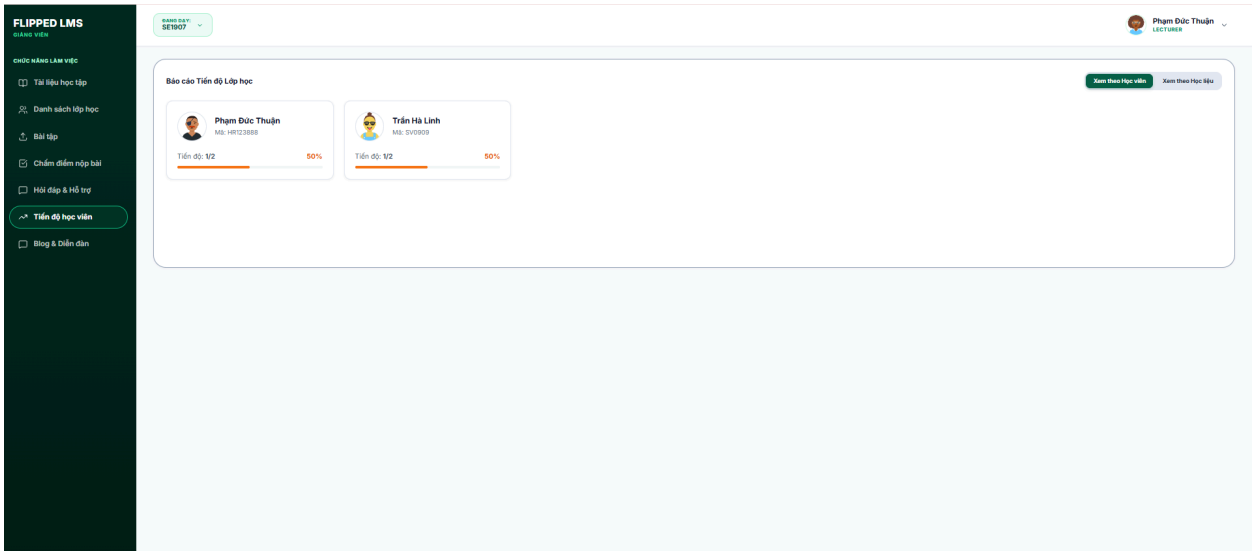
Field Description

Field Name	Description
Student Question Label	Read-only text box displaying the student's message.
Linked Material Link	Clickable link to view the material the student is asking about.
Response Input	Multi-line text field to write the response (Required).
Edit Button	Button to unlock the text area for a resolved ticket to edit the response.
Submit Reply Button	Button to save the response, changing the ticket status to Resolved.

Mapped Use Case: UC-37: Respond Support Forum Thread

5.7. Student Progress Tracking

5.7.1. Progress by Individual



This screen allows the User to:

- Track material completion status on a student-by-student basis.

On the screen, the person can also:

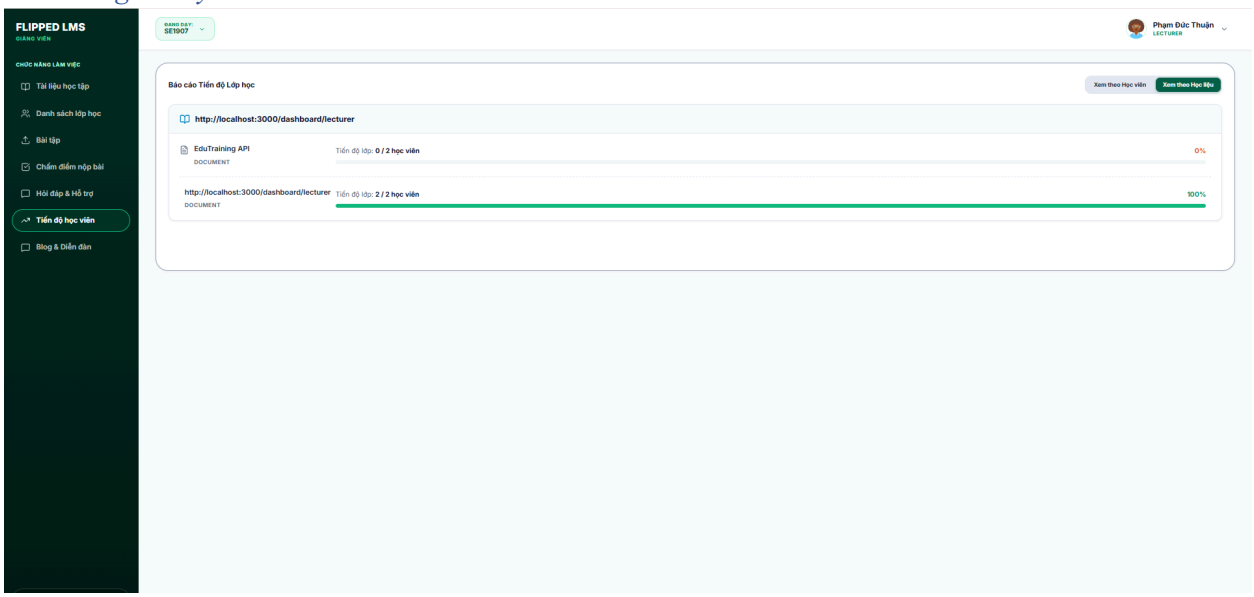
- Search for specific students and view their overall completion percentage.
- Click on a student to open a modal details listing Completed vs Pending materials.

Field Description

Field Name	Description
Student Search	Text input field to search students by name.
Progress Bar	Visual gauge showing the completion percentage of each student.
Completed Tab	Toggle sub-tab in details modal to view completed materials.
Pending Tab	Toggle sub-tab in details modal to view pending materials.

Mapped Use Case: UC-39: View Student Progress, UC-40: Track Student

5.7.2. Progress by Material



This screen allows the User to:

- Monitor completion rates of study materials across the entire class.

On the screen, the person can also:

- View each chapter and material alongside the percentage of students who completed it.

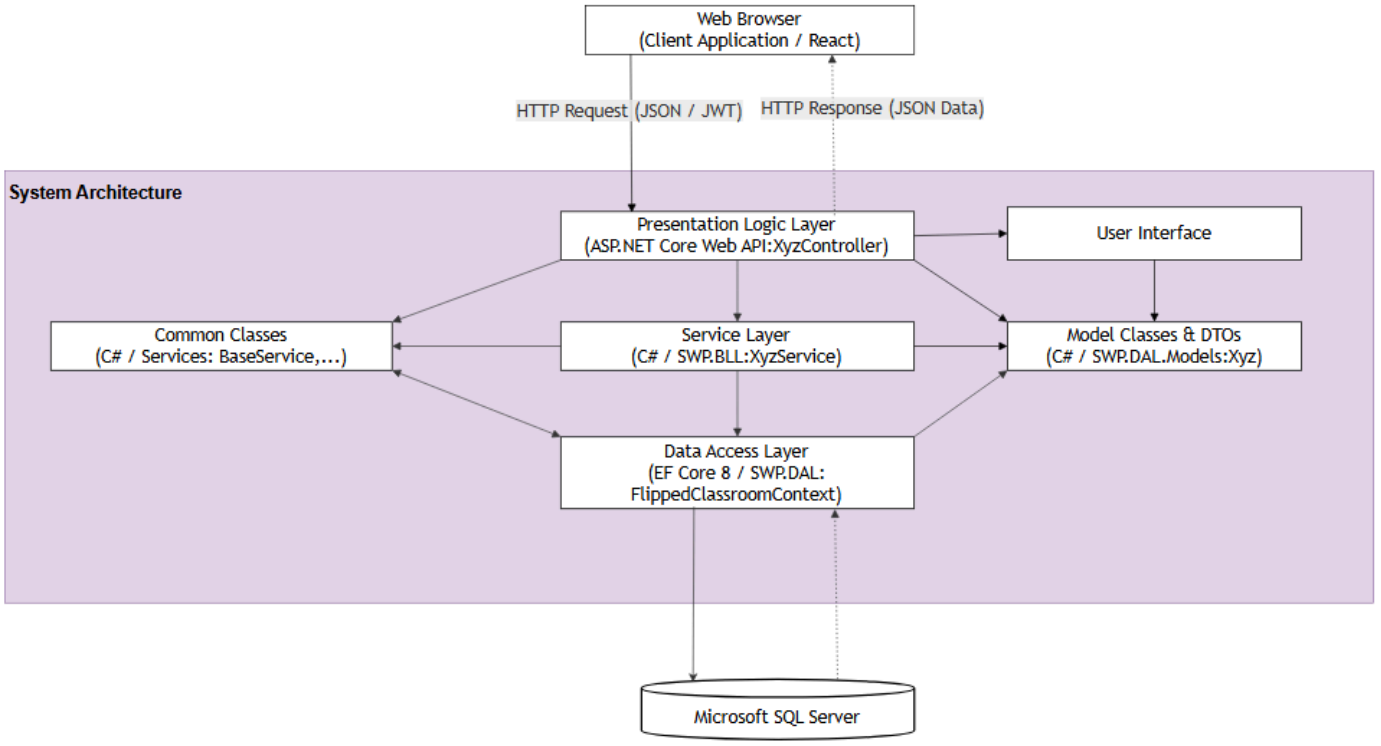
Field Description

Field Name	Description
Material Completion Rate	Visual gauge showing percentage of students who completed each material.
Completed Count Label	Text showing the exact ratio of completed students (e.g., 8/10 students).

Mapped Use Case: UC-39: View Student Progress

III. System Design

1. Software Architecture



1.1 Overall Architectural Diagram

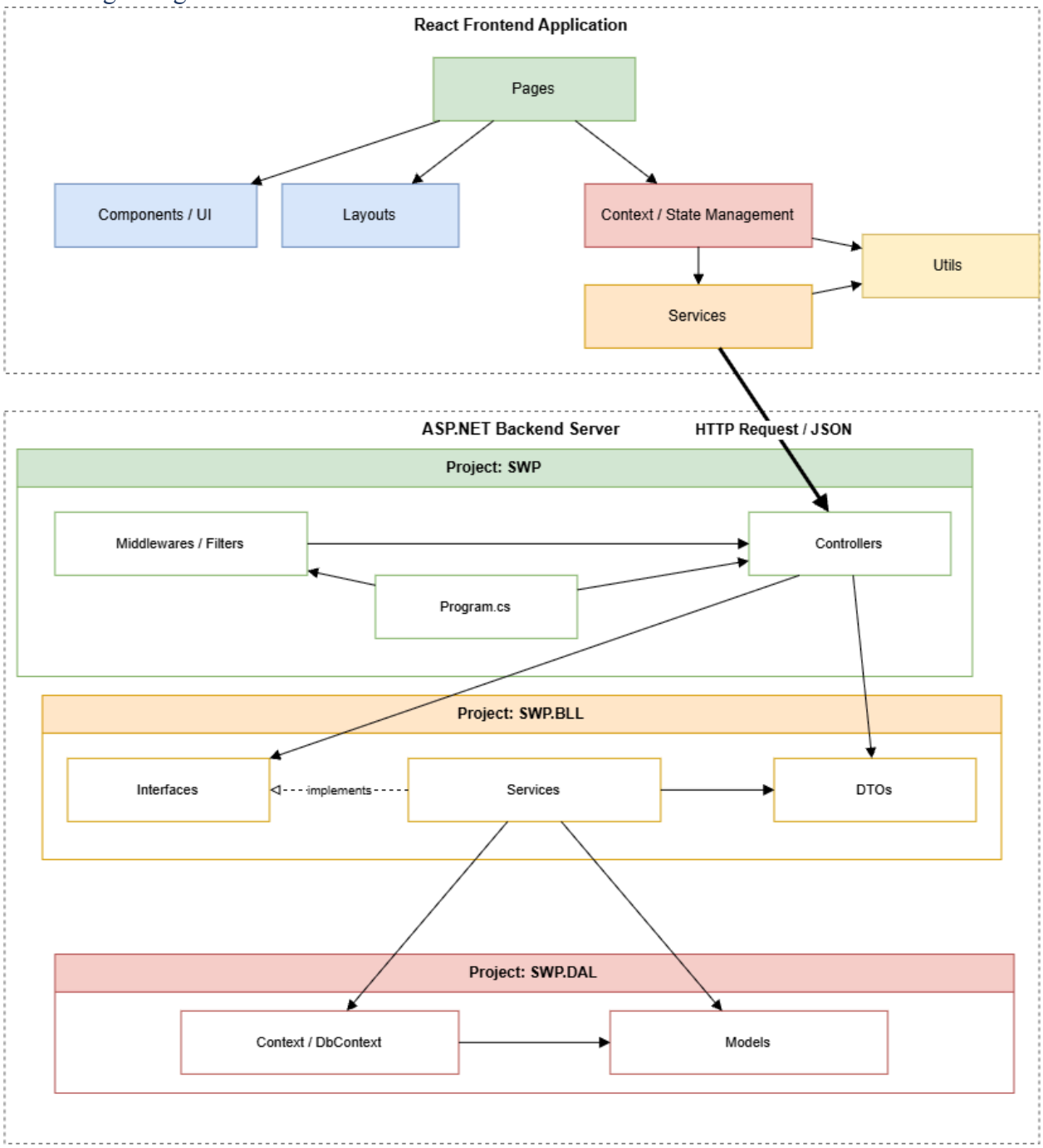
1.2 Component Descriptions

No	Component	Description
01	Web Browser	The client-side interface that initiates interactions by sending HTTP Requests and rendering HTTP Responses for the end-user.
02	Presentation Logic Layer	Acts as the API controller layer (ASP.NET Core Web API) that handles incoming HTTP requests, validates input, interacts with the Service layer, and returns JSON responses.
03	User Interface	The view component of the system (React SPA). It presents data to the user and receives information from the Model and Presentation layers.
04	Service Layer	Contains the core business logic of the application (SWP.BLL). It serves as an intermediary, coordinating data flow between the Presentation Logic and the Data Access Layer.
05	Model Classes & DTOs	Represents the data entities and Data Transfer Objects (SWP.DAL.Models & SWP.BLL.DTOs). These classes

		define the structure of the data used across the UI, Service, and Data Access layers.
06	Data Access Layer	Responsible for direct communication with the database. It uses Entity Framework Core (EF Core 8) to perform CRUD operations (Create, Read, Update, Delete).
07	Common Classes	A set of shared utility or helper classes providing reusable functionality (such as email service and JSON converters) accessed across the system layers.
08	Microsoft SQL Server	The relational database management system (RDBMS) where all persistent system data is stored and managed.

2. Code Package Design

2.1 Package Diagram

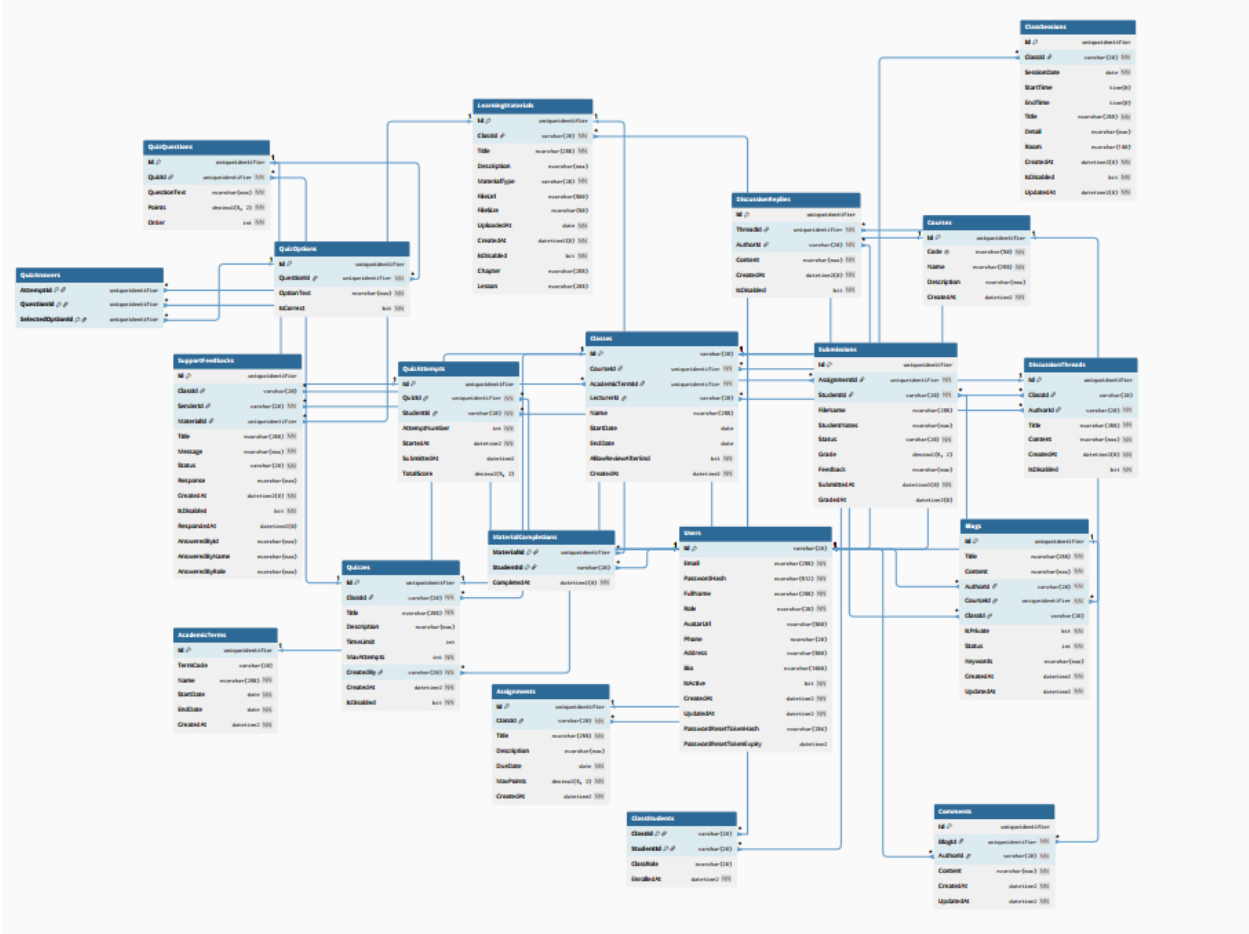


No	Package	Description
01	Pages	Contains the complete pages/screens of the React application (e.g., Home, Dashboard, Login). It acts as the coordinator that assembles child components and fetches data from the Context to render for the user.

02	Components	Contains small, independent, and highly reusable UI elements (e.g., Buttons, Inputs, Tables, Modals) shared across various pages.
03	Layouts	Defines the common structural layouts of the web application (e.g., sidebar, header, footer) that wrap the content of nested pages.
04	Context / State Management	Manages and stores global application states shared among components and pages (e.g., current logged-in user info, current class data), enabling smooth data flow without prop-drilling.
05	Services (FE)	Contains files responsible for declaring API call functions (using Axios/Fetch) to send HTTP network requests (GET, POST, PUT, DELETE) to the Backend Server.
06	Utils (FE)	Contains general helper and utility functions used across the Frontend project (e.g., saving/getting tokens from LocalStorage, currency formatting, date-time processing).
07	Program.cs	The entry point of the Backend .NET application. It configures the Web Host, registers services into the IoC Container (Dependency Injection), and sets up the HTTP request pipeline (Middlewares).
08	Middlewares / Filters	Contains intermediate request processing components executed before the requests hit the Controllers (e.g., JWT authentication/authorization filter, global exception handler).
09	Controllers	The primary entry points for receiving HTTP API requests from the Client (Frontend). They validate input DTOs, forward tasks to the Business Logic Layer (BLL), and return the results to the Client.
10	Interfaces	Declares the service contracts. They allow the API layer to interact with BLL services without knowing their internal implementation details, ensuring Loose Coupling.
11	Services (BLL)	The core of the Business Logic Layer (BLL). It processes system business logic, applies validation rules, performs computations, and interacts with the database through the Data Access Layer (DAL)
12	DTOs (Data Transfer Objects)	Defines the data structures transferred between Client and Server. They control exactly what information is input or output, preventing sensitive database fields from being exposed.
13	Context / DbContext (DAL)	Represents the database context in the application (using Entity Framework Core). It manages connections to the SQL Server database, maps entities to tables, and tracks transactions.
14	Models / Entities	Defines the entity data structures mapping directly to tables in the physical database (e.g., User, Course, ClassStudent) and their corresponding relationships.

3. Database Design

3.1 Database Schema



[Link Diagram]

3.2 Table Descriptions

3.2.1 Users

No	Field	PK	FK	UN	NN	Description
01	*ID*	*x*		*x*	*x*	*Primary key. Unique user identifier (varchar(20)).*
02	*Email*			*x*	*x*	*Login email address, unique across the system (nvarchar(255)).*
03	*PasswordHash*				*x*	*Hashed password for security (nvarchar(512)).*
04	*FullName*				*x*	*Full name of the user (nvarchar(255)).*
05	*Role*				*x*	*User role. Allowed values: 'admin', 'lecturer', 'student'.*

06	*AvatarUrl*					*URL path to the user's profile picture. Nullable (nvarchar(500)).*
07	*Phone*					*Phone number.*
08	*Address*					*Home address.*
09	*Bio*					*User biography.*
10	*IsActive*				*x*	*Account status. 1 = Active, 0 = Disabled/Soft-deleted. Default is 1.*
11	*CreatedAt*				*x*	*Timestamp when the account was created. Defaults to system datetime.*
12	*UpdatedAt*				*x*	*Timestamp when the account was last updated. Defaults to system datetime.*
13	*PasswordReset TokenHash*					*Password reset token hash.*
14	*PasswordReset TokenExpiry*					*Password reset token expiration.*

3.2.3 AcademicTerms

No	Field	PK	FK	UN	NN	Description
01	*ID*	*x*		*x*	*x*	*Primary key, auto-generated unique identifier (NEWSEQUENTIALID).*
02	*TermCode*			*x*		*Unique code assigned to the academic term (varchar(20)).*
03	*Name*				*x*	*Name of the academic term (nvarchar(255)).*
04	*StartDate*				*x*	*Starting date of the academic term (date).*
05	*EndDate*				*x*	*Ending date of the academic term (date).*
06	*CreatedAt*				*x*	*Date and time when the academic term was created (datetime2(7)).*

3.2.3 Courses

No	Field	PK	FK	UN	NN	Description
01	*ID*	*x*		*x*	*x*	*Primary key. Course identifier (uniqueidentifier).*
02	*Code*			*x*	*x*	*Unique course code.*

03	*Name*				*x*	*Course name.*
04	*Description*				*x*	*Detailed description of the course.*
05	*CreatedAt*				*x*	*Date and time when the course was created.*

3.2.4 Classes

No	Field	PK	FK	UN	NN	Description
01	*ID*	*x*		*x*	*x*	*Unique identifier of the class*
02	*CourseId*		*x*		*x*	*Foreign key referencing the Courses table.*
03	*AcademicTerm Id*		*x*		*x*	*Foreign key referencing the AcademicTerms table.*
04	*LecturerId*		*x*			*Foreign key referencing the Users table (the assigned lecturer).*
05	*Name*					*Specific name of the class.*
06	*StartDate*					*Start date of the class.*
07	*EndDate*					*End date of the class.*
08	*AllowReview AfterEnd*				*x*	*Indicates if students can review materials after the class ends*
09	*CreatedAt*				*x*	*Timestamp when the class record was created.*

3.2.5 ClassStudents

No	Field	PK	FK	UN	NN	Description
01	*ClassId*	*x*		*x*	*x*	*Foreign key referencing the Classes table. Combined with StudentId to form the composite primary key.*
02	*StudentId*	*x*		*x*	*x*	*Foreign key referencing the Users (or Students) table. Combined with ClassId to form the composite primary key.*
03	*ClassRole*				*x*	*The role of the user within this specific class. Allowed values: 'student' or 'assistant'.*
04	*EnrolledAt*				*x*	*The timestamp when the user joined the class. Automatically generated upon record creation.*

3.2.6 LearningMaterials

No	Field	PK	FK	UN	NN	Description
01	*Id*	*x*		*x*	*x*	*Primary key, unique identifier for each learning material record (typically using GUID/UniqueIdentifier format).*
02	*ClassId*		*x*		*x*	*Foreign key referencing the Classes table, indicating which specific class this material belongs to.*
03	*Title*				*x*	*The title or display name of the learning material (e.g., lecture topic or video title).*
04	*Description*				*x*	*Detailed summary or metadata of the material, stored as a JSON string format to support flexible properties.*
05	*MaterialType*				*x*	*The format classification of the material. Allowed values include: 'video', 'pdf', 'document', etc.*
06	*FileUrl*				*x*	*The storage path, URL link, or file reference to the uploaded material asset. Can be null if no file is attached.*
07	*FileSize*				*x*	*The total storage size of the uploaded file (e.g., displayed in MB or KB).*
08	*UploadedAt*				*x*	*The exact timestamp when the material file was successfully uploaded to the system.*
09	*CreatedAt*				*x*	*The database record creation timestamp, recording when the material entry was first generated.*
10	*IsDisabled*				*x*	*Status indicator for visibility. '1' means hidden/disabled; '0' means active and visible to students.*
11	*Chapter*				*x*	*The specific textbook chapter or course section name that this material is associated with.*
12	*Lesson*					*The specific lesson name, number, or sub-topic within a chapter that this material corresponds to. Can be null if the material covers the entire chapter.*

3.2.7 MaterialCompletions

No	Field	PK	FK	UN	NN	Description
01	MaterialId	x	x		x	Foreign key referencing the LearningMaterials table.
02	StudentId	x	x		x	Foreign key referencing the Users table (the student who completed it).
03	CompletedAt				x	Timestamp when the student completed the material.

3.2.8 QuizAnswers

No	Field	PK	FK	UN	NN	Description
01	*AttemptId*	*x*	*x*		*x*	*Foreign key referencing the QuizAttempts table.*
02	*QuestionId*	*x*	*x*		*x*	*Foreign key referencing the QuizQuestions table.*
03	*SelectedOptionId*	*x*	*x*		*x*	*Foreign key referencing the QuizOptions table (the option selected by the student).*

3.2.9 QuizAttempts

No	Field	PK	FK	UN	NN	Description
01	*Id*	*x*			*x*	*Unique identifier of the quiz attempt.*
02	*QuizId*		*x*		*x*	*Foreign key referencing the Quizzes table.*
03	*StudentId*		*x*		*x*	*Foreign key referencing the Users table (the student taking the quiz).*
04	*AttemptNumber*				*x*	*The sequential number of the attempt (1st, 2nd, etc.).*
05	*StartedAt*				*x*	*Timestamp when the attempt started.*
06	*SubmittedAt*					*Timestamp when the quiz was submitted.*
07	*TotalScore*					*The total score achieved for this attempt.*

3.2.10 QuizOptions

No	Field	PK	FK	UN	NN	Description
01	Id	x			x	Unique identifier of the quiz option.
02	QuestionId		x		x	Foreign key referencing the QuizQuestions table.
03	OptionText				x	The text/content of the answer option.
04	IsCorrect				x	Indicates if this option is the correct answer (True/False).

3.2.11 Quizzes

No	Field	PK	FK	UN	NN	Description
01	Id	x			x	Unique identifier of the quiz.
02	Title		x		x	Title of the quiz.
03	Description				x	Description or instructions for the quiz.
04	TimeLimit					Time limit for taking the quiz (in minutes).
05	MaxAttempts					Maximum number of attempts allowed for students.
06	CreatedBy		x		x	Foreign key referencing the Users table (the creator of the quiz).
07	CreatedAt				x	Timestamp when the quiz was created.
08	IsDisabled				x	Indicates if the quiz is hidden or disabled.

3.2.12 Submissions

No	Field	PK	FK	UN	NN	Description
01	*Id*	*x*		*x*	*x*	*Primary key, unique identifier for each submission record (typically using GUID/UniqueIdentifier format).*
02	*AssignmentId*		*x*	*x*	*x*	*Foreign key referencing the Assignments table, linking the submission to a specific task.*
03	*StudentId*		*x*		*x*	*Foreign key referencing the Users table, representing the student who owns this submission.*
04	*FileName*				*x*	*The file name, path, or reference URL of the uploaded student work. This field is nullable.*
05	*StudentNotes*				*x*	*Additional textual comments or notes provided by the student during submission. Nullable.*
06	*Status*				*x*	*The evaluation/workflow status of the submission (e.g., 'Submitted', 'Graded', 'Draft').*
07	*Grade*				*x*	*The score awarded by the lecturer. Nullable if the submission has not been graded yet.*
08	*Feedback*				*x*	*The review comments, corrections, or feedback provided by the lecturer. Nullable.*
09	*SubmittedAt*					*The precise database timestamp when the student officially submitted the work.*
10	*GradedAt*					*The timestamp indicating when the grade and feedback were saved by the lecturer. Nullable.*

3.2.13 SupportFeedbacks

No	Field	PK	FK	UN	NN	Description
01	*Id*	*x*			*x*	*Unique identifier of the support feedback/ticket.*
02	*ClassId*		*x*			*Foreign key referencing the Classes table (optional context).*

03	*SenderId*		*x*		*x*	*Foreign key referencing the Users table (the sender).*
04	*Title*				*x*	*Title or subject of the feedback/support request.*
05	*Message*				*x*	*Detailed message of the feedback.*
06	*Status*				*x*	*Current status of the feedback ('OPEN' or 'RESPONDED').*
07	*Response*					*The response message provided by support/admin.*
08	*CreatedAt*				*x*	*Timestamp when the feedback was submitted.*
09	*IsDisabled*				*x*	*Indicates if the feedback record is disabled/soft-deleted.*
10	*RespondedAt*					*Timestamp when the feedback received a response.*

3.2.14 Comments

No	Field	PK	FK	UN	NN	Description
01	*Id*	*x*			*x*	*Unique identifier of the comment.*
02	*BlogId*		*x*		*x*	*Foreign key referencing the Blogs table.*
03	*AuthorId*		*x*		*x*	*Foreign key referencing the Users table (the commenter).*
04	*Content*				*x*	*The text content of the comment.*
05	*CreatedAt*				*x*	*Timestamp when the comment was created.*
06	*UpdatedAt*				*x*	*Timestamp of the last update to the comment.*

3.2.15 DiscussionReplies

No	Field	PK	FK	UN	NN	Description
01	*Id*	*x*			*x*	*Unique identifier of the discussion reply.*
02	*ThreadId*		*x*		*x*	*Foreign key referencing the DiscussionThreads table.*

03	*AuthorId*		*x*		*x*	*Foreign key referencing the Users table (the replier).*
04	*Content*				*x*	*Content of the reply message.*
05	*CreatedAt*				*x*	*Timestamp when the reply was posted.*
06	*IsDisabled*				*x*	*Indicates if the reply is disabled/hidden.*

3.2.16 DiscussionThreads

No	Field	PK	FK	UN	NN	Description
01	*Id*	*x*			*x*	*Unique identifier of the discussion thread.*
02	*ClassId*		*x*			*Foreign key referencing the Classes table (optional context).*
03	*AuthorId*		*x*		*x*	*Foreign key referencing the Users table (the thread creator).*
04	*Title*				*x*	*Title of the discussion thread.*
05	*Content*				*x*	*Main content or starting message of the thread.*
06	*CreatedAt*				*x*	*Timestamp when the thread was created.*
07	*IsDisabled*				*x*	*Indicates if the thread is disabled/hidden.*

3.2.17 QuizQuestions

No	Field	PK	FK	UN	NN	Description
01	*Id*	*x*			*x*	*Unique identifier of the quiz question.*
02	*QuizId*		*x*		*x*	*Foreign key referencing the Quizzes table.*
03	*QuestionText*				*x*	*The actual text/content of the question.*
04	*Points*				*x*	*The point value assigned to this question.*
05	*Order*				*x*	*The display order of the question within the quiz.*

3.2.18 Assignments

No	Field	PK	FK	UN	NN	Description
01	Id	x		x	x	Primary key. Unique identifier of the assignment (uniqueidentifier).
02	ClassId		x		x	Foreign key. References the class to which the assignment belongs (varchar(20)).
03	Title				x	Title of the assignment (nvarchar(255)).
04	Description					Detailed description or instructions for the assignment (nvarchar(MAX)).
05	DueDate				x	Deadline for submitting the assignment (date).
06	MaxPoints				x	Maximum score that can be awarded for the assignment (decimal(5,2)).
07	CreatedAt				x	Date and time when the assignment was created (datetime2(7)).

3.2.19 Blogs

No	Field	PK	FK	UN	NN	Description
01	Id	x			x	Unique identifier of the blog post.
02	Title				x	Title of the blog post.
03	Content				x	Main content of the blog post.
04	AuthorId		x		x	Foreign key referencing the Users table (the author).
05	CourseId		x		x	Foreign key referencing the Courses table.
06	ClassId		x			Foreign key referencing the Classes table (optional).
07	IsPrivate				x	Indicates if the blog is private or public (True/False).
08	Status				x	Current status of the blog (e.g., draft, published).
09	Keywords					Keywords or tags associated with the blog.
10	CreatedAt				x	Timestamp when the blog was created.

11	*UpdatedAt*				*x*	*Timestamp of the last update to the blog.*

3.2.20 ClassSessions

No	Field	PK	FK	UN	NN	Description
01	*Id*	*x*			*x*	*Unique identifier of the class session.*
02	*ClassId*		*x*		*x*	*Foreign key referencing the Classes table.*
03	*SessionDate*				*x*	*The date the session takes place.*
04	*StartTime*					*The starting time of the session.*
05	*EndTime*					*The ending time of the session.*
06	*Title*				*x*	*Topic or title of the session.*
07	*Detail*					*Detailed content of the session.*
08	*Room*					*The room where the session is held.*
09	*CreatedAt*				*x*	*Timestamp when the session record was created.*
10	*IsDisabled*				*x*	*Indicates if the session is disabled/soft-deleted.*
11	*UpdatedAt*				*x*	*Timestamp of the last update to the session information.*